第二篇 线性代数与空间解析几何

第四章 矩阵和线性方程组

第五章 线性空间和线性变换

§1. 线性空间

习题. P.257.

1. 下列集合是否构成向量空间.

(1) $S=\{(x,y,z)\mid x=y=z\}$.

解: 对于任意的 $\vec{x}_1=(x_1,y_1,z_1)\in S$ $(x_1=y_1=z_1)$ 则有: $x_1+x_2=y_1+y_2=z_1+z_2$.

和 $\vec{x}_2=(x_2,y_2,z_2)\in S$ $(x_2=y_2=z_2)$ 即: $\vec{x}_1+\vec{x}_2\in S$.

而 $\lambda\vec{x}_1=(\lambda x_1,\lambda y_1,\lambda z_1)$ 满足 $\lambda x_1=\lambda y_1=\lambda z_1$. 即 $\lambda x_1\in S$.

$\therefore$ S 为 R^3 的子空间.

(2) $S=\{(x,y,z)\mid y=3x-2z\}$

解: 对于任意的 $\vec{x}_1=(x_1,y_1,z_1)\in S$: $y_1=3x_1-2z_1$

$\vec{x}_2=(x_2,y_2,z_2)\in S$: $y_2=3x_2-2z_2$

成立: $\vec{x}_1+\vec{x}_2=(x_1+x_2,y_1+y_2,z_1+z_2)$ 且 $y_1+y_2=3(x_1+x_2)-2(z_1+z_2)$.

即 $\vec{x}_1+\vec{x}_2\in S$.

$\lambda\vec{x}_1=(\lambda x_1,\lambda y_1,\lambda z_1)$. 且 $\lambda y_1=3\lambda x_1-2\lambda z_1$.

即 $\lambda\vec{x}_1\in S$.

$\therefore$ S 为 R^3 的子空间.

(3) $S=\{\vec{x}\mid A\vec{x}=a\vec{x},\ \vec{x}\in R^n,\ A$ 是给定的矩阵, a 是给定的常数$\}$.

解: 对于任意的 $\vec{x}_1\in S$: $A\vec{x}_1=a\vec{x}_1$. 成立: $A(\vec{x}_1+\vec{x}_2)=a(\vec{x}_1+\vec{x}_2)$

$\vec{x}_2\in S$: $A\vec{x}_2=a\vec{x}_2$. $A(\lambda\vec{x}_1)=a(\lambda\vec{x}_1)$

即: $\vec{x}_1+\vec{x}_2\in S$. $\lambda\vec{x}_1\in S$.

$\therefore$ S 为 R^n 的子空间.

(4) $S=\{\vec{x}\mid A\vec{x}=\vec{b},\ \vec{x}\in R^n,\ A$ 是给定的矩阵, a 是给定的常数$\}$

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解：对于任意的 $\vec{x_1} \in S$：$A\vec{x_1} = \vec{b_1}$ 且 $A(\vec{x_1}+\vec{x_2}) = 2\vec{b_1}$.

$\vec{x_2} \in S$：$A\vec{x_2} = \vec{b_1}$. 即 $\vec{x_1}+\vec{x_2} \notin S$

∴ S 不是 R^n 的子空间.

2. 对通常的加法与数乘运算，下列集合 V 是否构成线性空间：

(1) $V = \{ f \mid f(x) = A\sin(\omega x+\varphi),\ A, \omega, \varphi \in R \}$.

解：对于任意的 $f_1 \in V$. 设 $f_1(x) = A_1\sin(\omega_1 x+\varphi_1)$. $A_1, \omega_1, \varphi_1 \in R$.

$f_2 \in V$. 设 $f_2(x) = A_2\sin(\omega_2 x+\varphi_2)$.

则：$f_1(x)+f_2(x) = A_1\sin(\omega_1 x+\varphi_1) + A_2\sin(\omega_2 x+\varphi_2)$

$\neq A_3\sin(\omega_3 x+\varphi_3)$.

即 $f_1(x)+f_2(x) \notin V$. ∴ V 不是线性空间

(2) $V = \{ f \mid f$ 在 $[0,1]$ 连续 $\}$.

解：对于任意的 $f_1 \in V$. 有 f_1 在 $[0,1]$ 连续. $\Rightarrow f_1+f_2$ 在 $[0,1]$ 连续，αf_1 在 $[0,1]$ 连续.

$f_2 \in V$. 有 f_2 在 $[0,1]$ 连续.

即 $f_1+f_2 \in V$. $\alpha f_1 \in V$. ∴ V 是线性空间.

(3) $V = \{ f \mid$ 当 $x \to \infty$ 时，$f(x) \to a$，a 是常数 $\}$.

解：对于任意的 $f_1 \in V$. 有 $\lim\limits_{x\to\infty} f_1(x) = a$.

$f_2 \in V$. 有 $\lim\limits_{x\to\infty} f_2(x) = a$.

$\Rightarrow \lim\limits_{x\to\infty}[f_1(x)+f_2(x)] = 2a$.

$\lim\limits_{x\to\infty} \alpha f_1(x) = \lambda \lim\limits_{x\to\infty} f_1(x) = \lambda a$.

即 $f_1+f_2 \notin V$ $\lambda f_1 \notin V$

∴ V 不是线性空间. $(a \neq 0)$. 当 $a=0$ 时，V 是线性空间.

(4) $V = \{ f \mid$ 对于任意 x，$f(x) = f(1-x)$ 成立 $\}$.

解：对于任意的 $f_1 \in V$. 有 $f_1(x) = f_1(1-x)$

$f_2 \in V$. 有 $f_2(x) = f_2(1-x)$.

$\Rightarrow f_1(x)+f_2(x) = f_1(1-x)+f_2(1-x)$.

$\lambda f_1(x) = \lambda f_1(1-x)$

即 $f_1+f_2 \in V$. $\lambda f_1 \in V$ ∴ V 是线性空间.

3、证明定理 5.1.1 中的(3).

定理 5.1.1 中的(3)：若 V 是线性空间，则 $0\cdot\vec{x}=\vec{0}$，$(-1)\cdot\vec{x}=-\vec{x}$，$\lambda\cdot\vec{0}=\vec{0}$.

证明：由线性空间的定义可知：

$$0\cdot\vec{x}+\vec{x}=0\cdot\vec{x}+1\cdot\vec{x}=(0+1)\cdot\vec{x}=1\cdot\vec{x}=\vec{x}$$

两边加上 $(-\vec{x})$，则有：$0\cdot\vec{x}+\vec{x}+(-\vec{x})=0\cdot\vec{x}+[\vec{x}+(-\vec{x})]=0\cdot\vec{x}$

$\Rightarrow \vec{x}+(-\vec{x})=\vec{0}$（零元）$\Rightarrow 0\cdot\vec{x}=\vec{0}$（零元）

$$\vec{x}+(-1)\cdot\vec{x}=1\cdot\vec{x}+(-1)\cdot\vec{x}=[1+(-1)]\vec{x}=0\cdot\vec{x}=\vec{0}$$

由于 $\vec{x}$ 的负元唯一，$\therefore (-1)\cdot\vec{x}=-\vec{x}$.（负元）

$$\lambda\cdot\vec{0}+\lambda\cdot\vec{0}=\lambda(\vec{0}+\vec{0})=\lambda\cdot\vec{0}.$$

两边加上 $-\lambda\cdot\vec{0}$，则有：$\lambda\cdot\vec{0}+\lambda\cdot\vec{0}+(-\lambda\cdot\vec{0})=\lambda\cdot\vec{0}+(-\lambda\cdot\vec{0})=(\lambda-\lambda)\cdot\vec{0}=\vec{0}$

$\therefore \lambda\cdot\vec{0}=\vec{0}$，即 $\lambda\cdot$零元 = 零元.

4、求线性空间 $S=\{\vec{x}\mid \vec{x}\in R^3,\ x_1=x_2=x_3\}$ 的一组基及其维数.

解：任意的 $\vec{x}\in S$，有 $\vec{x}=\begin{pmatrix}x\\x\\x\end{pmatrix}=x\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\therefore \begin{pmatrix}1\\1\\1\end{pmatrix}$ 为 S 的一组基，$\dim S=1$.

5、设 S 为 R^n 中的 k 个向量 $\vec{a}_1,\vec{a}_2,\cdots\vec{a}_k$ 的所有线性组合构成的集合，$\vec{a}_1,\vec{a}_2,\cdots,\vec{a}_k$ 线性无关，记 R^n 的子空间 T 为 $T=\{\vec{x}\mid \vec{x}\in S,\ x_n=0\}$. 证明：$T$ 的维数等于 $k-1$ 或 k.

证明：$\because \vec{a}_1,\vec{a}_2,\cdots\vec{a}_k$ 线性无关 $\therefore \vec{a}_1,\vec{a}_2,\cdots\vec{a}_k$ 是 S 的一组基，$\dim S=k$

令 $A=(\vec{a}_1,\vec{a}_2,\cdots,\vec{a}_k)$，则 $A\in R^{n\times k}$，且有 $k\le n$，$\mathrm{rank}(A)=k$.

$\because T=\{\vec{x}\mid \vec{x}\in S,\ x_n=0\}$，令 $\vec{a}_i=(a_{1i},a_{2i},\cdots a_{n-1,i},a_{ni})^T$.

$$A=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1k}\\a_{21}&a_{22}&\cdots&a_{2k}\\\vdots&\vdots&&\vdots\\a_{n-1,1}&a_{n-1,2}&\cdots&a_{n-1,k}\\a_{n1}&a_{n2}&\cdots&a_{nk}\end{bmatrix}_{n\times k}=(\vec{a}_1,\vec{a}_2,\cdots\vec{a}_k),\qquad \widetilde{A}=\begin{bmatrix}a_{11}&a_{12}&\cdots&a_{1k}\\a_{21}&a_{22}&\cdots&a_{2k}\\\vdots&\vdots&&\vdots\\a_{n-1,1}&a_{n-1,2}&\cdots&a_{n-1,k}\\0&0&\cdots&0\end{bmatrix}_{n\times k}=(\widetilde{a}_1,\widetilde{a}_2,\cdots\widetilde{a}_k)$$

显然有：$A=\widetilde{A}+\begin{bmatrix}0&0&\cdots&0\\0&0&\cdots&0\\\vdots&\vdots&&\vdots\\0&0&\cdots&0\\a_{n1}&a_{n2}&\cdots&a_{nk}\end{bmatrix}=\widetilde{A}+\widetilde{B}$，且 $\mathrm{rank}(\widetilde{B})\le 1$.

则：$\mathrm{rank}(A)=\mathrm{rank}(\widetilde{A}+\widetilde{B})\le \mathrm{rank}(\widetilde{A})+\mathrm{rank}(\widetilde{B})$

$\because \mathrm{rank}(\widetilde{A})\ge \mathrm{rank}(A)-\mathrm{rank}(\widetilde{B})\ge k-1$，而 $\mathrm{rank}(\widetilde{A})\le k$

$\therefore k-1\le \mathrm{rank}(\widetilde{A})\le k$

$\therefore T$ 的维数等于 $k-1$ 或 k.

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6. 已知 $\vec{\varepsilon}_1=\begin{pmatrix}1\\-1\\1\end{pmatrix}$, $\vec{\varepsilon}_2=\begin{pmatrix}1\\-1\\0\end{pmatrix}$, $\vec{\varepsilon}_3=\begin{pmatrix}1\\2\\-1\end{pmatrix}$ 和 $\tilde{\vec{\varepsilon}}_1=\begin{pmatrix}1\\4\\0\end{pmatrix}$, $\tilde{\vec{\varepsilon}}_2=\begin{pmatrix}2\\-1\\3\end{pmatrix}$, $\tilde{\vec{\varepsilon}}_3=\begin{pmatrix}1\\-4\\1\end{pmatrix}$

是 R^3 中的两组基.

(1) 求由基 $\{\vec{e}_1,\vec{e}_2,\vec{e}_3\}$ 到 $\{\vec{\varepsilon}_1,\vec{\varepsilon}_2,\vec{\varepsilon}_3\}$ 的过渡矩阵;

(2) 求 $\{\vec{\varepsilon}_1,\vec{\varepsilon}_2,\vec{\varepsilon}_3\}$ 到 $\{\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3\}$ 的过渡矩阵;

(3) 求 $\vec{x}=(1,1,1)^T$ 在 $\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3$ 下的坐标;

(4) 求 $\vec{x}=3\vec{\varepsilon}_1-2\vec{\varepsilon}_2+\vec{\varepsilon}_3$ 在 $\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3$ 下的坐标.

解: 令 $E=(\vec{\varepsilon}_1,\vec{\varepsilon}_2,\vec{\varepsilon}_3)=\begin{bmatrix}1&1&1\\1&-1&2\\-1&0&-1\end{bmatrix}$ $\tilde{E}=(\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3)=\begin{bmatrix}1&2&1\\4&-1&-4\\0&3&1\end{bmatrix}$.

(1) 设 $\tilde{E}=(\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3)=(\vec{e}_1\ \vec{e}_2\ \vec{e}_3)T_1=I\cdot T_1=T_1$. $\therefore T_1=\tilde{E}=\begin{bmatrix}1&2&1\\4&-1&-4\\0&3&1\end{bmatrix}$.

(2) 设 $\tilde{E}=(\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3)=(\vec{\varepsilon}_1,\vec{\varepsilon}_2,\vec{\varepsilon}_3)T_2=ET_2$. $\therefore T_2=E^{-1}\cdot\tilde{E}$.

$$(E\mid I)=\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\1&-1&2&0&1&0\\1&0&-1&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&-2&1&-1&1&0\\0&1&-2&-1&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&1&-2&1&0&-1\\0&0&-5&-1&-1&2\end{array}\right)$$

$$\to\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&1&-2&1&0&-1\\0&0&1&\frac{1}{5}&\frac{1}{5}&-\frac{2}{5}\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&1&0&\frac{4}{5}&-\frac{1}{5}&\frac{2}{5}\\0&1&0&\frac{3}{5}&-\frac{2}{5}&-\frac{1}{5}\\0&0&1&\frac{1}{5}&\frac{1}{5}&-\frac{2}{5}\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&0&0&\frac{1}{5}&\frac{1}{5}&\frac{3}{5}\\0&1&0&\frac{3}{5}&-\frac{2}{5}&-\frac{1}{5}\\0&0&1&\frac{1}{5}&\frac{1}{5}&-\frac{2}{5}\end{array}\right)$$

即 $E^{-1}=\frac{1}{5}\begin{bmatrix}1&1&3\\3&-2&-1\\1&1&-2\end{bmatrix}$ $\therefore T_2=E^{-1}\tilde{E}=\frac{1}{5}\begin{bmatrix}1&1&3\\3&-2&-1\\1&1&-2\end{bmatrix}\begin{bmatrix}1&2&1\\4&-1&-4\\0&3&1\end{bmatrix}=\begin{bmatrix}1&2&0\\-1&1&2\\1&-1&-1\end{bmatrix}$

(3) 设 $\vec{x}=\begin{pmatrix}1\\1\\1\end{pmatrix}=(\tilde{\vec{\varepsilon}}_1,\tilde{\vec{\varepsilon}}_2,\tilde{\vec{\varepsilon}}_3)\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix}=\tilde{E}\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix}$ $\therefore\begin{pmatrix}\alpha_1\\\alpha_2\\\alpha_3\end{pmatrix}=\tilde{E}^{-1}\cdot\vec{x}$.

$$(\tilde{E}\mid I)=\left(\begin{array}{ccc|ccc}1&2&1&1&0&0\\4&-1&-4&0&1&0\\0&3&1&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&2&1&1&0&0\\0&-9&-8&-4&1&0\\0&3&1&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&2&1&1&0&0\\0&3&1&0&0&1\\0&0&5&4&1&-3\end{array}\right)$$

$$\to\left(\begin{array}{ccc|ccc}1&2&1&1&0&0\\0&1&\frac{1}{3}&0&0&\frac{1}{3}\\0&0&1&\frac{4}{5}&-\frac{1}{5}&-\frac{3}{5}\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&2&0&\frac{1}{5}&\frac{1}{5}&\frac{3}{5}\\0&1&0&-\frac{4}{15}&\frac{1}{15}&\frac{8}{15}\\0&0&1&\frac{4}{5}&-\frac{1}{5}&-\frac{3}{5}\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&0&0&\frac{11}{15}&\frac{1}{15}&-\frac{7}{15}\\0&1&0&-\frac{4}{15}&\frac{1}{15}&\frac{8}{15}\\0&0&1&\frac{4}{5}&-\frac{1}{5}&-\frac{3}{5}\end{array}\right)$$

即 $\tilde{E}^{-1}=\begin{pmatrix}\frac{11}{15} & \frac{1}{15} & -\frac{7}{15}\\ -\frac{4}{15} & \frac{1}{15} & \frac{8}{15}\\ \frac{4}{5} & -\frac{1}{5} & -\frac{3}{5}\end{pmatrix}=\frac{1}{15}\begin{bmatrix}11 & 1 & -7\\ -4 & 1 & 8\\ 12 & -3 & -9\end{bmatrix}$ $\therefore \begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix}=\tilde{E}^{-1}\cdot\vec{x}$

$$\begin{pmatrix}\alpha_1\\ \alpha_2\\ \alpha_3\end{pmatrix}=\frac{1}{15}\begin{bmatrix}11 & 1 & -7\\ -4 & 1 & 8\\ 12 & -3 & -9\end{bmatrix}\begin{pmatrix}1\\ 1\\ 1\end{pmatrix}=\frac{1}{15}\begin{pmatrix}5\\ 5\\ 0\end{pmatrix}=\begin{pmatrix}\frac{1}{3}\\ \frac{1}{3}\\ 0\end{pmatrix}.$$

(4) $\vec{x}=3\vec{\varepsilon}_1-2\vec{\varepsilon}_2+\vec{\varepsilon}_3=(\vec{\varepsilon}_1\ \vec{\varepsilon}_2\ \vec{\varepsilon}_3)\begin{pmatrix}3\\ -2\\ 1\end{pmatrix}=E\begin{pmatrix}3\\ -2\\ 1\end{pmatrix}=(\tilde{\varepsilon}_1\ \tilde{\varepsilon}_2\ \tilde{\varepsilon}_3)\begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\end{pmatrix}=\tilde{E}\begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\end{pmatrix}$

$$\therefore\begin{pmatrix}\beta_1\\ \beta_2\\ \beta_3\end{pmatrix}=\tilde{E}^{-1}\cdot E\begin{pmatrix}3\\ -2\\ 1\end{pmatrix}=\frac{1}{15}\begin{bmatrix}11 & 1 & -7\\ -4 & 1 & 8\\ 12 & -3 & -9\end{bmatrix}\begin{bmatrix}1 & 1 & 1\\ 1 & -1 & 2\\ 1 & 0 & -1\end{bmatrix}\begin{pmatrix}3\\ -2\\ 1\end{pmatrix}$$

$$=\frac{1}{15}\begin{bmatrix}5 & 10 & 20\\ 5 & -5 & -10\\ 0 & 15 & 15\end{bmatrix}\begin{pmatrix}3\\ -2\\ 1\end{pmatrix}=\frac{1}{15}\begin{pmatrix}15\\ 45\\ -15\end{pmatrix}=\begin{pmatrix}1\\ 3\\ -1\end{pmatrix}.$$

7. 设 S 为 R^3 的一个子空间，由基 $\vec{\varepsilon}_1=(1,0,-1)^T$，$\vec{\varepsilon}_2=(1,0,-2)^T$ 到基 $\{\tilde{\varepsilon}_1,\tilde{\varepsilon}_2\}$ 的过渡矩阵为 $\begin{pmatrix}3 & -2\\ 1 & -1\end{pmatrix}$. 求 $\tilde{\varepsilon}_1$ 和 $\tilde{\varepsilon}_2$，并求坐标变换公式.

解：设 $(\tilde{\varepsilon}_1,\tilde{\varepsilon}_2)=(\vec{\varepsilon}_1,\vec{\varepsilon}_2)\cdot T=(\vec{\varepsilon}_1,\vec{\varepsilon}_2)\begin{pmatrix}3 & -2\\ 1 & -1\end{pmatrix}=\begin{pmatrix}1 & 1\\ 0 & 0\\ -1 & -2\end{pmatrix}\begin{pmatrix}3 & -2\\ 1 & -1\end{pmatrix}=\begin{pmatrix}4 & -3\\ 0 & 0\\ -5 & 4\end{pmatrix}$

$\therefore \tilde{\varepsilon}_1=(4,0,-5)^T$，$\tilde{\varepsilon}_2=(-3,0,4)^T$

设 $\vec{x}=(\vec{\varepsilon}_1,\vec{\varepsilon}_2)\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}=(\tilde{\varepsilon}_1,\tilde{\varepsilon}_2)\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix}=(\vec{\varepsilon}_1\ \vec{\varepsilon}_2)\cdot T\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix}$

$$\therefore\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}=T\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix}=\begin{pmatrix}3 & -2\\ 1 & -1\end{pmatrix}\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix}=\begin{pmatrix}3\beta_1-2\beta_2\\ \beta_1-\beta_2\end{pmatrix}\Rightarrow\begin{cases}\alpha_1=3\beta_1-2\beta_2\\ \alpha_2=\beta_1-\beta_2\end{cases}$$

$$\begin{pmatrix}\beta_1\\ \beta_2\end{pmatrix}=T^{-1}\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}=\begin{pmatrix}1 & -2\\ 1 & -3\end{pmatrix}\begin{pmatrix}\alpha_1\\ \alpha_2\end{pmatrix}=\begin{pmatrix}\alpha_1-2\alpha_2\\ \alpha_1-3\alpha_2\end{pmatrix}\Rightarrow\begin{cases}\beta_1=\alpha_1-2\alpha_2\\ \beta_2=\alpha_1-3\alpha_2\end{cases}.$$

8. 设 P_2 为次数不超过 2 的多项式全体构成的线性空间.

(1) 证明 P_2 的基可取为 $\varepsilon_1=1$，$\varepsilon_2=x$，$\varepsilon_3=x^2$，也可取为 $\tilde{\varepsilon}_1=1$，$\tilde{\varepsilon}_2=x+2$，$\tilde{\varepsilon}_3=(x+2)^2$，

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(1) 求 $\{\varepsilon_1, \varepsilon_2, \varepsilon_3\}$ 到 $\{\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3\}$ 的过渡矩阵;

(2) 求向量 x^2+x+1 在 $\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3$ 下的坐标.

证明: (1) $P_2 = \{a_0 + a_1 x + a_2 x^2 \mid a_0, a_1, a_2 \in R\}$.

$\because 0 = a_0 + a_1 x + a_2 x^2 \iff a_0 = a_1 = a_2 = 0$ $\therefore \{1, x, x^2\}$ 线性无关

对于 P_2 中任一元素 $a_0 + a_1 x + a_2 x^2 = (1, x, x^2)\begin{pmatrix} a_0 \\ a_1 \\ a_2 \end{pmatrix}$ $\therefore \{1, x, x^2\}$ 是 P_2 中的一组基.

同样: $0 = b_0 + b_1(x+2) + b_2(x+2)^2$

$= (b_0 + 2b_1 + 4b_2) + (b_1 + 4b_2)x + b_2 x^2$ $\iff \begin{cases} b_0 + 2b_1 + 4b_2 = 0 \\ b_1 + 4b_2 = 0 \\ b_2 = 0 \end{cases} \Rightarrow \begin{cases} b_0 = 0 \\ b_1 = 0 \\ b_2 = 0 \end{cases}$

$\therefore \{1, x+2, (x+2)^2\}$ 线性无关.

对于 P_2 中任一元素 $b_0 + b_1 x + b_2 x^2 = (1, x+2, (x+2)^2)\begin{pmatrix} b_0 - 2b_1 + 4b_2 \\ b_1 - 4b_2 \\ b_2 \end{pmatrix}$

$\therefore \{1, x+2, (x+2)^2\}$ 是 P_2 中的一组基

(2) $(\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3) = (\varepsilon_1, \varepsilon_2, \varepsilon_3)T \Rightarrow (1, x+2, (x+2)^2) = (1, x, x^2)T$.

而 $\{1, x+2, (x+2)^2\} = \{1, x+2, x^2+4x+4\} = (1, x, x^2)\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$

$\therefore T = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$.

(3) $x^2+x+1 = (1, x, x^2)\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = (\tilde{\varepsilon}_1\ \tilde{\varepsilon}_2\ \tilde{\varepsilon}_3)\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = (\varepsilon_1\ \varepsilon_2\ \varepsilon_3)\cdot T\begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix}$.

$\therefore \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = T^{-1}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

$(T \mid I) = \left[\begin{array}{ccc|ccc} 1 & 2 & 4 & 1 & 0 & 0 \\ 0 & 1 & 4 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 2 & 0 & 1 & 0 & -4 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right] \to \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & -2 & 4 \\ 0 & 1 & 0 & 0 & 1 & -4 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right]$

即 $T^{-1} = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$. $\therefore \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = T^{-1}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 3 \\ -3 \\ 1 \end{pmatrix}$.

§2. 线性变换及其矩阵表示

习题. P208

1. 下列变换是否为线性变换，若线性变换则给出它在给定基下 $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ 的表示矩阵.

(1) $\mathcal{A}(x_1, x_2, x_3)^T = (x_1+x_2, x_2+x_3, x_3+x_1)^T$

解: 设 $\vec{x} = (x_1, x_2, x_3)^T$ $\vec{y} = (y_1, y_2, y_3)^T$.

$$\mathcal{A}(\lambda\vec{x}+\mu\vec{y}) = \mathcal{A}(\lambda x_1+\mu y_1, \lambda x_2+\mu y_2, \lambda x_3+\mu y_3)^T$$

$$= (\lambda(x_1+x_2)+\mu(y_1+y_2), \lambda(x_2+x_3)+\mu(y_2+y_3), \lambda(x_3+x_1)+\mu(y_3+y_1))^T$$

$$= \lambda(x_1+x_2, x_2+x_3, x_3+x_1)^T + \mu(y_1+y_2, y_2+y_3, y_3+y_1)^T$$

$$= \lambda\mathcal{A}(\vec{x}) + \mu\mathcal{A}(\vec{y}). \quad \therefore \text{是线性变换.}$$

$$\mathcal{A}(\vec{x}) = \begin{pmatrix} x_1+x_2 \\ x_2+x_3 \\ x_3+x_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \quad \therefore \text{在 } \{\vec{e}_1, \vec{e}_2, \vec{e}_3\} \text{ 下. } A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}.$$

(2) $\mathcal{A}(x_1, x_2, x_3)^T = (x_1x_3, x_1+x_3, x_2+x_3)^T$.

解: $\mathcal{A}(\lambda\vec{x}+\mu\vec{y}) = \mathcal{A}(\lambda x_1+\mu y_1, \lambda x_2+\mu y_2, \lambda x_3+\mu y_3)^T$

$$= ((\lambda x_1+\mu y_1)(\lambda x_3+\mu y_3), \lambda(x_1+x_3)+\mu(y_1+y_3), \lambda(x_2+x_3)+\mu(y_2+y_3))^T.$$

$$\lambda\mathcal{A}(\vec{x}) + \mu\mathcal{A}(\vec{y}) = \lambda(x_1x_3, x_1+x_3, x_2+x_3)^T + \mu(y_1y_3, y_1+y_3, y_2+y_3)^T$$

$$= (\lambda x_1x_3 + \mu y_1y_3, \lambda(x_1+x_3)+\mu(y_1+y_3), \lambda(x_2+x_3)+\mu(y_2+y_3))^T.$$

$$\because (\lambda x_1+\mu y_1)(\lambda x_3+\mu y_3) = \lambda^2 x_1x_3 + \mu^2 y_1y_3 + \lambda\mu(x_1y_3+y_1x_3) \neq \lambda x_1x_3 + \mu y_1y_3.$$

$$\therefore \mathcal{A}(\lambda\vec{x}+\mu\vec{y}) \neq \lambda\mathcal{A}(\vec{x}) + \mu\mathcal{A}(\vec{y}). \quad \therefore \mathcal{A} \text{ 不是线性变换.}$$

(3) $\mathcal{A}(x_1, x_2, x_3)^T = (x_1, 2x_2, x_3)^T$

解: $\mathcal{A}(\lambda\vec{x}+\mu\vec{y}) = \mathcal{A}(\lambda x_1+\mu y_1, \lambda x_2+\mu y_2, \lambda x_3+\mu y_3)^T$

$$= (\lambda x_1+\mu y_1, 2\lambda x_2+2\mu y_2, \lambda x_3+\mu y_3)^T$$

$$= \lambda(x_1, 2x_2, x_3)^T + \mu(y_1, 2y_2, y_3)^T$$

$$= \lambda\mathcal{A}(\vec{x}) + \mu\mathcal{A}(\vec{y}). \quad \therefore \mathcal{A} \text{ 是线性变换.}$$

$$\mathcal{A}(\vec{x}) = \begin{pmatrix} x_1 \\ 2x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}. \quad \therefore \text{在 } \{\vec{e}_1, \vec{e}_2, \vec{e}_3\} \text{ 下. } A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

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2. 设 $\{\vec{\varepsilon}_1, \vec{\varepsilon}_2\}$ 是线性空间 V 的基，V 上的线性变换 $\mathcal{A}$ 满足

$$\mathcal{A}(\vec{\varepsilon}_1+\vec{\varepsilon}_2)=3\vec{\varepsilon}_1+9\vec{\varepsilon}_2,\quad \mathcal{A}(3\vec{\varepsilon}_1+2\vec{\varepsilon}_2)=7\vec{\varepsilon}_1+23\vec{\varepsilon}_2.$$

求 $\mathcal{A}$ 在基 $\{\vec{\varepsilon}_1, \vec{\varepsilon}_2\}$ 下的表示矩阵.

解:
$$\left.\begin{aligned}\mathcal{A}(\vec{\varepsilon}_1+\vec{\varepsilon}_2)&=\mathcal{A}(\vec{\varepsilon}_1)+\mathcal{A}(\vec{\varepsilon}_2)=3\vec{\varepsilon}_1+9\vec{\varepsilon}_2\\ \mathcal{A}(3\vec{\varepsilon}_1+2\vec{\varepsilon}_2)&=3\mathcal{A}(\vec{\varepsilon}_1)+2\mathcal{A}(\vec{\varepsilon}_2)=7\vec{\varepsilon}_1+23\vec{\varepsilon}_2\end{aligned}\right\}\Rightarrow (\mathcal{A}(\vec{\varepsilon}_1)\ \mathcal{A}(\vec{\varepsilon}_2))\begin{pmatrix}1&3\\1&2\end{pmatrix}=(\vec{\varepsilon}_1\ \vec{\varepsilon}_2)\begin{pmatrix}3&7\\9&23\end{pmatrix}$$

设 $(\mathcal{A}(\vec{\varepsilon}_1)\ \mathcal{A}(\vec{\varepsilon}_2))=(\vec{\varepsilon}_1\ \vec{\varepsilon}_2)A$. 则有 $(\vec{\varepsilon}_1\ \vec{\varepsilon}_2)A\begin{bmatrix}1&3\\1&2\end{bmatrix}=(\vec{\varepsilon}_1\ \vec{\varepsilon}_2)\begin{bmatrix}3&7\\9&23\end{bmatrix}$.

$\because \{\vec{\varepsilon}_1\ \vec{\varepsilon}_2\}$ 是一组基. $\therefore A=\begin{bmatrix}3&7\\9&23\end{bmatrix}\begin{bmatrix}1&3\\1&2\end{bmatrix}^{-1}=\begin{bmatrix}3&7\\9&23\end{bmatrix}\begin{bmatrix}-2&3\\1&-1\end{bmatrix}=\begin{bmatrix}1&2\\5&4\end{bmatrix}$

3. 设 R^3 上的线性变换 $\mathcal{A}$ 在基 $\left\{\begin{pmatrix}-1\\1\\1\end{pmatrix},\begin{pmatrix}1\\0\\-1\end{pmatrix},\begin{pmatrix}0\\1\\1\end{pmatrix}\right\}$ 下的表示矩阵为 $\begin{bmatrix}1&0&1\\1&1&0\\-1&2&1\end{bmatrix}$，求 $\mathcal{A}$ 在自然基下 $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ 的表示矩阵.

解: 设 $\{\vec{\varepsilon}_1, \vec{\varepsilon}_2, \vec{\varepsilon}_3\}=\left\{\begin{pmatrix}-1\\1\\1\end{pmatrix},\begin{pmatrix}1\\0\\-1\end{pmatrix},\begin{pmatrix}0\\1\\1\end{pmatrix}\right\}=\{\vec{e}_1,\vec{e}_2,\vec{e}_3\}\begin{bmatrix}-1&1&0\\1&0&1\\1&-1&1\end{bmatrix}=\{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}T$.

即 $\{\vec{e}_1, \vec{e}_2, \vec{e}_3\}$ 与 $\{\vec{\varepsilon}_1, \vec{\varepsilon}_2, \vec{\varepsilon}_3\}$ 的过渡矩阵 $T=\begin{bmatrix}-1&1&0\\1&0&1\\1&-1&1\end{bmatrix}$.

设 $\{\mathcal{A}(\vec{\varepsilon}_1)\ \mathcal{A}(\vec{\varepsilon}_2)\ \mathcal{A}(\vec{\varepsilon}_3)\}=\{\vec{\varepsilon}_1\ \vec{\varepsilon}_2\ \vec{\varepsilon}_3\}A$ $\quad A=\begin{bmatrix}1&0&1\\1&1&0\\-1&2&1\end{bmatrix}$.

$\{\mathcal{A}(\vec{e}_1)\ \mathcal{A}(\vec{e}_2)\ \mathcal{A}(\vec{e}_3)\}=\{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}B$.

$\because \{\vec{\varepsilon}_1\ \vec{\varepsilon}_2\ \vec{\varepsilon}_3\}=\{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}T$.

两边作线性变换 $\mathcal{A}$: $\{\mathcal{A}(\vec{\varepsilon}_1)\ \mathcal{A}(\vec{\varepsilon}_2)\ \mathcal{A}(\vec{\varepsilon}_3)\}=\{\mathcal{A}(\vec{e}_1)\ \mathcal{A}(\vec{e}_2)\ \mathcal{A}(\vec{e}_3)\}T$

即: $\{\vec{\varepsilon}_1\ \vec{\varepsilon}_2\ \vec{\varepsilon}_3\}A=\{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}BT$

$\Rightarrow \{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}TA=\{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}BT$. 考虑到 $\{\vec{e}_1\ \vec{e}_2\ \vec{e}_3\}$ 是一组基. $\therefore TA=BT$.

$$\therefore B=TAT^{-1}=\begin{bmatrix}-1&1&0\\1&0&1\\1&-1&1\end{bmatrix}\begin{bmatrix}1&0&1\\1&1&0\\-1&2&1\end{bmatrix}\begin{bmatrix}-1&1&0\\1&0&1\\1&-1&1\end{bmatrix}^{-1}$$

$$=\begin{bmatrix}0&1&-1\\0&2&2\\-1&1&2\end{bmatrix}\begin{bmatrix}-1&1&-1\\0&1&-1\\1&0&1\end{bmatrix}=\begin{bmatrix}-1&1&-2\\2&2&0\\3&0&2\end{bmatrix}$$

4. 对次数不超过2的实系数多项式全体构成的线性空间 P_2，定义变换

$\mathcal{A}(f(x)) = f(x-2)$, $f(x) \in P_2$.

(1) 证明 $\mathcal{A}$ 为 P_2 上的线性变换;

(2) 求 $\mathcal{A}$ 在基 $\varepsilon_1 = 1$, $\varepsilon_2 = x$, $\varepsilon_3 = x^2$ 下的表示矩阵;

(3) 利用基 $\varepsilon_1 = 1$, $\varepsilon_2 = x$, $\varepsilon_3 = x^2$ 到基 $\tilde{\varepsilon}_1 = 1$, $\tilde{\varepsilon}_2 = x+2$, $\tilde{\varepsilon}_3 = (x+2)^2$ 的过渡矩阵, 导出 $\mathcal{A}$ 在基 $\{\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3\}$ 下的表示矩阵;

(4) 直接求 $\mathcal{A}$ 在基 $\{\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3\}$ 下的表示矩阵, 并与(3)的结果比较.

解: (1) $\mathcal{A}(\lambda f(x) + \mu g(x)) = \lambda f(x-2) + \mu g(x-2) = \lambda \mathcal{A}(f(x)) + \mu \mathcal{A}(g(x))$

$\therefore$ $\mathcal{A}$ 为 P_2 上的线性变换 $\quad f(x), g(x) \in P_2$.

(2) $(\mathcal{A}(1), \mathcal{A}(x), \mathcal{A}(x^2)) = (1, x-2, (x-2)^2) = (1, x-2, x^2-4x+4)$

$$= (1, x, x^2)\begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}.$$

$\therefore$ $\mathcal{A}$ 在基 $\{1, x, x^2\}$ 下的表示矩阵 $A = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$.

(3) $\{\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3\} = \{1, x+2, (x+2)^2\} = \{1, x+2, x^2+4x+4\}$

$$= \{1, x, x^2\}\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}.$$

即 $\{1, x, x^2\}$ 到 $\{1, x+2, (x+2)^2\}$ 的过渡矩阵为 $T = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$.

$$= \{\varepsilon_1, \varepsilon_2, \varepsilon_3\} T.$$

$\therefore$ $\{1, x+2, (x+2)^2\} = \{1, x, x^2\} T$.

两边作用线性变换 $\mathcal{A}$, 则有:

$$\{\mathcal{A}(1), \mathcal{A}(x+2), \mathcal{A}(x+2)^2\} = \{\mathcal{A}(1), \mathcal{A}(x), \mathcal{A}(x^2)\} T.$$

$$\Rightarrow \quad \{1, x+2, (x+2)^2\} B = \{1, x, x^2\} AT$$

$$\Rightarrow \quad \{1, x, x^2\} TB = \{1, x, x^2\} AT. \quad \Rightarrow \quad B = T^{-1}AT.$$

$$\therefore B = \begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}^{-1}\begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 2 & 4 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}.$$

(4) $(\mathcal{A}(\tilde{\varepsilon}_1), \mathcal{A}(\tilde{\varepsilon}_2), \mathcal{A}(\tilde{\varepsilon}_3)) = \{\mathcal{A}(1), \mathcal{A}(x+2), \mathcal{A}((x+2)^2)\} = \{1, x, x^2\}$

$$= \{1, (x+2)-2, (x+2)^2-4(x+2)+4\} = \{1, x+2, (x+2)^2\}\begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}.$$

$\therefore$ $\mathcal{A}$ 在基 $\{\tilde{\varepsilon}_1, \tilde{\varepsilon}_2, \tilde{\varepsilon}_3\} = \{1, x+2, (x+2)^2\}$ 下的表示矩阵为 $\begin{bmatrix} 1 & -2 & 4 \\ 0 & 1 & -4 \\ 0 & 0 & 1 \end{bmatrix}$.

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5. 证明 n阶矩阵之间的相似关系满足如下性质：

(1) 自反性，即 $A \sim A$；

(2) 对称性，即若 $A \sim B$，则 $B \sim A$；

(3) 传递性，即若 $A \sim B$，$B \sim C$，则 $A \sim C$.

证明：(1) $\because A = I^{-1} A I$，I 可逆，$\therefore A \sim A$.

(2) 若 $A \sim B$，则存在可逆方阵 T，使得 $B = T^{-1} A T$.

进一步地有：$TBT^{-1} = A$.

即 $A = TBT^{-1} = (T^{-1})^{-1} B (T^{-1})$. $\because T$ 可逆，$\therefore T^{-1}$ 也可逆.

$\therefore B \sim A$.

(3) 若 $A \sim B$，$B \sim C$，则存在可逆方阵 T_1，T_2，使得：$B = T_1^{-1} A T_1$，$C = T_2^{-1} B T_2$.

进一步地有：$C = T_2^{-1}(T_1^{-1} A T_1) T_2 = (T_1 T_2)^{-1} A (T_1 T_2)$.

$\because T_1$ 和 T_2 均可逆且同阶. $\therefore T_1 T_2$ 可逆.

$\therefore A \sim C$.

6. 设 $\mathscr{A}$ 是线性空间 U 上的线性变换，$\{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_k\}$ 是 U 的一组基. 证明 $\mathscr{A}$ 为可逆变换的充要条件是：$\{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots, \mathscr{A}(\vec{a}_k)\}$ 也是 U 的一组基.

证明：$\Rightarrow$ 必要性. $\mathscr{A}$ 为可逆变换，即存在 $\mathscr{A}^{-1}$，使得 $\mathscr{A}^{-1}(\mathscr{A}(\vec{x})) = \vec{x}$.

假设 $\{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots \mathscr{A}(\vec{a}_k)\}$ 线性相关，则存在不全为 0 的 $\lambda_1, \lambda_2, \cdots \lambda_k$，使得：

$$\mathscr{A}(\vec{0}) = \lambda_1 \mathscr{A}(\vec{a}_1) + \lambda_2 \mathscr{A}(\vec{a}_2) + \cdots + \lambda_k \mathscr{A}(\vec{a}_k)$$

等式两边作用 $\mathscr{A}^{-1}$，则有：

$$\mathscr{A}^{-1}(\mathscr{A}(\vec{0})) = \lambda_1 \mathscr{A}^{-1}(\mathscr{A}(\vec{a}_1)) + \lambda_2 \mathscr{A}^{-1}(\mathscr{A}(\vec{a}_2)) + \cdots + \lambda_k \mathscr{A}^{-1}(\mathscr{A}(\vec{a}_k))$$

即：$\vec{0} = \lambda_1 \vec{a}_1 + \lambda_2 \vec{a}_2 + \cdots + \lambda_k \vec{a}_k$.

由 $\lambda_1, \lambda_2, \cdots \lambda_k$ 不全为 0，可知 $\{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_k\}$ 线性相关，矛盾.

$\therefore \{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots, \mathscr{A}(\vec{a}_k)\}$ 线性无关，也是 U 的一组基.

$\Leftarrow$ 充分性. 若 $\{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots, \mathscr{A}(\vec{a}_k)\}$ 是 U 的一组基

则 $\vec{a}_1, \vec{a}_2, \cdots \vec{a}_k$ 均可由 $\{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots \mathscr{A}(\vec{a}_k)\}$ 线性表出，即：

$$\{\vec{a}_1, \vec{a}_2, \cdots \vec{a}_k\} = \{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots \mathscr{A}(\vec{a}_k)\} D$$

$$= \{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_k\} A D \quad (A \text{ 为 } \mathscr{A} \text{ 在 } \{\vec{a}_1, \cdots, \vec{a}_k\} \text{ 下的表示矩阵})$$

$\therefore AD = I$，$D = A^{-1}$ 设 D 为变换 $\mathscr{D}$ 在基 $\{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots \mathscr{A}(\vec{a}_k)\}$ 下的表示矩阵.

则：$\{\mathscr{D}(\mathscr{A}(\vec{a}_1)), \mathscr{D}(\mathscr{A}(\vec{a}_2)), \cdots, \mathscr{D}(\mathscr{A}(\vec{a}_k))\} = \{\mathscr{A}(\vec{a}_1), \mathscr{A}(\vec{a}_2), \cdots, \mathscr{A}(\vec{a}_k)\} D$

$= \{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_k\} AD = \{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_k\}$.

即 $\mathcal{D}(\mathcal{A}(\vec{a}_i)) = \vec{a}_i$. ∴ $\mathcal{D}$ 为 $\mathcal{A}$ 的可逆变换.

§3. 特征值问题

习题 P228

1. 求下列矩阵的特征值及相应的特征向量:

(1) $\begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}$

解: $|A-\lambda I| = \begin{vmatrix} -1-\lambda & 1 & 0 \\ -4 & 3-\lambda & 0 \\ 1 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)\begin{vmatrix} -1-\lambda & 1 \\ -4 & 3-\lambda \end{vmatrix} = (2-\lambda)\left[(\lambda+1)(\lambda-3)+4\right]$

$= (2-\lambda)(\lambda^2-2\lambda+1) = (2-\lambda)(\lambda-1)^2 = 0$. ∴ $\lambda_1 = \lambda_2 = 1, \lambda_3 = 2$.

对 $\lambda_1 = \lambda_2 = 1$, 解齐次方程组:

$(A-\lambda_1 I)\vec{x} = \begin{bmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ 1 & 0 & -1 \end{bmatrix}\vec{x} = \vec{0}$.

$(A-\lambda_1 I) = \begin{pmatrix} -2 & 1 & 0 \\ -4 & 2 & 0 \\ 1 & 0 & -1 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ -2 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{pmatrix}$.

∵ $\operatorname{rank}(A-\lambda_1 I) = 2$. ∴ $(A-\lambda_1 I)\vec{x} = \vec{0}$ 的基础解系只有1个解向量 $\vec{\xi}_1$.

$\vec{\xi}_1 = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$. 此时有: $(A-\lambda_1 I)\vec{\xi}_1 = \vec{0}$. $\vec{\xi}_1$ 为 λ_1 对应的特征向量.

对 $\lambda_3 = 2$, $(A-\lambda_3 I)\vec{x} = \begin{bmatrix} -3 & 1 & 0 \\ -4 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}\vec{x} = \vec{0}$. ⇒ 特征向量 $\vec{\xi}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$

∴ $\lambda_1 = \lambda_2 = 1$. $\vec{\xi}_1 = \begin{pmatrix} -1 \\ -2 \\ 1 \end{pmatrix}$.

$\lambda_3 = 2$. $\vec{\xi}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$.

虽然 A 不能对角化, 但可化为 Jordan 标准形!

事实上, 求解 $(A-\lambda_1 I)\vec{\xi}_2 = \vec{\xi}_1$

得特解 $\vec{\xi}_2 = \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$. 记 $T = (\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3)$.

$T^{-1}AT = \begin{bmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}^{-1}\begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}\begin{bmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$.

$= \begin{bmatrix} 1 & -1 & 0 \\ 2 & -1 & 0 \\ -1 & 1 & 1 \end{bmatrix}\begin{bmatrix} -1 & 1 & 0 \\ -4 & 3 & 0 \\ 1 & 0 & 2 \end{bmatrix}\begin{bmatrix} -1 & 1 & 0 \\ -2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} \lambda_1 & 1 & 0 \\ 0 & \lambda_1 & 0 \\ 0 & 0 & \lambda_3 \end{bmatrix}$

(2) $\begin{bmatrix} 2 & 1 & 0 & 0 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 \\ 0 & 0 & 0 & 2 \end{bmatrix}$.

解: $|A-\lambda I| = \begin{vmatrix} 2-\lambda & 1 & 0 & 0 \\ 0 & 2-\lambda & 1 & 0 \\ 0 & 0 & 2-\lambda & 1 \\ 0 & 0 & 0 & 2-\lambda \end{vmatrix} = (2-\lambda)^4 = 0$. ∴ $\lambda_1 = \lambda_2 = \lambda_3 = \lambda_4 = 2$.

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对于 $\lambda_1=\lambda_2=\lambda_3=\lambda_4=2$，求解齐次方程组：

$(A-\lambda_1 I)\vec{x}=\vec{0}$ 即 $\begin{bmatrix}0&1&0&0\\0&0&1&0\\0&0&0&1\\0&0&0&0\end{bmatrix}\vec{x}=\vec{0}$ $\Rightarrow$ $\vec{x}=\begin{pmatrix}1\\0\\0\\0\end{pmatrix}$ 此即对应于 $\lambda=2$ 的特征向量.

(A 事实证明不可对角化 Jordan 块)

(3) $\begin{bmatrix}5&-6&-6\\-1&4&2\\3&-6&-4\end{bmatrix}$.

解: $|A-\lambda I|=\begin{vmatrix}5-\lambda&-6&-6\\-1&4-\lambda&2\\3&-6&-4-\lambda\end{vmatrix}=(\lambda-2)^2(1-\lambda)=0$. $\therefore \lambda_1=1,\ \lambda_2=\lambda_3=2$.

对 $\lambda_1=1$. $(A-\lambda_1 I)\vec{x}=\begin{pmatrix}4&-6&-6\\-1&3&2\\3&-6&-5\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$ $\Rightarrow$ $\vec{x}_1=\begin{pmatrix}3\\-1\\3\end{pmatrix}$ —— 对应 $\lambda_1=1$ 的特征向量.

对 $\lambda_2=\lambda_3=2$. $(A-\lambda_2 I)\vec{x}=\begin{pmatrix}3&-6&-6\\-1&2&2\\3&-6&-6\end{pmatrix}\vec{x}=\vec{0}$

$(A-\lambda_2 I)=\begin{pmatrix}3&-6&-6\\-1&2&2\\3&-6&-6\end{pmatrix}\to\begin{pmatrix}1&-2&-2\\0&0&0\\0&0&0\end{pmatrix}$ $\because \operatorname{rank}(A-\lambda_2 I)=1$.

即 $(A-\lambda_2 I)\vec{x}=\vec{0}$ 的基础解系含有2个的向量.

$\vec{x}_2=\begin{pmatrix}2\\1\\0\end{pmatrix}$ $\vec{x}_3=\begin{pmatrix}2\\0\\1\end{pmatrix}$ —— 对应 $\lambda_2=\lambda_3=2$ 的特征向量.

令 $(\vec{x}_1,\vec{x}_2,\vec{x}_3)=T$. 则有: $T^{-1}AT=\Lambda=\begin{bmatrix}1&&\\&2&\\&&2\end{bmatrix}$.

A 可对角化!

(4) $\begin{bmatrix}2&5&-6\\4&6&-9\\3&6&-8\end{bmatrix}$.

解: $|A-\lambda I|=\begin{vmatrix}2-\lambda&5&-6\\4&6-\lambda&-9\\3&6&-8-\lambda\end{vmatrix}=-\lambda^3+1=-(\lambda-1)(\lambda^2+\lambda+1)=0$. $\therefore \lambda_1=1,\ \lambda_{2,3}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$.

对于 $\lambda_1=1$. $(A-\lambda_1 I)\vec{x}=\begin{bmatrix}1&5&-6\\4&5&-9\\3&6&-9\end{bmatrix}\vec{x}=\vec{0}$ $\Rightarrow$ $\vec{x}_1=\begin{pmatrix}1\\1\\1\end{pmatrix}$ —— 对应于 $\lambda_1=1$ 的特征向量

对于 $\lambda_{2,3}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$, $(A-\lambda I)\vec{x}=\begin{bmatrix}2-\left(-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i\right)&5&-6\\4&6-\left(-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i\right)&-9\\3&6&-8-\left(-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i\right)\end{bmatrix}\vec{x}=\vec{0}$

$\Rightarrow$ $\vec{x}_{2,3}=\left(1,\ \frac{11\pm5\sqrt{3}i}{14},\ \frac{15\pm3\sqrt{3}i}{14}\right)^T$ —— 对应于 $\lambda_{2,3}=-\frac{1}{2}\pm\frac{\sqrt{3}}{2}i$ 的特征向量.

2. 设 A 为实方阵，$A^2=I$ 且 A 的特征值全为 1，试证 $A=I$.

证明：$\because A^2=I \Rightarrow A^2-I=0$，$\operatorname{rank}(A^2-I)=0$

即 $0=\operatorname{rank}(A^2-I)=\operatorname{rank}[(A+I)(A-I)]$

$\geq \operatorname{rank}(A+I)+\operatorname{rank}(A-I)-n$.

$\because A$ 的特征值全为 1，$\therefore |A+I|\neq 0$（否则 -1 也是 A 的特征值）.

$\therefore \operatorname{rank}(A+I)=n$

即 $0\geq \operatorname{rank}(A+I)+\operatorname{rank}(A-I)-n=n+\operatorname{rank}(A-I)-n=\operatorname{rank}(A-I)$.

$\therefore \operatorname{rank}(A-I)=0$. $\therefore A-I=0$ $\therefore A=I$.

3. 将下列矩阵化为 Jordan 标准形：

(1) $\begin{bmatrix} 4 & -3 & 3 \\ -1 & 3 & -2 \\ -3 & 4 & -3 \end{bmatrix}$

解：$|A-\lambda I|=\begin{vmatrix} 4-\lambda & -3 & 3 \\ -1 & 3-\lambda & -2 \\ -3 & 4 & -3-\lambda \end{vmatrix}=\begin{vmatrix} 4-\lambda & 0 & 3 \\ -1 & 1-\lambda & -2 \\ -3 & 1-\lambda & -3-\lambda \end{vmatrix}=(1-\lambda)\begin{vmatrix} 4-\lambda & 0 & 3 \\ -1 & 1 & -2 \\ -3 & 1 & -3-\lambda \end{vmatrix}$

$=(1-\lambda)\begin{vmatrix} 4-\lambda & 0 & 3 \\ -1 & 1 & -2 \\ -2 & 0 & -1-\lambda \end{vmatrix}=(1-\lambda)\begin{vmatrix} 4-\lambda & 3 \\ -2 & -1-\lambda \end{vmatrix}=-(\lambda-1)[(\lambda+1)(4-\lambda)+6]$

$=-(\lambda-1)(\lambda^2-3\lambda-4+6)=-(\lambda-1)(\lambda-1)(\lambda-2)$

$=-(\lambda-1)^2(\lambda-2)=0$. $\therefore \lambda_1=\lambda_2=1$，$\lambda_3=2$.

对于 $\lambda_1=\lambda_2=1$，$(A-I)\vec{x}=\begin{pmatrix} 3 & -3 & 3 \\ -1 & 2 & -2 \\ -3 & 4 & -4 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$.

$(A-I)=\begin{pmatrix} 3 & -3 & 3 \\ -1 & 2 & -2 \\ -3 & 4 & -4 \end{pmatrix}\to\begin{pmatrix} 1 & -2 & 2 \\ 0 & 3 & -3 \\ 0 & -2 & 2 \end{pmatrix}\to\begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}\to\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$.

$\operatorname{rank}(A-I)=2$. $\therefore (A-I)\vec{x}=\vec{0}$ 只有 1 个线性无关的解向量：$\vec{\eta}_1=\begin{pmatrix} 0 \\ +1 \\ 1 \end{pmatrix}$

求解方程：$(A-I)^2\vec{x}=\vec{0}$. 即解 $(A-I)\vec{x}=\vec{\eta}_1$

$(A-I\,|\,\vec{\eta}_1)=\left(\begin{array}{ccc|c} 3 & -3 & 3 & 0 \\ -1 & 2 & -2 & +1 \\ -3 & 4 & -4 & 1 \end{array}\right)\to\left(\begin{array}{ccc|c} 1 & -2 & 2 & -1 \\ 0 & 3 & -3 & +3 \\ 0 & -2 & 2 & -2 \end{array}\right)\to\left(\begin{array}{ccc|c} 1 & 0 & 0 & +1 \\ 0 & 1 & -1 & +1 \\ 0 & 0 & 0 & 0 \end{array}\right)$ 特解 $\vec{\eta}_2=\begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}$.

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即有: $\begin{cases}(A-\lambda_1 I)\vec{\xi}_1=0\\(A-\lambda_1 I)\vec{\xi}_2=\vec{\xi}_1\end{cases}\Rightarrow\begin{cases}A\vec{\xi}_1=\lambda_1\cdot\vec{\xi}_1\\A\vec{\xi}_2=\lambda_1\cdot\vec{\xi}_2+\vec{\xi}_1\end{cases}\Rightarrow A(\vec{\xi}_1,\vec{\xi}_2)=(\vec{\xi}_1,\vec{\xi}_2)\begin{bmatrix}\lambda_1&1\\0&\lambda_1\end{bmatrix}$.

对于 $\lambda_3=2$. 的方程: $(A-\lambda_3 I)\vec{x}=\vec{0}$ 即 $\begin{pmatrix}2&-3&3\\-1&1&-2\\-3&4&-5\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix}$

$A-\lambda_3 I=\begin{bmatrix}2&-3&3\\-1&1&-2\\-3&4&-5\end{bmatrix}\to\begin{bmatrix}1&-1&2\\0&-1&-1\\0&1&1\end{bmatrix}\to\begin{bmatrix}1&-1&2\\0&1&1\\0&0&0\end{bmatrix}\to\begin{bmatrix}1&0&3\\0&1&1\\0&0&0\end{bmatrix}$. ∴可取 $\vec{\xi}_3=\begin{pmatrix}-3\\-1\\1\end{pmatrix}$.

即有 $A\vec{\xi}_3=\lambda_3\vec{\xi}_3$.

$\therefore A(\vec{\xi}_1,\vec{\xi}_2,\vec{\xi}_3)=(\vec{\xi}_1,\vec{\xi}_2,\vec{\xi}_3)\begin{bmatrix}\lambda_1&1&0\\0&\lambda_1&0\\0&0&\lambda_3\end{bmatrix}=(\vec{\xi}_1,\vec{\xi}_2,\vec{\xi}_3)\begin{bmatrix}1&1&0\\0&1&0\\0&0&2\end{bmatrix}$.

若 $AP=PJ$ 则有 $P=(\vec{\xi}_1,\vec{\xi}_2,\vec{\xi}_3)=\begin{bmatrix}0&1&-3\\1&1&-1\\1&0&1\end{bmatrix}$. $J=\begin{bmatrix}1&1&0\\0&1&0\\0&0&2\end{bmatrix}$

(P 矩阵不唯一! 如 $P=\begin{bmatrix}3&0&-2\\-1&-2&-1\\-1&-2&1\end{bmatrix}$. $P^{-1}AP=J$)

(2) $\begin{bmatrix}3&1&0&0\\-4&-1&0&0\\6&1&2&1\\-14&-5&-1&0\end{bmatrix}$

解:

$|A-\lambda I|=\begin{vmatrix}3-\lambda&1&0&0\\-4&-1-\lambda&0&0\\6&1&2-\lambda&1\\-14&-5&-1&-\lambda\end{vmatrix}=\begin{vmatrix}3-\lambda&1&0&0\\-4&-1-\lambda&0&0\\6&1&2-\lambda&1\\6\lambda-14&\lambda-5&2\lambda-\lambda^2-1&0\end{vmatrix}=-\begin{vmatrix}3-\lambda&1&0\\-4&-1-\lambda&0\\6\lambda-14&\lambda-5&-\lambda^3+2\lambda+1\end{vmatrix}$

$=-(\lambda-1)^2\begin{vmatrix}3-\lambda&1\\-4&-1-\lambda\end{vmatrix}=(\lambda-1)^2(\lambda^2-2\lambda-3+4)=(\lambda-1)^4=0$.

$\therefore\lambda_1=\lambda_2=\lambda_3=\lambda_4=1$.

$(A-\lambda I)=\begin{pmatrix}2&1&0&0\\-4&-2&0&0\\6&1&1&1\\-14&-5&-1&-1\end{pmatrix}\to\begin{pmatrix}2&1&0&0\\0&0&0&0\\0&2&-1&-1\\0&2&-1&-1\end{pmatrix}\to\begin{pmatrix}2&1&0&0\\0&2&-1&-1\\0&0&0&0\\0&0&0&0\end{pmatrix}\to\begin{pmatrix}1&0&\frac14&\frac14\\0&1&-\frac12&-\frac12\\0&0&0&0\\0&0&0&0\end{pmatrix}$

由 $(A-\lambda I)\vec{x}=\vec{0}$ 可得两个线性无关的特征向量: $\vec{\xi}_1=\begin{pmatrix}-1\\2\\4\\0\end{pmatrix}$ $\vec{\xi}_3=\begin{pmatrix}-1\\-2\\0\\4\end{pmatrix}$.

解方程: $(A-\lambda I)\vec{\xi}_2=\vec{\xi}_1$

$\left(A-\lambda I\mid\vec{\xi}_1\right)=\left(\begin{array}{cccc|c}2&1&0&0&-1\\-4&-2&0&0&2\\6&1&1&1&4\\-14&-5&-1&-1&0\end{array}\right)\to\left(\begin{array}{cccc|c}2&1&0&0&-1\\0&0&0&0&0\\0&2&-1&-1&-7\\0&2&-1&-1&-7\end{array}\right)\to\left(\begin{array}{cccc|c}2&1&0&0&-1\\0&2&-1&-1&-7\\0&0&0&0&0\\0&0&0&0&0\end{array}\right)$

$$\to \begin{pmatrix} 1 & \frac{1}{2} & 0 & 0 & -\frac{1}{4} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & -\frac{7}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{5}{4} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & -\frac{7}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \therefore \text{可取 } \vec{\xi}_2 = \begin{pmatrix} \frac{5}{4} \\ -\frac{7}{2} \\ 0 \\ 0 \end{pmatrix}$$

即 $A\vec{\xi}_2 = \lambda\vec{\xi}_2 + \vec{\xi}_1$. $A\vec{\xi}_1 = \lambda\vec{\xi}_1$. $\therefore A(\vec{\xi}_1, \vec{\xi}_2) = (\vec{\xi}_1, \vec{\xi}_2)\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$.

解方程: $(A-\lambda I)\vec{\xi}_4 = \vec{\xi}_3$.

$$(A-\lambda I \mid \vec{\xi}_3) = \begin{pmatrix} 2 & 1 & 0 & 0 & -1 \\ -4 & -2 & 0 & 0 & 2 \\ 6 & 1 & 1 & 1 & 0 \\ -14 & -5 & -1 & -1 & 4 \end{pmatrix} \to \begin{pmatrix} 2 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 2 & -1 & -1 & -3 \\ 0 & 2 & -1 & -1 & -3 \end{pmatrix} \to \begin{pmatrix} 2 & 1 & 0 & 0 & -1 \\ 0 & 2 & -1 & -1 & -3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$

$$\to \begin{pmatrix} 1 & \frac{1}{2} & 0 & 0 & -\frac{1}{2} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & \frac{1}{4} & \frac{1}{4} & \frac{1}{4} \\ 0 & 1 & -\frac{1}{2} & -\frac{1}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix} \quad \therefore \text{可取 } \vec{\xi}_4 = \begin{pmatrix} \frac{1}{4} \\ -\frac{3}{2} \\ 0 \\ 0 \end{pmatrix}$$

即 $A\vec{\xi}_4 = \lambda\vec{\xi}_4 + \vec{\xi}_3$. $A\vec{\xi}_3 = \lambda\vec{\xi}_3$. $\therefore A(\vec{\xi}_3, \vec{\xi}_4) = (\vec{\xi}_3, \vec{\xi}_4)\begin{bmatrix} \lambda & 1 \\ 0 & \lambda \end{bmatrix}$

$$\therefore A(\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3, \vec{\xi}_4) = (\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3, \vec{\xi}_4)\begin{bmatrix} \lambda & 1 & 0 & 0 \\ 0 & \lambda & 0 & 0 \\ 0 & 0 & \lambda & 1 \\ 0 & 0 & 0 & \lambda \end{bmatrix}.$$

若 $AP = PJ$, 则有:

$$P = (\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3, \vec{\xi}_4) = \begin{bmatrix} -1 & \frac{5}{4} & -1 & \frac{1}{4} \\ 2 & -\frac{7}{2} & 2 & -\frac{3}{2} \\ -4 & -0 & -0 & 0 \\ 0 & 0 & 4 & 0 \end{bmatrix} \quad J = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}.$$

(P 矩阵必可逆! 如: $P = \begin{bmatrix} 2 & 1 & 0 & 0 \\ -4 & 0 & 0 & 0 \\ 7 & 1 & 1 & 1 \\ -15 & 0 & -1 & 0 \end{bmatrix}$. $P^{-1}AP = J$)

4. 设 A 是 3 阶方阵, 它的三个特征值为 $\lambda_1 = 1$, $\lambda_2 = -1$, $\lambda_3 = 2$, 设 $B = A^3 - 5A^2$, 求 $|B|$ 和 $|A-5I|$.

解: 设 $P^{-1}AP = \Lambda = \begin{bmatrix} -1 & & \\ & 1 & \\ & & 2 \end{bmatrix}$. 则 $A = P\Lambda P^{-1}$.

$$|A| = |P||\Lambda||P^{-1}| = |\Lambda| = \begin{vmatrix} -1 & & \\ & 1 & \\ & & 2 \end{vmatrix} = -2.$$

$|A-\lambda I| = (\lambda_1-\lambda)(\lambda_2-\lambda)(\lambda_3-\lambda) = (-1-\lambda)(1-\lambda)(2-\lambda)$. 则 $|A-5I| = (-6)(-4)(-3) = -72$.

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$$|B| = |A^3 - 5A^2| = |A \times A \times (A-5I)| = |A| \cdot |A| \cdot |A-5I| = (-2)(-2)(-12) = -288.$$

或者，由 $A = P^{-1}\Lambda P \Rightarrow A^3 = P^{-1}\Lambda^3 P$, $A^2 = P^{-1}\Lambda^2 P$.

$$B = A^3 - 5A^2 = P^{-1}\Lambda^3 P - 5P^{-1}\Lambda^2 P = P^{-1}\begin{bmatrix} -8 & & \\ & -4 & \\ & & -12 \end{bmatrix} P.$$

$$\therefore |B| = |\Lambda^3 - 5\Lambda^2| = -288.$$

$$|A-5I| = \frac{|B|}{|A| \cdot |A|} = \frac{-288}{(-2)(-2)} = -72.$$

5. 设 A 是 n 阶方阵.

(1) 若 $A^2 = I$，问 $8I - A$ 是否可逆？

(2) 设 λ 是 A 的特征值，且 $\lambda \neq \pm 1$，问 $A \pm I$ 是否可逆？

解：(1) $(8I-A)(8I+A) = 64I - A^2 = 63I$.

两边取行列式：$|8I-A| \cdot |8I+A| = |63I| = 63^n \neq 0$. $\therefore |8I-A| \neq 0$.

即 $8I - A$ 可逆.

(2) $\because$ A 的特征值 $\lambda \neq \pm 1$. 即 $|A - \lambda I| = |A \pm I| \neq 0$. $\therefore$ $A \pm I$ 可逆.

6. 设方阵 $A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & 1 & 2 \end{bmatrix}$. 已知 A^{-1} 的一个特征向量为 $\vec{x} = (1, k, 1)^T$，求 k.

解：$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 1 & 1 & 2-\lambda \end{vmatrix} = -(\lambda-4)(\lambda-1)^2 = 0$. $\therefore \lambda_1 = \lambda_2 = 1$. $\lambda_3 = 4$.

对于 $\lambda_1 = \lambda_2 = 1$. $(A - \lambda_1 I) = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. 两个线性无关特征向量：$\vec{\xi}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$, $\vec{\xi}_2 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix}$.

对于 $\lambda_3 = 4$. $(A - \lambda_3 I) = \begin{pmatrix} -2 & 1 & 1 \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{pmatrix} \to \begin{pmatrix} 1 & -2 & 1 \\ 0 & -3 & 3 \\ 0 & 3 & -3 \end{pmatrix} \to \begin{pmatrix} 1 & -2 & 1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$ $\vec{\xi}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$

$\therefore A(\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3) = (\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3)\begin{bmatrix} 1 & & \\ & 1 & \\ & & 4 \end{bmatrix}$. $P = (\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3) = \begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

则有：$P^{-1}AP = \Lambda$. $A = P\Lambda P^{-1} \Rightarrow A^{-1} = P\Lambda^{-1}P^{-1} = P\begin{bmatrix} 1 & & \\ & 1 & \\ & & \frac{1}{4} \end{bmatrix}P^{-1}$.

A^{-1} 的三个特征值 $1, 1, \frac{1}{4}$，对应的特征向量分别为 $\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3$.

显然 $\vec{x}$ 只能是 $\vec{x}_3=\frac{1}{k}$ 对应的 $\vec{x}_3$ 的非零倍. 即 $\vec{x}=c\vec{x}_3$

$$\begin{pmatrix}1\\k\\1\end{pmatrix}=c\begin{pmatrix}1\\1\\1\end{pmatrix}\Rightarrow c=1,\ k=1.\quad\therefore\ k=1.$$

7. 设 A 是 n 阶方阵, 它的 n 个特征值为 $2,4,\cdots,2n$. 计算 $|A-3I|$.

解: $\because |A-\lambda I|=(\lambda_1-\lambda)(\lambda_2-\lambda)\cdots(\lambda_n-\lambda)=(2-\lambda)(4-\lambda)\cdots(2n-\lambda)$.

$\therefore |A-3I|=(2-3)(4-3)\cdots(2n-3)=(-1)\cdot 1\cdot 3\cdots(2n-3)=-1\times 3\times\cdots\times(2n-3)$

$=-(2n-3)!!$

8. 设 3 阶方阵 A 的特征值为 $\lambda_1=1,\ \lambda_2=2,\ \lambda_3=3$. 与它们对应的特征向量依次为

$$\vec{x}_1=\begin{pmatrix}1\\1\\1\end{pmatrix},\quad \vec{x}_2=\begin{pmatrix}1\\2\\4\end{pmatrix},\quad \vec{x}_3=\begin{pmatrix}1\\3\\9\end{pmatrix},\quad \text{设 } \vec{y}=\begin{pmatrix}1\\1\\3\end{pmatrix}.$$

(1) 将 $\vec{y}$ 用 $\vec{x}_1,\vec{x}_2,\vec{x}_3$ 线性表示;

(2) 求 $A^n\vec{y}$. (n 为正整数).

解: (1) 设 $\vec{y}=c_1\vec{x}_1+c_2\vec{x}_2+c_3\vec{x}_3=(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}$.

$\because \lambda_1\neq\lambda_2\neq\lambda_3$. $\therefore (\vec{x}_1\ \vec{x}_2\ \vec{x}_3)$ 线性无关.

$$\therefore \begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}=(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)^{-1}\vec{y}=\begin{bmatrix}1&1&1\\1&2&3\\1&4&9\end{bmatrix}^{-1}\begin{pmatrix}1\\1\\3\end{pmatrix}=\begin{bmatrix}3&-\frac{5}{2}&\frac{1}{2}\\-3&4&-1\\1&-\frac{3}{2}&\frac{1}{2}\end{bmatrix}\begin{pmatrix}1\\1\\3\end{pmatrix}=\begin{pmatrix}2\\-2\\1\end{pmatrix}$$

$$\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\1&2&3&0&1&0\\1&4&9&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&1&2&-1&1&0\\0&3&8&-1&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&1&1&1&0&0\\0&1&2&-1&1&0\\0&0&2&2&-3&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&0&0&3&-\frac{5}{2}&\frac{1}{2}\\0&1&0&-3&4&-1\\0&0&1&1&-\frac{3}{2}&\frac{1}{2}\end{array}\right)$$

$\therefore \vec{y}=2\vec{x}_1-2\vec{x}_2+\vec{x}_3$.

(2) $A=P\Lambda P^{-1}=(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)\begin{bmatrix}1&&\\&2&\\&&3\end{bmatrix}(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)^{-1}$.

$$A^n\vec{y}=P\Lambda^nP^{-1}\cdot\vec{y}=(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)\begin{bmatrix}1&&\\&2^n&\\&&3^n\end{bmatrix}\begin{pmatrix}c_1\\c_2\\c_3\end{pmatrix}=(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)\begin{pmatrix}c_1\\c_2\cdot 2^n\\c_3\cdot 3^n\end{pmatrix}$$

$=c_1\vec{x}_1+c_2 2^n\vec{x}_2+c_3 3^n\vec{x}_3=2\vec{x}_1-2^{n+1}\vec{x}_2+3^n\vec{x}_3$.

$$=2\begin{pmatrix}1\\1\\1\end{pmatrix}-2^{n+1}\begin{pmatrix}1\\2\\4\end{pmatrix}+3^n\begin{pmatrix}1\\3\\9\end{pmatrix}=\begin{pmatrix}2-2^{n+1}+3^n\\2-2^{n+2}+3^{n+1}\\2-2^{n+3}+3^{n+2}\end{pmatrix}$$

9. 设 $A=\begin{bmatrix}4&2&2\\0&4&0\\0&-2&2\end{bmatrix}$. 求 A^n. (n为正整数).

解: $|A-\lambda I|=\begin{vmatrix}4-\lambda&2&2\\0&4-\lambda&0\\0&-2&2-\lambda\end{vmatrix}=(4-\lambda)\begin{vmatrix}4-\lambda&0\\-2&2-\lambda\end{vmatrix}=(4-\lambda)(4-\lambda)(2-\lambda)$. $\therefore \lambda_1=2,\ \lambda_2=\lambda_3=4$.

对应于$\lambda_1=2$. $(A-\lambda_1 I)=\begin{pmatrix}2&2&2\\0&2&0\\0&-2&0\end{pmatrix}\to\begin{pmatrix}1&1&1\\0&1&0\\0&0&0\end{pmatrix}\to\begin{pmatrix}1&0&1\\0&1&0\\0&0&0\end{pmatrix}$ $\therefore \vec{\xi}_1=\begin{pmatrix}-1\\0\\1\end{pmatrix}$.

对应于$\lambda_2=\lambda_3=4$. $(A-\lambda_2 I)=\begin{pmatrix}0&2&2\\0&0&0\\0&-2&-2\end{pmatrix}\to\begin{pmatrix}0&1&1\\0&0&0\\0&0&0\end{pmatrix}$ $\therefore \vec{\xi}_2=\begin{pmatrix}1\\0\\0\end{pmatrix}$, $\vec{\xi}_3=\begin{pmatrix}0\\-1\\1\end{pmatrix}$.

即: $A(\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3)=(\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3)\begin{bmatrix}2&&\\&4&\\&&4\end{bmatrix}$. $\therefore P=(\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3)=\begin{bmatrix}-1&1&0\\0&0&-1\\1&0&1\end{bmatrix}$

则有: $AP=P\Lambda$ $A=P\Lambda P^{-1}$.

$\therefore A^n=P\Lambda^n P^{-1}=\begin{bmatrix}-1&1&0\\0&0&-1\\1&0&1\end{bmatrix}\begin{bmatrix}2^n&&\\&4^n&\\&&4^n\end{bmatrix}\begin{bmatrix}-1&1&0\\0&0&-1\\1&0&1\end{bmatrix}^{-1}$

$=\begin{bmatrix}-2^n&4^n&0\\0&0&-4^n\\2^n&0&4^n\end{bmatrix}\begin{bmatrix}0&1&1\\1&1&1\\0&-1&0\end{bmatrix}=\begin{bmatrix}4^n&2^n(2^n-1)&2^n(2^n-1)\\0&4^n&0\\0&-2^n(2^n-1)&2^n\end{bmatrix}$

$=2^n\begin{bmatrix}2^n&2^n-1&2^n-1\\0&2^n&0\\0&-2^n+1&1\end{bmatrix}=4^n\begin{bmatrix}1&1&1\\0&1&0\\0&-1&0\end{bmatrix}+2^n\begin{bmatrix}0&-1&-1\\0&0&0\\0&1&1\end{bmatrix}$

10. 设 $\begin{cases}x_n=2x_{n-1}-y_{n-1}\\y_n=\frac{3}{2}x_{n-1}-\frac{1}{2}y_{n-1}\end{cases}$, 且 $x_0=-1$, $y_0=+1$. 求 $\lim\limits_{n\to\infty}x_n$ 和 $\lim\limits_{n\to\infty}y_n$.

解: $\begin{pmatrix}x_n\\y_n\end{pmatrix}=\begin{bmatrix}2&-1\\\frac{3}{2}&-\frac{1}{2}\end{bmatrix}\begin{pmatrix}x_{n-1}\\y_{n-1}\end{pmatrix}=\begin{bmatrix}2&-1\\\frac{3}{2}&-\frac{1}{2}\end{bmatrix}^n\begin{pmatrix}x_0\\y_0\end{pmatrix}=A^n\begin{pmatrix}-1\\+1\end{pmatrix}$ $A=\begin{bmatrix}2&-1\\\frac{3}{2}&-\frac{1}{2}\end{bmatrix}$

$|A-\lambda I|=\begin{vmatrix}2-\lambda&-1\\\frac{3}{2}&-\frac{1}{2}-\lambda\end{vmatrix}=(2-\lambda)(-\frac{1}{2}-\lambda)+\frac{3}{2}=(\lambda-\frac{1}{2})(\lambda-1)=0$ $\therefore \lambda_1=\frac{1}{2}$, $\lambda_2=1$.

对应于$\lambda_1=\frac{1}{2}$. $(A-\lambda_1 I)=\begin{pmatrix}\frac{3}{2}&-1\\\frac{3}{2}&-1\end{pmatrix}\to\begin{pmatrix}1&-\frac{2}{3}\\0&0\end{pmatrix}$ $\therefore \vec{\xi}_1=\begin{pmatrix}\frac{2}{3}\\1\end{pmatrix}$.

对应于$\lambda_2=1$. $(A-\lambda_2 I)=\begin{pmatrix}1&-1\\\frac{3}{2}&-\frac{3}{2}\end{pmatrix}\to\begin{pmatrix}1&-1\\0&0\end{pmatrix}$ $\therefore \vec{\xi}_2=\begin{pmatrix}1\\1\end{pmatrix}$.

即: $A(\vec{\xi}_1, \vec{\xi}_2) = (\vec{\xi}_1, \vec{\xi}_2)\begin{bmatrix} \frac{1}{2} & \\ & 1 \end{bmatrix}$ $\quad A = (\vec{\xi}_1, \vec{\xi}_2)\begin{bmatrix} \frac{1}{2} & \\ & 1 \end{bmatrix}(\vec{\xi}_1, \vec{\xi}_2)^{-1}$.

$$A^n = (\vec{\xi}_1, \vec{\xi}_2)\begin{bmatrix} \frac{1}{2^n} & \\ & 1 \end{bmatrix}(\vec{\xi}_1, \vec{\xi}_2)^{-1} = \begin{bmatrix} \frac{2}{3} & 1 \\ 1 & 1 \end{bmatrix}\begin{bmatrix} \frac{1}{2^n} & 0 \\ 0 & 1 \end{bmatrix}\begin{bmatrix} \frac{2}{3} & 1 \\ 1 & 1 \end{bmatrix}^{-1}$$

$$= \begin{bmatrix} \frac{1}{3}\cdot\frac{1}{2^{n-1}} & 1 \\ \frac{1}{2^n} & 1 \end{bmatrix}\begin{bmatrix} -3 & 3 \\ 3 & -2 \end{bmatrix} = \begin{bmatrix} 3-\frac{1}{2^{n-1}} & \frac{1}{2^{n-1}}-2 \\ 3(1-\frac{1}{2^n}) & \frac{3}{2^n}-2 \end{bmatrix}.$$

$$\therefore \begin{pmatrix} x_n \\ y_n \end{pmatrix} = A^n\begin{pmatrix} -1 \\ +1 \end{pmatrix} = \begin{bmatrix} 3-\frac{1}{2^{n-1}} & \frac{1}{2^{n-1}}-2 \\ 3-\frac{3}{2^n} & \frac{3}{2^n}-2 \end{bmatrix}\begin{pmatrix} -1 \\ +1 \end{pmatrix} = \begin{pmatrix} -5+\frac{4}{2^n} \\ -5+\frac{6}{2^n} \end{pmatrix}.$$

$$\therefore \lim_{n\to\infty} x_n = \lim_{n\to\infty}\left(-5+\frac{4}{2^n}\right) = -5.$$

$$\lim_{n\to\infty} y_n = \lim_{n\to\infty}\left(-5+\frac{6}{2^n}\right) = -5.$$

§4. 内积和正交变换

习题. P286.

1、分别计算内积 $(\vec{x}, \vec{y})$ 和模 $\|\vec{x}\|$

(1) $\vec{x} = (-1, 2, -1)^T$. $\quad \vec{y} = (2, 1, -1)^T$.

解: $(\vec{x}, \vec{y}) = (\vec{x}^T\vec{y}) = (1, 2, -1)\begin{pmatrix} 2 \\ 1 \\ -1 \end{pmatrix} = 2+2+1 = 5.$

$$\|\vec{x}\| = (\vec{x}^T\vec{x})^{\frac{1}{2}} = \sqrt{(1, 2, -1)\begin{pmatrix} 1 \\ 2 \\ -1 \end{pmatrix}} = \sqrt{6}.$$

(2) $\vec{x} = (1+i, 1)^T$, $\vec{y} = (1, -1+i)^T$.

解: ~~$(\vec{x}, \vec{y}) = \vec{x}^T\vec{y} = (1+i, 1)\begin{pmatrix} 1 \\ -1+i \end{pmatrix} = 1+i-1+i = 2i$.~~

$$(\vec{x}, \vec{y}) = \vec{y}^H\vec{x} = (1, -1-i)\begin{pmatrix} 1+i \\ 1 \end{pmatrix} = 1+i-1-i = 0.$$

$$\|\vec{x}\| = \sqrt{(\vec{x}, \vec{x})} = \sqrt{\vec{x}^H\vec{x}} = \sqrt{(1-i, 1)\begin{pmatrix} 1+i \\ 1 \end{pmatrix}} = \sqrt{3}.$$

(3) $\vec{x} = (1+2i, 1-2i)^T$, $\quad \vec{y} = (2i, i-1)$

解: $(\vec{x}, \vec{y}) = \vec{y}^H\vec{x} = (-2i, -i-1)\begin{pmatrix} 1+2i \\ 1-2i \end{pmatrix} = -2i+4-i-1-2+2i = 1-i.$

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$$\|\vec{x}\| = \sqrt{\vec{x}^H \vec{x}} = \sqrt{(1-2i,\ 1+2i)\begin{pmatrix}1+2i\\1-2i\end{pmatrix}} = \sqrt{1+4+1+4} = \sqrt{10}.$$

2. 将下列各组向量正交化：

(1) $\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}1\\0\\1\end{pmatrix}$

解：设 $\vec{a}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix}$, $\vec{a}_2 = \begin{pmatrix}1\\0\\1\end{pmatrix}$. 令 $\vec{b}_1 = \vec{a}_1 = \begin{pmatrix}1\\1\\1\end{pmatrix}$. $(\vec{a}_2, \vec{b}_1) = (1\ 0\ 1)\begin{pmatrix}1\\1\\1\end{pmatrix} = 2$. $\|\vec{b}_1\|^2 = 3$.

$$\vec{b}_2 = \vec{a}_2 - \frac{(\vec{a}_2, \vec{b}_1)}{\|\vec{b}_1\|^2}\vec{b}_1 = \begin{pmatrix}1\\0\\1\end{pmatrix} - \frac{2}{3}\begin{pmatrix}1\\1\\1\end{pmatrix} = \begin{pmatrix}\frac{1}{3}\\-\frac{2}{3}\\\frac{1}{3}\end{pmatrix} = \frac{1}{3}\begin{pmatrix}1\\-2\\1\end{pmatrix}.$$

标准化：$\vec{\eta}_1 = \frac{1}{\sqrt{3}}\vec{b}_1 = \begin{pmatrix}\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\end{pmatrix}$ $\vec{\eta}_2 = \frac{\sqrt{3}}{2}\vec{b}_2 = \begin{pmatrix}\frac{\sqrt{3}}{6}\\-\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{6}\end{pmatrix}$.

$\therefore (\vec{a}_1, \vec{a}_2)$ 标准正交化为 $(\vec{\eta}_1, \vec{\eta}_2) = \left(\begin{pmatrix}\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\end{pmatrix}, \begin{pmatrix}\frac{\sqrt{3}}{6}\\-\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{6}\end{pmatrix}\right)$. (结果不唯一！)

(2) $\begin{pmatrix}1\\1\\0\\1\end{pmatrix}, \begin{pmatrix}1\\0\\2\\1\end{pmatrix}$

解：设 $\vec{a}_1 = \begin{pmatrix}1\\1\\0\\1\end{pmatrix}$, $\vec{a}_2 = \begin{pmatrix}1\\0\\2\\1\end{pmatrix}$. 令 $\vec{b}_1 = \vec{a}_1 = \begin{pmatrix}1\\1\\0\\1\end{pmatrix}$. $(\vec{a}_2, \vec{b}_1) = (1\ 0\ 2\ 1)\begin{pmatrix}1\\1\\0\\1\end{pmatrix} = 2$. $\|\vec{b}_1\|^2 = 3$.

$$\vec{b}_2 = \vec{a}_2 - \frac{(\vec{a}_2, \vec{b}_1)}{\|\vec{b}_1\|^2}\vec{b}_1 = \begin{pmatrix}1\\0\\2\\1\end{pmatrix} - \frac{2}{3}\begin{pmatrix}1\\1\\0\\1\end{pmatrix} = \begin{pmatrix}\frac{1}{3}\\-\frac{2}{3}\\2\\\frac{1}{3}\end{pmatrix} = \frac{1}{3}\begin{pmatrix}1\\-2\\6\\1\end{pmatrix}.$$

标准化：$\vec{\eta}_1 = \frac{1}{\sqrt{3}}\vec{b}_1$ $\vec{\eta}_2 = \frac{\sqrt{3}}{2}\vec{b}_2$.

$\therefore (\vec{a}_1, \vec{a}_2)$ 标准正交化为 $(\vec{\eta}_1, \vec{\eta}_2) = \left(\begin{pmatrix}\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\\0\\\frac{\sqrt{3}}{3}\end{pmatrix}, \begin{pmatrix}\frac{\sqrt{3}}{6}\\-\frac{\sqrt{3}}{3}\\\sqrt{3}\\\frac{\sqrt{3}}{6}\end{pmatrix}\right)$ (结果不唯一！)

3. 将下列向量标准正交化：

(1) $\begin{pmatrix}1\\1\\1\end{pmatrix}, \begin{pmatrix}-1\\-1\\1\end{pmatrix}$

解：令 $\vec{a}_1=\begin{pmatrix}1\\1\\1\end{pmatrix}$ $\vec{a}_2=\begin{pmatrix}-1\\-1\\1\end{pmatrix}$ 令 $\vec{b}_1=\vec{a}_1=\begin{pmatrix}1\\1\\1\end{pmatrix}$ $(\vec{a}_2,\vec{b}_1)=(-1,-1,1)\begin{pmatrix}1\\1\\1\end{pmatrix}=-1$ $\|\vec{b}_1\|^2=3$

$$\vec{b}_2=\vec{a}_2-\frac{(\vec{a}_2,\vec{b}_1)}{\|\vec{b}_1\|^2}\cdot\vec{b}_1=\begin{pmatrix}-1\\-1\\1\end{pmatrix}-\frac{-1}{3}\begin{pmatrix}1\\1\\1\end{pmatrix}=\begin{pmatrix}-\frac{4}{3}\\-\frac{2}{3}\\\frac{2}{3}\end{pmatrix}=\frac{2}{3}\begin{pmatrix}-2\\-1\\1\end{pmatrix}$$

标准化：$\vec{\eta}_1=\frac{1}{\sqrt{3}}\vec{b}_1$ $\vec{\eta}_2=\frac{\sqrt{6}}{4}\vec{b}_2$

$\therefore (\vec{a}_1,\vec{a}_2)$ 标准正交化为为 $(\vec{\eta}_1,\vec{\eta}_2)=\left(\begin{pmatrix}\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\\\frac{\sqrt{3}}{3}\end{pmatrix},\begin{pmatrix}-\frac{\sqrt{6}}{3}\\\frac{\sqrt{6}}{6}\\\frac{\sqrt{6}}{6}\end{pmatrix}\right)$ (结果不唯一!)

(2) $\begin{pmatrix}1\\1\\0\\0\end{pmatrix}, \begin{pmatrix}0\\1\\1\\0\end{pmatrix}, \begin{pmatrix}0\\0\\-1\\1\end{pmatrix}$

解：$\vec{a}_1=\begin{pmatrix}1\\1\\0\\0\end{pmatrix}$ $\vec{a}_2=\begin{pmatrix}0\\1\\1\\0\end{pmatrix}$ $\vec{a}_3=\begin{pmatrix}0\\0\\-1\\1\end{pmatrix}$ 令 $\vec{b}_1=\vec{a}_1=\begin{pmatrix}1\\1\\0\\0\end{pmatrix}$ $\|\vec{b}_1\|^2=2$

$(\vec{a}_2,\vec{b}_1)=(0,1,1,0)\begin{pmatrix}1\\1\\0\\0\end{pmatrix}=1$ $(\vec{a}_3,\vec{b}_1)=(0,0,-1,1)\begin{pmatrix}1\\1\\0\\0\end{pmatrix}=0$

$$\vec{b}_2=\vec{a}_2-\frac{(\vec{a}_2,\vec{b}_1)}{\|\vec{b}_1\|^2}\cdot\vec{b}_1=\begin{pmatrix}0\\1\\1\\0\end{pmatrix}-\frac{1}{2}\begin{pmatrix}1\\1\\0\\0\end{pmatrix}=\begin{pmatrix}-\frac{1}{2}\\\frac{1}{2}\\1\\0\end{pmatrix}=\frac{1}{2}\begin{pmatrix}-1\\1\\2\\0\end{pmatrix}\quad \|\vec{b}_2\|^2=\left(\frac{\sqrt{6}}{2}\right)^2=\frac{3}{2}$$

$(\vec{a}_3,\vec{b}_2)=(0,0,-1,1)\begin{pmatrix}-1\\1\\2\\0\end{pmatrix}\cdot\frac{1}{2}=1$

$$\vec{b}_3=\vec{a}_3-\frac{(\vec{a}_3,\vec{b}_1)}{\|\vec{b}_1\|^2}\vec{b}_1-\frac{(\vec{a}_3,\vec{b}_2)}{\|\vec{b}_2\|^2}\cdot\vec{b}_2=\begin{pmatrix}0\\0\\-1\\1\end{pmatrix}-0\cdot\begin{pmatrix}1\\1\\0\\0\end{pmatrix}-\frac{1}{\frac{3}{2}}\begin{pmatrix}-1\\1\\2\\0\end{pmatrix}\cdot\frac{1}{2}=\frac{\sqrt{6}}{3}\cdot\frac{\sqrt{6}}{3}\cdot\frac{\sqrt{6}}{3}\cdot\frac{1}{3}\begin{pmatrix}1\\-1\\-1\\3\end{pmatrix}$$

$\|\vec{b}_3\|^2=\frac{4}{3}$ 标准化：$\vec{\eta}_1=\frac{1}{\sqrt{2}}\vec{b}_1$ $\vec{\eta}_2=\frac{\sqrt{6}}{3}\vec{b}_2$ $\vec{\eta}_3=\frac{\sqrt{3}}{2}\vec{b}_3$

$\therefore (\vec{a}_1,\vec{a}_2,\vec{a}_3)$ 标准正交化为为 $(\vec{\eta}_1,\vec{\eta}_2,\vec{\eta}_3)=\left(\begin{pmatrix}\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{2}}\\0\\0\end{pmatrix},\begin{pmatrix}\frac{1}{\sqrt{6}}\\-\frac{1}{\sqrt{6}}\\\frac{2}{\sqrt{6}}\\0\end{pmatrix},\begin{pmatrix}\frac{1}{\sqrt{12}}\\-\frac{1}{\sqrt{12}}\\-\frac{1}{\sqrt{12}}\\\frac{3}{\sqrt{12}}\end{pmatrix}\right)$

(结果不唯一!)

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4. 设 $\{\vec{\varepsilon}_1, \vec{\varepsilon}_2, \cdots \vec{\varepsilon}_n\}$ 是 R^n 的一组标准正交基，R^n 中向量 $\vec{x} = \sum_{i=1}^{n} c_i \vec{\varepsilon}_i$，证明：

$c_i = (\vec{x}, \vec{\varepsilon}_i)$，$(i = 1, 2, \cdots, n)$.

证明：$\vec{x} = c_1\vec{\varepsilon}_1 + c_2\vec{\varepsilon}_2 + \cdots + c_n\vec{\varepsilon}_n$.

两边与 $\vec{\varepsilon}_i$ 作内积：$(\vec{x}, \vec{\varepsilon}_i) = c_1(\vec{\varepsilon}_1, \vec{\varepsilon}_i) + c_2(\vec{\varepsilon}_2, \vec{\varepsilon}_i) + \cdots + c_i(\vec{\varepsilon}_i, \vec{\varepsilon}_i) + \cdots + c_n(\vec{\varepsilon}_n, \vec{\varepsilon}_i)$

$\because \{\vec{\varepsilon}_1, \vec{\varepsilon}_2, \cdots \vec{\varepsilon}_n\}$ 是 R^n 的一组标准正交基. 即 $(\vec{\varepsilon}_i, \vec{\varepsilon}_j) = \begin{cases} 1, & i = j \\ 0, & i \neq j \end{cases}$

$\therefore$ $c_i = (\vec{x}, \vec{\varepsilon}_i)$.

5. 已知 S 是 R^4 中由 $\vec{a}_1 = (1, 1, 1, 1)^T$，$\vec{a}_2 = (1, -2, 0, 0)^T$ 张成的向量全体构成的空间.

(1) 求 S 的一组标准正交基；

(2) 扩充 S 的基，使之成为 R^4 的一组标准正交基.

解：(1) 设 $\vec{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{pmatrix} \in S$. 则有：$\begin{cases} \vec{x}^T\vec{a}_1 = 0 \\ \vec{x}^T\vec{a}_2 = 0 \end{cases} \Rightarrow \begin{cases} x_1 + x_2 + x_3 + x_4 = 0 \\ x_1 - 2x_2 = 0 \end{cases}$

$$\begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & -2 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 3 & 1 & 1 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & \frac{1}{3} & \frac{1}{3} \end{pmatrix} \to \begin{pmatrix} 1 & 0 & \frac{2}{3} & \frac{2}{3} \\ 0 & 1 & \frac{1}{3} & \frac{1}{3} \end{pmatrix}$$

基础解系：$\vec{\xi}_1 = \begin{pmatrix} 2 \\ 1 \\ -3 \\ 0 \end{pmatrix}$ $\vec{\xi}_2 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ -3 \end{pmatrix}$.

正交化：令 $\vec{\eta}_1 = \vec{\xi}_1 = \begin{pmatrix} 2 \\ 1 \\ -3 \\ 0 \end{pmatrix}$ $\|\vec{\eta}_1\|^2 = 14$. $(\vec{\eta}_1, \vec{\xi}_2) = (2, 1, -3, 0)\begin{pmatrix} 2 \\ 1 \\ 0 \\ -3 \end{pmatrix} = 5$.

$$\vec{\eta}_2 = \vec{\xi}_2 - \frac{(\vec{\xi}_2, \vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\vec{\eta}_1 = \begin{pmatrix} 2 \\ 1 \\ 0 \\ -3 \end{pmatrix} - \frac{5}{14}\begin{pmatrix} 2 \\ 1 \\ -3 \\ 0 \end{pmatrix} = \frac{3}{14}\begin{pmatrix} 6 \\ 3 \\ 5 \\ -14 \end{pmatrix} \quad \|\vec{\eta}_2\| = \frac{3\sqrt{66}}{14}$$

标准化：$\vec{\varepsilon}_1 = \frac{1}{\|\vec{\eta}_1\|}\vec{\eta}_1 = \frac{1}{\sqrt{14}}\begin{pmatrix} 2 \\ 1 \\ -3 \\ 0 \end{pmatrix}$ $\vec{\varepsilon}_2 = \frac{1}{\|\vec{\eta}_2\|}\vec{\eta}_2 = \frac{1}{\sqrt{266}}\begin{pmatrix} 6 \\ 3 \\ 5 \\ -14 \end{pmatrix}$

$\therefore$ S 的一组标准正交基 $(\vec{\varepsilon}_1, \vec{\varepsilon}_2) = \left(\frac{1}{\sqrt{14}}\begin{pmatrix} 2 \\ 1 \\ -3 \\ 0 \end{pmatrix}, \frac{1}{\sqrt{266}}\begin{pmatrix} 6 \\ 3 \\ 5 \\ -14 \end{pmatrix}\right)$.

(2) 取 S 的一组基 $\vec{\xi}_1=\begin{pmatrix}2\\1\\-3\\0\end{pmatrix}$, $\vec{\xi}_2=\begin{pmatrix}2\\1\\0\\-3\end{pmatrix}$ 与它们线性无关的向量满足:

$$\begin{cases}\vec{\xi}_1^T\vec{x}=0\\ \vec{\xi}_2^T\vec{x}=0\end{cases}\Rightarrow\begin{cases}2x_1+x_2-3x_3=0\\ 2x_1+x_2-3x_4=0\end{cases}$$

$$\begin{pmatrix}2&1&-3&0\\2&1&0&-3\end{pmatrix}\to\begin{pmatrix}2&1&-3&0\\0&0&3&-3\end{pmatrix}\to\left(\begin{pmatrix}2\\0\end{pmatrix}\ \begin{pmatrix}1&0\\0&1\end{pmatrix}\ \begin{pmatrix}-3\\-1\end{pmatrix}\right)\quad(-\beta_1,\ I_2,\ -\beta_2)$$

可取 $\vec{\xi}_3=\begin{pmatrix}1\\-2\\0\\0\end{pmatrix}$, $\vec{\xi}_4=\begin{pmatrix}0\\3\\1\\1\end{pmatrix}$.

则有 $(\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3\ \vec{\xi}_4)$ 线性无关, 构成 S 的一组基.

正交化: $\vec{\eta}_1=\vec{\xi}_1=\begin{pmatrix}2\\1\\-3\\0\end{pmatrix}$. $\|\vec{\eta}_1\|^2=14$. $(\vec{\xi}_2,\vec{\eta}_1)=5$. $(\vec{\xi}_3,\vec{\eta}_1)=0$. $(\vec{\xi}_4,\vec{\eta}_1)=0$.

$$\vec{\eta}_2=\vec{\xi}_2-\frac{(\vec{\xi}_2,\vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\vec{\eta}_1=\begin{pmatrix}2\\1\\0\\-3\end{pmatrix}-\frac{5}{14}\begin{pmatrix}2\\1\\-3\\0\end{pmatrix}=\frac{3}{14}\begin{pmatrix}6\\3\\5\\-14\end{pmatrix}\qquad \|\vec{\eta}_2\|^2=\frac{3\sqrt{266}}{14}$$

$(\vec{\xi}_3,\vec{\eta}_2)=0$, $(\vec{\xi}_4,\vec{\eta}_2)=0$

$$\vec{\eta}_3=\vec{\xi}_3-\frac{(\vec{\xi}_3,\vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\vec{\eta}_1-\frac{(\vec{\xi}_3,\vec{\eta}_2)}{\|\vec{\eta}_2\|^2}\vec{\eta}_2=\begin{pmatrix}1\\-2\\0\\0\end{pmatrix}-\begin{pmatrix}0\\0\\0\\0\end{pmatrix}-\begin{pmatrix}0\\0\\0\\0\end{pmatrix}=\begin{pmatrix}1\\-2\\0\\0\end{pmatrix}\qquad \|\vec{\eta}_3\|^2=5,\ (\vec{\xi}_4,\vec{\eta}_3)=-6$$

$$\vec{\eta}_4=\vec{\xi}_4-\frac{(\vec{\xi}_4,\vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\vec{\eta}_1-\frac{(\vec{\xi}_4,\vec{\eta}_2)}{\|\vec{\eta}_2\|^2}\vec{\eta}_2-\frac{(\vec{\xi}_4,\vec{\eta}_3)}{\|\vec{\eta}_3\|^2}\vec{\eta}_3=\begin{pmatrix}0\\3\\1\\1\end{pmatrix}-\frac{-6}{5}\begin{pmatrix}1\\-2\\0\\0\end{pmatrix}=\frac{1}{5}\begin{pmatrix}6\\3\\5\\5\end{pmatrix}$$

标准化: $\vec{\eta}_1=\frac{\vec{\eta}_1}{\|\vec{\eta}_1\|}$ $\vec{\eta}_2=\frac{\vec{\eta}_2}{\|\vec{\eta}_2\|}$ $\vec{\eta}_3=\frac{\vec{\eta}_3}{\|\vec{\eta}_3\|}$ $\vec{\eta}_4=\frac{\vec{\eta}_4}{\|\vec{\eta}_4\|}$.

∴ 四维一组标准正交基为 $(\vec{\eta}_1,\vec{\eta}_2,\vec{\eta}_3,\vec{\eta}_4)=\left(\frac{1}{\sqrt{14}}\begin{pmatrix}2\\1\\-3\\0\end{pmatrix},\frac{1}{\sqrt{266}}\begin{pmatrix}6\\3\\5\\-14\end{pmatrix},\frac{1}{\sqrt{5}}\begin{pmatrix}1\\-2\\0\\0\end{pmatrix},\frac{1}{\sqrt{105}}\begin{pmatrix}6\\3\\5\\5\end{pmatrix}\right)$.

6. 设 $\{\vec{a}_1,\vec{a}_2,\cdots,\vec{a}_r\}$ 是 r 个线性无关的 n 维向量, 试证存在 n−r 个线性无关的非零向量与它们正交.

证明: 设 $A=\begin{pmatrix}\vec{a}_1^T\\ \vec{a}_2^T\\ \vdots\\ \vec{a}_r^T\end{pmatrix}_{r\times n}$. ∵ $\{\vec{a}_1,\vec{a}_2,\cdots\vec{a}_r\}$ 线性无关. ∴ $\mathrm{rank}(A)=r$.

设 $\vec{x}$ 与 $\{\vec{a}_1,\vec{a}_2,\cdots,\vec{a}_r\}$ 正交. 则满足: $A\vec{x}=0$.

该齐次方程的基础解系共有 n−r 个解向量, 它们都与 $\{\vec{a}_1,\cdots\vec{a}_r\}$ 正交.

即存在 n−r 个线性无关的非零向量与它们正交.

7. 设 $\{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m\}$ 和 $\vec{b}$ 是给定的向量. 证明：对任何与 $\{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m\}$ 也正交的向量 $\vec{x}$ 恒有 $(\vec{x}, \vec{b}) = 0$ 的充分必要条件为：$\vec{b}$ 是 $\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m$ 的线性组合.

证明：$\Rightarrow$ 必要性.

记由 $\{\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m\}$ 生成的子空间为 A. 其正交补为 $A^\perp$.

若 $(\vec{x}, \vec{a}_i) = 0$, $i = 1, 2, \cdots, m$. 则必有 $\vec{x} \in A^\perp$.

而 $\forall \vec{x} \in A^\perp$ 有 $(\vec{x}, \vec{b}) = 0$. 则必有 $\vec{b} \in (A^\perp)^\perp = A$. $\Rightarrow \vec{b} \in A$.

即 $\vec{b}$ 可由 A 的基线性表出. $\Rightarrow$ $\vec{b}$ 也可由 $\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m$ 线性表出.

$\therefore$ $\vec{b}$ 是 $\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m$ 的线性组合.

$\Leftarrow$ 充分性.

若 $\vec{b} = \sum_{i=1}^{m} k_i \vec{a}_i$.

$\forall \vec{x} \in A^\perp$, $(\vec{x}, \vec{b}) = \sum_{i=1}^{m} k_i (\vec{x}, \vec{a}_i) = 0$. $\therefore$ 恒有 $(\vec{x}, \vec{b}) = 0$.

8. 设 Q 是正交矩阵，证明：

(1) Q^{-1} 也是正交矩阵；

(2) $\det(Q) = \pm 1$；

(3) 若 Q_1 和 Q_2 都是正交矩阵，则 $Q_1 Q_2$ 也是正交矩阵.

证明：(1) $Q^T Q = I$. 两边求逆：$Q^{-1} \cdot Q^{-T} = I^{-1} = I$.

即 $Q^{-1} \cdot (Q^{-1})^T = I$. $\therefore$ Q^{-1} 也是正交矩阵.

(2) $Q^T Q = I$. 两边求行列式：$\det(Q^T) \cdot \det(Q) = \det(I)$

即 $[\det(Q)]^2 = 1$. $\therefore$ $\det(Q) = \pm 1$.

(3) $Q_1^T Q_1 = I$. $Q_2^T Q_2 = I$. 则有：

$(Q_1 Q_2)^T (Q_1 Q_2) = Q_2^T (Q_1^T Q_1) Q_2 = Q_2^T Q_2 = I$. $\therefore$ $Q_1 Q_2$ 也是正交矩阵.

9. 设 A 为 n 阶方阵，且满足 $A^2 - 4A + 3I = 0$, $A^T = A$，证明 $A - 2I$ 为正交矩阵.

证明：$(A-2I)^T (A-2I) = (A^T - 2I)(A-2I) = (A-2I)^2 = A^2 - 4A + 4I = A^2 - 4A + 3I + I = I$.

$\therefore$ $A - 2I$ 为正交矩阵.

10. 试证 $A \in C^{n \times n}$ 为酉阵的充分必要条件是：A 的行向量组为 C^n 中的标准正交向量组.

证明：$A \in C^{n \times n}$ 为酉矩阵 $\Leftrightarrow$ A 的 n 个列向量是 C^n 中的一个标准正交向量组

$$\Leftrightarrow A^HA=\begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}^H \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} = \begin{bmatrix} \bar{c}_{11} & \bar{c}_{21} & \cdots & \bar{c}_{n1} \\ \bar{c}_{12} & \bar{c}_{22} & \cdots & \bar{c}_{n2} \\ \vdots & \vdots & & \vdots \\ \bar{c}_{1n} & \bar{c}_{2n} & \cdots & \bar{c}_{nn} \end{bmatrix} \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{i=1}^n \bar{c}_{i1}c_{i1} & \sum_{i=1}^n \bar{c}_{i1}c_{i2} & \cdots & \sum_{i=1}^n \bar{c}_{i1}c_{in} \\ \sum_{i=1}^n \bar{c}_{i2}c_{i1} & \sum_{i=1}^n \bar{c}_{i2}c_{i2} & \cdots & \sum_{i=1}^n \bar{c}_{i2}c_{in} \\ \vdots & \vdots & & \vdots \\ \sum_{i=1}^n \bar{c}_{in}c_{i1} & \sum_{i=1}^n \bar{c}_{in}c_{i2} & \cdots & \sum_{i=1}^n \bar{c}_{in}c_{in} \end{bmatrix} = I. \quad \text{即} \sum_{k=1}^n \bar{c}_{ki}c_{kj} = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$$

~~$\Leftrightarrow$ A 的 n 个行向量也是 $\mathbb{C}^n$ 中的一个标准正交向量组~~

$$\Leftrightarrow A^H = A^{-1}$$

$$\Leftrightarrow AA^H = \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \vdots & \vdots & & \vdots \\ c_{n1} & c_{n2} & \cdots & c_{nn} \end{bmatrix} \begin{bmatrix} \bar{c}_{11} & \bar{c}_{21} & \cdots & \bar{c}_{n1} \\ \bar{c}_{12} & \bar{c}_{22} & \cdots & \bar{c}_{n2} \\ \vdots & \vdots & & \vdots \\ \bar{c}_{1n} & \bar{c}_{2n} & \cdots & \bar{c}_{nn} \end{bmatrix}$$

$$= \begin{bmatrix} \sum_{k=1}^n c_{1k}\bar{c}_{1k} & \sum_{k=1}^n c_{1k}\bar{c}_{2k} & \cdots & \sum_{k=1}^n c_{1k}\bar{c}_{nk} \\ \sum_{k=1}^n c_{2k}\bar{c}_{1k} & \sum_{k=1}^n c_{2k}\bar{c}_{2k} & \cdots & \sum_{k=1}^n c_{2k}\bar{c}_{nk} \\ \vdots & \vdots & & \vdots \\ \sum_{k=1}^n c_{nk}\bar{c}_{1k} & \sum_{k=1}^n c_{nk}\bar{c}_{2k} & \cdots & \sum_{k=1}^n c_{nk}\bar{c}_{nk} \end{bmatrix} = I. \quad \text{即} \sum_{k=1}^n c_{ik}\bar{c}_{jk} = \begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$$

$\Leftrightarrow$ A 的 n 个行向量也是 $\mathbb{C}^n$ 中的标准正交向量组.

11. 下列矩阵是否为正交阵或酉阵:

(1) $\begin{pmatrix} e^{ia} & 0 \\ 0 & e^{ib} \end{pmatrix}$

解: $A^HA = \begin{pmatrix} e^{-ia} & 0 \\ 0 & e^{-ib} \end{pmatrix}\begin{pmatrix} e^{ia} & 0 \\ 0 & e^{ib} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$. ∴是酉阵.

$A^TA = \begin{pmatrix} e^{ia} & 0 \\ 0 & e^{ib} \end{pmatrix}\begin{pmatrix} e^{ia} & 0 \\ 0 & e^{ib} \end{pmatrix} = \begin{pmatrix} e^{2ia} & 0 \\ 0 & e^{2ib} \end{pmatrix} \neq I_2$. ∴不是正交阵.

(2) $\frac{1}{\sqrt{3}}\begin{pmatrix} 1+i & 1 \\ 1 & -1+i \end{pmatrix}$

解: $A^HA = \frac{1}{\sqrt{3}}\begin{pmatrix} 1-i & 1 \\ 1 & -1-i \end{pmatrix}\cdot\frac{1}{\sqrt{3}}\begin{pmatrix} 1+i & 1 \\ 1 & -1+i \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I_2$ ∴是酉阵.

$A^TA = \frac{1}{\sqrt{3}}\begin{pmatrix} 1+i & 1 \\ 1 & -1+i \end{pmatrix}\cdot\frac{1}{\sqrt{3}}\begin{pmatrix} 1+i & 1 \\ 1 & -1+i \end{pmatrix} = \frac{1}{3}\begin{pmatrix} 2i+1 & 2i \\ 2i & 1-2i \end{pmatrix} \neq I_2$. ∴不是正交阵.

B1 $\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}$

解: $A^HA = A^TA = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}\begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$.

∴ 酉矩阵也是正交阵.

B4 $\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix}$ (第1列与第2列不正交)

解: $A^HA = A^TA = \begin{bmatrix} \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{6}} & \frac{2}{\sqrt{6}} & -\frac{1}{\sqrt{6}} \end{bmatrix}\begin{bmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & \frac{1}{\sqrt{6}} \\ 0 & \frac{1}{\sqrt{3}} & \frac{2}{\sqrt{6}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{3}} & -\frac{1}{\sqrt{6}} \end{bmatrix} = \begin{bmatrix} 1 & -\frac{2}{\sqrt{6}} & 0 \\ -\frac{2}{\sqrt{6}} & 1 & \frac{2}{\sqrt{18}} \\ 0 & \frac{2}{\sqrt{18}} & 1 \end{bmatrix} \neq I_3$.

∴ 酉矩阵也不是正交阵.

12. 定义向量组 $\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m$ 的Gram行列式为

$$G[\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m] = \det((\vec{a}_i, \vec{a}_j)) = \begin{vmatrix} (\vec{a}_1, \vec{a}_1) & (\vec{a}_1, \vec{a}_2) & \cdots & (\vec{a}_1, \vec{a}_m) \\ (\vec{a}_2, \vec{a}_1) & (\vec{a}_2, \vec{a}_2) & \cdots & (\vec{a}_2, \vec{a}_m) \\ \vdots & \vdots & & \vdots \\ (\vec{a}_m, \vec{a}_1) & (\vec{a}_m, \vec{a}_2) & \cdots & (\vec{a}_m, \vec{a}_m) \end{vmatrix}$$

证明: $\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m$ 线性相关的充分必要条件是 $G[\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m] = 0$

证明: $\Rightarrow$ 必要性. $\vec{a}_1, \vec{a}_2, \cdots, \vec{a}_m$ 线性相关 $\Leftrightarrow$ 存在不全为0的系数 $\lambda_1, \lambda_2, \cdots, \lambda_m$ 使得

$\lambda_1\vec{a}_1 + \lambda_2\vec{a}_2 + \cdots + \lambda_m\vec{a}_m = \vec{0}$. 不妨设 $\lambda_1 \neq 0$

则 $G[\vec{a}_1, \cdots \vec{a}_m] = \begin{vmatrix} (\sum_{k=1}^m \lambda_k\vec{a}_k, \vec{a}_1) & (\sum_{k=1}^m \lambda_k\vec{a}_k, \vec{a}_2) & \cdots & (\sum_{k=1}^m \lambda_k\vec{a}_k, \vec{a}_m) \\ (\vec{a}_2, \vec{a}_1) & (\vec{a}_2, \vec{a}_2) & \cdots & (\vec{a}_2, \vec{a}_m) \\ \vdots & \vdots & & \vdots \\ (\vec{a}_m, \vec{a}_1) & (\vec{a}_m, \vec{a}_2) & \cdots & (\vec{a}_m, \vec{a}_m) \end{vmatrix} \cdot \frac{1}{\lambda_1}$

(将第1行乘上 λ_k 加到第1行)

$= \begin{vmatrix} (\vec{0}, \vec{a}_1) & (\vec{0}, \vec{a}_2) & \cdots & (\vec{0}, \vec{a}_m) \\ (\vec{a}_2, \vec{a}_1) & (\vec{a}_2, \vec{a}_2) & \cdots & (\vec{a}_2, \vec{a}_m) \\ \vdots & \vdots & & \vdots \\ (\vec{a}_m, \vec{a}_1) & (\vec{a}_m, \vec{a}_2) & \cdots & (\vec{a}_m, \vec{a}_m) \end{vmatrix} \cdot \frac{1}{\lambda_1} = 0$.

⇐ 充分性. 由 $G[\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m]=0$ 可得 G 的 m 个列向量线性相关.

即存在不全为0的系数 $\lambda_1, \lambda_2, \cdots, \lambda_m$ 成立:

$$\lambda_1 \begin{pmatrix} (\vec{a}_1,\vec{a}_1) \\ (\vec{a}_2,\vec{a}_1) \\ \vdots \\ (\vec{a}_m,\vec{a}_1) \end{pmatrix} + \lambda_2 \begin{pmatrix} (\vec{a}_1,\vec{a}_2) \\ (\vec{a}_2,\vec{a}_2) \\ \vdots \\ (\vec{a}_m,\vec{a}_2) \end{pmatrix} + \cdots + \lambda_m \begin{pmatrix} (\vec{a}_1,\vec{a}_m) \\ (\vec{a}_2,\vec{a}_m) \\ \vdots \\ (\vec{a}_m,\vec{a}_m) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

即:

$$\begin{pmatrix} (\vec{a}_1,\lambda_1\vec{a}_1) \\ (\vec{a}_2,\lambda_1\vec{a}_1) \\ \vdots \\ (\vec{a}_m,\lambda_1\vec{a}_1) \end{pmatrix} + \begin{pmatrix} (\vec{a}_1,\lambda_2\vec{a}_2) \\ (\vec{a}_2,\lambda_2\vec{a}_2) \\ \vdots \\ (\vec{a}_m,\lambda_2\vec{a}_2) \end{pmatrix} + \cdots + \begin{pmatrix} (\vec{a}_1,\lambda_m\vec{a}_m) \\ (\vec{a}_2,\lambda_m\vec{a}_m) \\ \vdots \\ (\vec{a}_m,\lambda_m\vec{a}_m) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix}$$

由此得: $(\vec{a}_j, \lambda_1\vec{a}_1) + (\vec{a}_j, \lambda_2\vec{a}_2) + \cdots + (\vec{a}_j, \lambda_m\vec{a}_m) = 0$.

$(\vec{a}_j, \lambda_1\vec{a}_1 + \lambda_2\vec{a}_2 + \cdots + \lambda_m\vec{a}_m) = 0$. $j=1, 2, \cdots, m$.

进一步得: $(\lambda_1\vec{a}_1 + \lambda_2\vec{a}_2 + \cdots + \lambda_m\vec{a}_m, \lambda_1\vec{a}_1 + \lambda_2\vec{a}_2 + \cdots + \lambda_m\vec{a}_m) = 0$.

$\therefore \lambda_1\vec{a}_1 + \lambda_2\vec{a}_2 + \cdots + \lambda_m\vec{a}_m = 0$. 且 $\lambda_1, \lambda_2 \cdots \lambda_m$ 不全为0.

$\therefore \vec{a}_1, \vec{a}_2, \cdots \vec{a}_m$ 线性相关

13. 设 A 为 $n\times m$ 阵, $\text{rank}(A)=m$, 试证明 $\text{rank}(A^TA)=m$.

证明: 设 $A=(\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m)$ $\vec{a}_j \in R^n$. $j=1, 2, \cdots, m$.

$\because \text{rank}(A)=m$ $\therefore (\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m)$ 线性无关.

则 $G[\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m] \neq 0$.

而 $$A^TA = \begin{pmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_m^T \end{pmatrix} (\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m) = \begin{pmatrix} \vec{a}_1^T\vec{a}_1 & \vec{a}_1^T\vec{a}_2 & \cdots & \vec{a}_1^T\vec{a}_m \\ \vec{a}_2^T\vec{a}_1 & \vec{a}_2^T\vec{a}_2 & \cdots & \vec{a}_2^T\vec{a}_m \\ \vdots & \vdots & & \vdots \\ \vec{a}_m^T\vec{a}_1 & \vec{a}_m^T\vec{a}_2 & \cdots & \vec{a}_m^T\vec{a}_m \end{pmatrix}_{m\times m}$$

$|A^TA| = G[\vec{a}_1, \vec{a}_2, \cdots \vec{a}_m) \neq 0$. $\therefore \text{rank}(A^TA)=m$.

14. 证明 $\text{rank}(A^TA) = \text{rank}(AA^T) = \text{rank}(A)$

证明: 设 A 为 $n\times m$ 矩阵. 则 A^TA 为 $m\times m$ 阵, AA^T 为 $n\times n$ 阵.

首先证明齐次线性方程组 $A\vec{x}=\vec{0}$ 与 $A^TA\vec{x}=\vec{0}$ 同解 (即同解).

若 $\vec{x}_1$ 是 $A\vec{x}=\vec{0}$ 的解, 则有 $A\vec{x}_1=\vec{0}$ 显然 $A^TA\vec{x}_1=\vec{0}$ 也成立. $\therefore \vec{x}_1$ 也是 $A^TA\vec{x}=\vec{0}$ 的解.

若 $\vec{x}_1$ 是 $A^TA\vec{x}=\vec{0}$ 的解，则有 $A^TA\vec{x}_1=\vec{0}$ 左乘 $\vec{x}_1^T$ 有 $\vec{x}_1^TA^TA\vec{x}_1=\vec{0}$ 即

$(A\vec{x}_1)^T(A\vec{x}_1)=\vec{0}$ $\Rightarrow$ $A\vec{x}_1=\vec{0}$，即 $\vec{x}_1$ 也是 $A\vec{x}=\vec{0}$ 的解.

$\therefore$ $A^TA\vec{x}=\vec{0}$ 与 $A\vec{x}=\vec{0}$ 同解.

$\therefore$ $\text{rank}(A^TA)=\text{rank}(A)$

同样可证，$AA^T\vec{x}=\vec{0}$ 与 $A^T\vec{x}=\vec{0}$ 同解. 即 $\text{rank}(AA^T)=\text{rank}(A^T)$

$\because \text{rank}(A)=\text{rank}(A^T)$.

$\therefore \text{rank}(A^TA)=\text{rank}(AA^T)=\text{rank}(A)$.

§5 正交相似和酉相似.

习题. P.294.

1. 用正交相似变换将下列矩阵化成对角矩阵：

(1) $\begin{pmatrix} 9 & 12 \\ 12 & 16 \end{pmatrix}$.

解：$|A-\lambda I|=\begin{vmatrix} 9-\lambda & 12 \\ 12 & 16-\lambda \end{vmatrix}=\lambda^2-25\lambda+144-144=\lambda(\lambda-25)=0 \Rightarrow \lambda_1=0,\ \lambda_2=25.$

对于 $\lambda_1=0$，$(A-\lambda_1 I)=A=\begin{pmatrix} 9 & 12 \\ 12 & 16 \end{pmatrix}\to\begin{pmatrix} 1 & \frac{4}{3} \\ 1 & \frac{4}{3} \end{pmatrix}\to\begin{pmatrix} 1 & \frac{4}{3} \\ 0 & 0 \end{pmatrix}$. 可取 $\vec{\xi}_1=\begin{pmatrix} -\frac{4}{3} \\ 1 \end{pmatrix}$.

对于 $\lambda_2=25$，$(A-\lambda_2 I)=\begin{bmatrix} -16 & 12 \\ 12 & -9 \end{bmatrix}\to\begin{pmatrix} 1 & -\frac{3}{4} \\ 1 & -\frac{3}{4} \end{pmatrix}\to\begin{pmatrix} 1 & -\frac{3}{4} \\ 0 & 0 \end{pmatrix}$. 可取 $\vec{\xi}_2=\begin{pmatrix} \frac{3}{4} \\ 1 \end{pmatrix}$.

$\because \lambda_1\neq\lambda_2$，$\therefore$ $\vec{\xi}_1,\vec{\xi}_2$ 线性无关且正交.

标准化：$\vec{\eta}_1=\frac{1}{\|\vec{\xi}_1\|}\vec{\xi}_1=\begin{pmatrix} \frac{4}{5} \\ -\frac{3}{5} \end{pmatrix}$. $\vec{\eta}_2=\frac{1}{\|\vec{\xi}_2\|}\vec{\xi}_2=\begin{pmatrix} \frac{3}{5} \\ \frac{4}{5} \end{pmatrix}$

$\therefore$ 取 $S=(\vec{\eta}_1\ \vec{\eta}_2)=\begin{bmatrix} \frac{4}{5} & \frac{3}{5} \\ -\frac{3}{5} & \frac{4}{5} \end{bmatrix}$. 则有 $S^TS=I$

且 $S^TAS=\Lambda=\begin{bmatrix} 0 & 0 \\ 0 & 25 \end{bmatrix}$.

(2) $\begin{bmatrix} 1 & 3 & 0 \\ 3 & -2 & -1 \\ 0 & -1 & 1 \end{bmatrix}$

解：$|A-\lambda I|=\begin{vmatrix} 1-\lambda & 3 & 0 \\ 3 & -2-\lambda & -1 \\ 0 & -1 & 1-\lambda \end{vmatrix}=-(\lambda-1)(\lambda-3)(\lambda+4)=0 \Rightarrow \lambda_1=-4,\ \lambda_2=1,\ \lambda_3=3.$

$\lambda_1=-4$. $(A-\lambda_1 I)=\begin{pmatrix}5&3&0\\3&2&-1\\0&-1&5\end{pmatrix}\to\begin{pmatrix}1&\frac{3}{5}&0\\0&\frac{1}{5}&-1\\0&-1&5\end{pmatrix}\to\begin{pmatrix}1&\frac{3}{5}&0\\0&1&-5\\0&0&0\end{pmatrix}$

$\to\begin{pmatrix}1&0&3\\0&1&-5\\0&0&0\end{pmatrix}$ $\therefore$ 可取 $\vec{\xi}_1=\begin{pmatrix}-3\\5\\1\end{pmatrix}$ $\vec{\eta}_1=\frac{\vec{\xi}_1}{\|\vec{\xi}_1\|}=\frac{1}{\sqrt{35}}\begin{pmatrix}-3\\5\\1\end{pmatrix}$.

$\lambda_2=1$. $(A-\lambda_2 I)=\begin{pmatrix}0&3&0\\3&-2&-1\\0&-1&0\end{pmatrix}\to\begin{pmatrix}1&-\frac{2}{3}&-\frac{1}{3}\\0&1&0\\0&0&0\end{pmatrix}\to\begin{pmatrix}1&0&-\frac{1}{3}\\0&1&0\\0&0&0\end{pmatrix}$.

$\therefore$ 可取 $\vec{\xi}_2=\begin{pmatrix}\frac{1}{3}\\0\\1\end{pmatrix}$. $\vec{\eta}_2=\frac{1}{\|\vec{\xi}_2\|}\vec{\xi}_2=\frac{1}{\sqrt{10}}\begin{pmatrix}1\\0\\3\end{pmatrix}$.

$\lambda_3=3$. $(A-\lambda_3 I)=\begin{pmatrix}-2&3&0\\3&-5&-1\\0&-1&-2\end{pmatrix}\to\begin{pmatrix}1&-\frac{3}{2}&0\\0&-\frac{1}{2}&-1\\0&-1&-2\end{pmatrix}\to\begin{pmatrix}1&-\frac{3}{2}&0\\0&1&2\\0&0&0\end{pmatrix}\to\begin{pmatrix}1&0&3\\0&1&2\\0&0&0\end{pmatrix}$.

$\therefore$ 可取 $\vec{\xi}_3=\begin{pmatrix}-3\\-2\\1\end{pmatrix}$. $\vec{\eta}_3=\frac{1}{\|\vec{\xi}_3\|}\vec{\xi}_3=\frac{1}{\sqrt{14}}\begin{pmatrix}-3\\-2\\1\end{pmatrix}$.

$\therefore$ 取 $S=(\vec{\eta}_1,\vec{\eta}_2,\vec{\eta}_3)=\begin{pmatrix}\frac{-3}{\sqrt{35}}&\frac{1}{\sqrt{10}}&-\frac{3}{\sqrt{14}}\\\frac{5}{\sqrt{35}}&0&-\frac{2}{\sqrt{14}}\\\frac{1}{\sqrt{35}}&\frac{3}{\sqrt{10}}&\frac{1}{\sqrt{14}}\end{pmatrix}$ 则有: $S^TS=I$.

且: $S^TAS=\Lambda=\begin{bmatrix}-4&0&0\\0&1&0\\0&0&3\end{bmatrix}$.

(3) $\begin{bmatrix}1&2&2\\2&1&2\\2&2&1\end{bmatrix}$

解: $|A-\lambda I|=\begin{vmatrix}1-\lambda&2&2\\2&1-\lambda&2\\2&2&1-\lambda\end{vmatrix}=-(\lambda+1)^2(\lambda-5)=0 \Rightarrow \lambda_1=\lambda_2=-1,\ \lambda_2=5$.

$\lambda_1=\lambda_2=-1$: $(A-\lambda_1 I)=\begin{pmatrix}2&2&2\\2&2&2\\2&2&2\end{pmatrix}\to\begin{pmatrix}1&1&1\\0&0&0\\0&0&0\end{pmatrix}$. $\therefore$ 可取 $\vec{\xi}_1=\begin{pmatrix}-1\\1\\0\end{pmatrix}$. $\vec{\xi}_2=\begin{pmatrix}-1\\0\\1\end{pmatrix}$.

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正交化: $\vec{\xi}_1 = \vec{\eta}_1 = \begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$.

$\vec{\xi}_2 = \vec{\eta}_2 - \frac{(\vec{\eta}_2, \vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\vec{\eta}_1 = \begin{pmatrix} -1 \\ 0 \\ 1 \end{pmatrix} - \frac{1}{2}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} -\frac{1}{2} \\ -\frac{1}{2} \\ 1 \end{pmatrix}$.

标准化: $\vec{\eta}_1 = \frac{1}{\|\vec{\xi}_1\|}\vec{\xi}_1 = \frac{1}{\sqrt{2}}\begin{pmatrix} -1 \\ 1 \\ 0 \end{pmatrix}$. $\vec{\eta}_2 = \frac{1}{\|\vec{\xi}_2\|}\vec{\xi}_2 = \frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ 1 \\ -2 \end{pmatrix}$.

$\lambda_3 = 5$. $(A - \lambda_3 I) = \begin{pmatrix} -4 & 2 & 2 \\ 2 & -4 & 2 \\ 2 & 2 & -4 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & -2 \\ 0 & 6 & -6 \\ 0 & 8 & -8 \end{pmatrix} \to \begin{pmatrix} 1 & 1 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{pmatrix}$

$\therefore$ 可取 $\vec{\eta}_3 = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$. $\vec{\eta}_3 = \frac{1}{\sqrt{3}}\begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}$.

$\therefore$ 取 $S = (\vec{\eta}_1, \vec{\eta}_2, \vec{\eta}_3) = \begin{bmatrix} -\frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{6}} & \frac{1}{\sqrt{3}} \\ 0 & -\frac{2}{\sqrt{6}} & \frac{1}{\sqrt{3}} \end{bmatrix}$ 则有 $S^TS = I_3$.

则 $S^TAS = \Lambda = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 5 \end{bmatrix}$

(4) $\begin{bmatrix} 2 & 2 & -2 \\ 2 & 5 & -4 \\ -2 & -4 & 5 \end{bmatrix}$.

解: $|A - \lambda I| = \begin{vmatrix} 2-\lambda & 2 & -2 \\ 2 & 5-\lambda & -4 \\ -2 & -4 & 5-\lambda \end{vmatrix} = -(\lambda-1)^2(\lambda-10) = 0 \Rightarrow \lambda_1 = \lambda_2 = 1, \lambda_3 = 10$.

$\lambda_1 = \lambda_2 = 1$. $(A - \lambda_1 I) = \begin{pmatrix} 1 & 2 & -2 \\ 2 & 4 & -4 \\ -2 & -4 & 4 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & -2 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$. $\therefore$ 可取 $\vec{\eta}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$, $\vec{\eta}_2 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix}$.

正交化: $\vec{\xi}_1 = \vec{\eta}_1 = \begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$.

$\vec{\xi}_2 = \vec{\eta}_2 - \frac{(\vec{\eta}_2, \vec{\xi}_1)}{\|\vec{\xi}_1\|^2}\vec{\xi}_1 = \begin{pmatrix} 2 \\ 0 \\ 1 \end{pmatrix} - \frac{-4}{5}\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2}{5} \\ \frac{4}{5} \\ 1 \end{pmatrix}$.

标准化: $\vec{\eta}_1 = \frac{1}{\|\vec{\xi}_1\|}\vec{\xi}_1 = \frac{1}{\sqrt{5}}\begin{pmatrix} -2 \\ 1 \\ 0 \end{pmatrix}$. $\vec{\eta}_2 = \frac{1}{\|\vec{\xi}_2\|}\vec{\xi}_2 = \frac{1}{3\sqrt{5}}\begin{pmatrix} 2 \\ 4 \\ 5 \end{pmatrix}$.

$\lambda_3 = 10$. $(A - \lambda_3 I) = \begin{pmatrix} -8 & 2 & -2 \\ 2 & -5 & -4 \\ -2 & -4 & -5 \end{pmatrix} \to \begin{pmatrix} 2 & -4 & 5 \\ 0 & -9 & -9 \\ 0 & -18 & -18 \end{pmatrix} \to \begin{pmatrix} 1 & 2 & \frac{5}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix} \to \begin{pmatrix} 1 & 0 & +\frac{1}{2} \\ 0 & 1 & 1 \\ 0 & 0 & 0 \end{pmatrix}$.

∴ 可取 $\vec{\eta}_3 = \begin{pmatrix} -\frac{1}{2} \\ -1 \\ 1 \end{pmatrix}$ $\vec{\eta}_3 = \frac{1}{3}\begin{pmatrix} -1 \\ -2 \\ 2 \end{pmatrix}$

∴ 取 $S = (\vec{\eta}_1\ \vec{\eta}_2\ \vec{\eta}_3) = \begin{bmatrix} -\frac{2}{\sqrt{5}} & \frac{2\sqrt{5}}{15} & -\frac{1}{3} \\ \frac{1}{\sqrt{5}} & \frac{4\sqrt{5}}{15} & -\frac{2}{3} \\ 0 & \frac{\sqrt{5}}{3} & \frac{2}{3} \end{bmatrix}$ 则有 $S^TS = I_3$.

且: $S^TAS = \Lambda = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 10 \end{bmatrix}$

2. 用酉相似变换将下列矩阵化成对角矩阵:

1) $\begin{pmatrix} 1 & 2i \\ -2i & 3 \end{pmatrix}$

解:

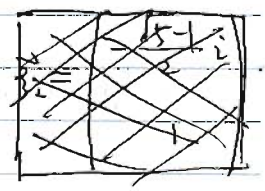
$|A-\lambda I| = \begin{vmatrix} 1-\lambda & 2i \\ -2i & 3-\lambda \end{vmatrix} = \lambda^2 - 4\lambda + 3 - 4 = \lambda^2 - 4\lambda + 7 = 0 \Rightarrow \lambda_{1,2} = 2 \pm \sqrt{3}i$.

$(A-\lambda_1 I) = \begin{pmatrix} -1-\sqrt{3}i & 2i \\ -2i & 1-\sqrt{3}i \end{pmatrix}$ $(A-\lambda_1 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1-\sqrt{3}i & 2i \\ -2i & 1-\sqrt{3}i \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$

$\Rightarrow \vec{\xi}_1 = \begin{pmatrix} 2 \\ +\sqrt{3}-i \end{pmatrix}$ $\vec{\xi}_2 = \begin{pmatrix} 2 \\ -\sqrt{3}+i \end{pmatrix}$

$|A-\lambda I| = \begin{vmatrix} 1-\lambda & 2i \\ -2i & 3-\lambda \end{vmatrix} = \lambda^2 - 4\lambda + 3 - 4 = \lambda^2 - 4\lambda - 1 = 0 \Rightarrow \lambda_{1,2} = 2 \pm \sqrt{5}$.

$(A-\lambda_1 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1-\sqrt{5} & 2i \\ -2i & 1-\sqrt{5} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_1 = \begin{pmatrix} \frac{\sqrt{5}-1}{2}i \\ 1 \end{pmatrix}$



$(A-\lambda_2 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1+\sqrt{5} & 2i \\ -2i & 1+\sqrt{5} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_2 = \begin{pmatrix} -\frac{\sqrt{5}+1}{2}i \\ 1 \end{pmatrix}$

$\|\vec{\xi}_1\| = (\vec{\xi}_1^H \vec{\xi}_1)^{\frac{1}{2}} = \frac{1}{2}\sqrt{10-2\sqrt{5}}$ $\|\vec{\xi}_2\| = (\vec{\xi}_2^H \vec{\xi}_2)^{\frac{1}{2}} = \frac{1}{2}\sqrt{10+2\sqrt{5}}$

∴ 取 $U = \left(\frac{\vec{\xi}_1}{\|\vec{\xi}_1\|}, \frac{\vec{\xi}_2}{\|\vec{\xi}_2\|}\right) = \begin{bmatrix} \frac{\sqrt{5}-1}{\sqrt{10-2\sqrt{5}}}i & -\frac{\sqrt{5}+1}{\sqrt{10+2\sqrt{5}}}i \\ \frac{2}{\sqrt{10-2\sqrt{5}}} & \frac{2}{\sqrt{10+2\sqrt{5}}} \end{bmatrix}$ 则有: $U^HAU = \Lambda$

$= \begin{bmatrix} 2+\sqrt{5} & 0 \\ 0 & 2-\sqrt{5} \end{bmatrix}$

(2) $\begin{pmatrix} 2 & -3i \\ 3i & 4 \end{pmatrix}$ $\begin{pmatrix} 2 & -2i \\ 3i & 4 \end{pmatrix}$ 不是正规矩阵 ($AA^H \neq A^HA$). 不必研究对角化.

解: $|A-\lambda I| = \begin{vmatrix} 2-\lambda & -2i \\ 3i & 4-\lambda \end{vmatrix} = \lambda^2 - 6\lambda + 8 + 6i^2 = \lambda^2 - 6\lambda + 2 = 0 \Rightarrow \lambda_1 = 3+\sqrt{7},\ \lambda_2 = 3-\sqrt{7}$.

$(A-\lambda_1 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1-\sqrt{7} & -2i \\ 3i & 1-\sqrt{7} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_1 = \begin{pmatrix} -2i \\ 1+\sqrt{7} \end{pmatrix}$ $\|\vec{\xi}_1\| = \sqrt{12+2\sqrt{7}}$

$(A-\lambda_2 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1+\sqrt{7} & -2i \\ 3i & 1+\sqrt{7} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_2 = \begin{pmatrix} -2i \\ 1-\sqrt{7} \end{pmatrix}$ $\|\vec{\xi}_2\| = \sqrt{12-2\sqrt{7}}$

正交化: $\vec{\eta}_1 = \vec{\xi}_1 = \begin{pmatrix} -2i \\ 1+\sqrt{7} \end{pmatrix}$ $(\vec{\xi}_2, \vec{\eta}_1) = \vec{\xi}_2^H \vec{\eta}_1 = -2$. $\|\vec{\eta}_1\|^2 = \|\vec{\xi}_1\|^2 = 12+2\sqrt{7}$

$\vec{\eta}_2 = \vec{\xi}_2 - \frac{(\vec{\xi}_2, \vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\vec{\eta}_1 = \begin{pmatrix} -2i \\ 1-\sqrt{7} \end{pmatrix} - \frac{-2}{12+2\sqrt{7}}\begin{pmatrix} -2i \\ 1+\sqrt{7} \end{pmatrix} = -\frac{2\sqrt{7}}{6+\sqrt{7}}\begin{pmatrix} (1+\sqrt{7})i \\ 2 \end{pmatrix}$

标准化: $\vec{\varepsilon}_1 = \frac{\vec{\eta}_1}{\|\vec{\eta}_1\|} = \frac{1}{\sqrt{12+2\sqrt{7}}}\begin{pmatrix} -2i \\ 1+\sqrt{7} \end{pmatrix}$. $\vec{\varepsilon}_2 = \frac{\vec{\eta}_2}{\|\vec{\eta}_2\|} = \frac{1}{\sqrt{12+2\sqrt{7}}}\begin{pmatrix} (1+\sqrt{7})i \\ 2 \end{pmatrix}$.

$|A-\lambda I| = \begin{vmatrix} 2-\lambda & -3i \\ 3i & 4-\lambda \end{vmatrix} = \lambda^2 - 6\lambda + 8 + 9i^2 = \lambda^2 - 6\lambda - 1 = 0 \Rightarrow \lambda_1 = 3+\sqrt{10},\ \lambda_2 = 3-\sqrt{10}$.

$(A-\lambda_1 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1-\sqrt{10} & -3i \\ 3i & 1-\sqrt{10} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_1 = \begin{pmatrix} -3i \\ 1+\sqrt{10} \end{pmatrix}$ $\|\vec{\xi}_1\| = \sqrt{20+2\sqrt{10}}$

$(A-\lambda_2 I)\vec{x} = \vec{0} \Rightarrow \begin{pmatrix} -1+\sqrt{10} & -3i \\ 3i & 1+\sqrt{10} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_2 = \begin{pmatrix} -3i \\ 1-\sqrt{10} \end{pmatrix}$ $\|\vec{\xi}_2\| = \sqrt{20-2\sqrt{10}}$

$\therefore$ 取 $U = \left(\frac{\vec{\xi}_1}{\|\vec{\xi}_1\|}, \frac{\vec{\xi}_2}{\|\vec{\xi}_2\|}\right) = \begin{bmatrix} \frac{-3i}{\sqrt{20+2\sqrt{10}}} & \frac{-3i}{\sqrt{20-2\sqrt{10}}} \\ \frac{1+\sqrt{10}}{\sqrt{20+2\sqrt{10}}} & \frac{1-\sqrt{10}}{\sqrt{20-2\sqrt{10}}} \end{bmatrix}$. 则有: $U^HAU = \Lambda$

$= \begin{bmatrix} 3+\sqrt{10} & 0 \\ 0 & 3-\sqrt{10} \end{bmatrix}$.

3. 作为符合下述要求的三阶对称阵:

(1) 特征值 1, −2, 3, 分别有对应的特征向量 $(2,1,-2)^T$, $(1,2,2)^T$, $(2,-2,1)^T$;

(2) 特征值 1, 1, −1, 且特征值 1 有对应的特征向量 $(1,1,1)^T$ 和 $(2,2,1)^T$.

解: (1) $\lambda_1 = 1$. $\vec{\xi}_1 = \begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$. 标准化: $\vec{\eta}_1 = \frac{1}{3}\begin{pmatrix} 2 \\ 1 \\ -2 \end{pmatrix}$

$\lambda_2 = -2$. $\vec{\xi}_2 = \begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$. $\therefore$ $\vec{\eta}_2 = \frac{1}{3}\begin{pmatrix} 1 \\ 2 \\ 2 \end{pmatrix}$.

(实对称阵: $\vec{\xi}_1, \vec{\xi}_2, \vec{\xi}_3$ 互相垂直.)

$\lambda_3=3$: $\vec{\xi}_3=\begin{pmatrix}2\\-2\\1\end{pmatrix}$. 标准化: $\vec{\eta}_3=\frac{1}{3}\begin{pmatrix}2\\-2\\1\end{pmatrix}$.

取 $S=(\vec{\eta}_1\ \vec{\eta}_2\ \vec{\eta}_3)$. 则有: $S^TAS=\Lambda=\begin{bmatrix}\lambda_1&&\\&\lambda_2&\\&&\lambda_3\end{bmatrix}=\begin{bmatrix}1&&\\&-2&\\&&3\end{bmatrix}$ (正交变换)

$S^T=S^{-1}$.

$$\therefore A=S\Lambda S^T=\frac{1}{3}\begin{bmatrix}2&1&2\\1&2&-2\\-2&2&1\end{bmatrix}\begin{bmatrix}1&&\\&-2&\\&&3\end{bmatrix}\frac{1}{3}\begin{bmatrix}2&1&-2\\1&2&2\\2&-2&1\end{bmatrix}=\frac{1}{9}\begin{bmatrix}14&-14&-2\\-14&5&-16\\-2&-16&-1\end{bmatrix}$$

(或: 取 $T=(\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3)$. 则有: $T^{-1}AT=\Lambda=\begin{bmatrix}1&&\\&-2&\\&&3\end{bmatrix}$ (相似变换)

$$A=T\Lambda T^{-1}=\begin{bmatrix}2&1&2\\1&2&-2\\-2&2&1\end{bmatrix}\begin{bmatrix}1&&\\&-2&\\&&3\end{bmatrix}\begin{bmatrix}2&1&2\\1&2&-2\\-2&2&1\end{bmatrix}^{-1}=\frac{1}{9}\begin{bmatrix}14&-14&-2\\-14&5&-16\\-2&-16&-1\end{bmatrix}$$

)

(2) 对应于 (-1) 的特征向量也与对应于 1 的特征向量正交 ($\because A$ 为对称阵).

设 $\vec{\eta}_1=\begin{pmatrix}1\\1\\1\end{pmatrix}$ ($\lambda_1=1$). $\vec{\eta}_2=\begin{pmatrix}2\\2\\1\end{pmatrix}$ ($\lambda_2=1$). $\vec{\eta}_3=\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}$ ($\lambda_3=-1$). 则有: $\begin{cases}\vec{\eta}_1^T\vec{\eta}_3=0\\ \vec{\eta}_2^T\vec{\eta}_3=0\end{cases}\Rightarrow\begin{cases}x_1+x_2+x_3=0\\2x_1+2x_2+x_3=0\end{cases}\Rightarrow\vec{\eta}_3=\begin{pmatrix}1\\-1\\0\end{pmatrix}$

正交化: $\vec{\xi}_1=\vec{\eta}_1=\begin{pmatrix}1\\1\\1\end{pmatrix}$. $\vec{\xi}_2=\vec{\eta}_2-\frac{(\vec{\eta}_2,\vec{\xi}_1)}{\|\vec{\xi}_1\|^2}\vec{\xi}_1=\frac{1}{3}\begin{pmatrix}1\\1\\-2\end{pmatrix}$. $\vec{\xi}_3=\vec{\eta}_3=\begin{pmatrix}1\\-1\\0\end{pmatrix}$.

标准化: $\vec{\gamma}_1=\frac{1}{\sqrt{3}}\vec{\xi}_1=\frac{1}{\sqrt{3}}\begin{pmatrix}1\\1\\1\end{pmatrix}$. $\vec{\gamma}_2=\frac{3}{\sqrt{6}}\vec{\xi}_2=\frac{1}{\sqrt{6}}\begin{pmatrix}1\\1\\-2\end{pmatrix}$. $\vec{\gamma}_3=\frac{1}{\sqrt{2}}\vec{\xi}_3=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\\0\end{pmatrix}$.

取 $S=[\vec{\gamma}_1\ \vec{\gamma}_2\ \vec{\gamma}_3)=\begin{bmatrix}\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{6}}&\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{6}}&-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}}&-\frac{2}{\sqrt{6}}&0\end{bmatrix}$. 则有: $S^TAS=\Lambda=\begin{bmatrix}1&&\\&1&\\&&-1\end{bmatrix}$.

$S^T=S^{-1}$

$$\therefore A=S\Lambda S^T=\begin{bmatrix}\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{6}}&\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{6}}&-\frac{1}{\sqrt{2}}\\\frac{1}{\sqrt{3}}&-\frac{2}{\sqrt{6}}&0\end{bmatrix}\begin{bmatrix}1&&\\&1&\\&&-1\end{bmatrix}\begin{bmatrix}\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{6}}&\frac{1}{\sqrt{6}}&-\frac{2}{\sqrt{6}}\\\frac{1}{\sqrt{2}}&-\frac{1}{\sqrt{2}}&0\end{bmatrix}$$

$$=\begin{bmatrix}0&1&0\\1&0&0\\0&0&1\end{bmatrix}$$

4. 证明：若矩阵 A 既是三角矩阵，又是正交矩阵，则 A 必是对角矩阵.

证明：不妨设 A 为下三角阵：

$$A=\begin{pmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ a_{21} & a_{22} & 0 & \cdots & 0 \\ a_{31} & a_{32} & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \cdots & a_{nn} \end{pmatrix} \quad \text{其中 } a_{ii}\neq 0,\ i=1,2,\cdots,n.$$

$$=(\vec{a}_1\ \vec{a}_2\ \vec{a}_3\ \cdots\ \vec{a}_n) \quad \text{其中 } \vec{a}_j=\begin{pmatrix} a_{1j} \\ a_{2j} \\ \vdots \\ a_{nj} \end{pmatrix},\ j=1,2,\cdots,n.$$

$\because A^TA=I$

$$\therefore \begin{pmatrix} \vec{a}_1^T \\ \vec{a}_2^T \\ \vdots \\ \vec{a}_n^T \end{pmatrix}(\vec{a}_1\ \vec{a}_2\ \cdots\ \vec{a}_n)=\begin{pmatrix} \vec{a}_1^T\vec{a}_1 & \vec{a}_1^T\vec{a}_2 & \cdots & \vec{a}_1^T\vec{a}_n \\ \vec{a}_2^T\vec{a}_1 & \vec{a}_2^T\vec{a}_2 & \cdots & \vec{a}_2^T\vec{a}_n \\ \vdots & \vdots & \ddots & \vdots \\ \vec{a}_n^T\vec{a}_1 & \vec{a}_n^T\vec{a}_2 & \cdots & \vec{a}_n^T\vec{a}_n \end{pmatrix}=I. \quad \text{即 } \vec{a}_i^T\vec{a}_j=\begin{cases} 1 & i=j \\ 0 & i\neq j \end{cases}$$

取 $j=n$，则 $\vec{a}_n=(0,0,\cdots,0,a_{nn})^T$. 由 $\vec{a}_i^T\vec{a}_n=\begin{cases} 1 & i=n \\ 0 & i\neq n \end{cases}\Rightarrow a_{nj}=0,$

$j=1,2,\cdots,n-1.$

取 $j=n-1$，则 $\vec{a}_{n-1}=(0,0,\cdots,a_{n-1,n-1},0)^T$. 由 $\vec{a}_i^T\vec{a}_{n-1}=\begin{cases} 1 & i=n-1 \\ 0 & i\neq n-1 \end{cases}\Rightarrow a_{n-1,j}=0,$

$j=1,2,\cdots,n-2.$

$\vdots$

取 $j=2$，则 $\vec{a}_2=(0,a_{22},0,\cdots 0,0)^T$. 由 $\vec{a}_1^T\vec{a}_2=0\Rightarrow a_{22}=0.$

即 $\forall i>j$，成立 $a_{ij}=0$. $\therefore A$ 必是对角阵.

5. 设 $\vec{a}$ 为 n 阶实列向量，且满足 $\vec{a}^T\vec{a}=1$. 证明 $A=I-2\vec{a}\vec{a}^T$ 是对称的正交矩阵.

证明：$A^T=(I-2\vec{a}\vec{a}^T)^T=I-2\vec{a}\vec{a}^T=A.$ $\therefore A$ 为对称阵.

$$A^TA=(I-2\vec{a}\vec{a}^T)(I-2\vec{a}\vec{a}^T)=I-4\vec{a}\vec{a}^T+4\vec{a}(\vec{a}^T\vec{a})\vec{a}^T$$

$$=I-4\vec{a}\vec{a}^T+4\vec{a}\vec{a}^T=I. \quad \therefore A \text{ 为正交阵.}$$

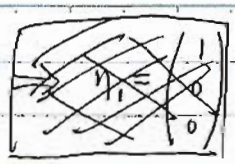
即 $A=I-2\vec{a}\vec{a}^T$ 是对称的正交矩阵.

6. 对于矩阵序列 $A_n=(a_{ij}^{(n)})$，$n=1,2,\cdots$，如果每个 $\lim\limits_{n\to\infty}a_{ij}^{(n)}$ 存在，定义 $\lim\limits_{n\to\infty}A_n=\left(\lim\limits_{n\to\infty}a_{ij}^{(n)}\right)$.

若 $A=\begin{pmatrix} \frac{1}{2} & 1 & 2 \\ 0 & \frac{1}{3} & 1 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$，求 $\lim\limits_{n\to\infty}A^n$.

解：$|A-\lambda I|=\begin{vmatrix} \frac{1}{2}-\lambda & 1 & 2 \\ 0 & \frac{1}{3}-\lambda & 1 \\ 0 & 0 & \frac{1}{4}-\lambda \end{vmatrix}=\left(\frac{1}{2}-\lambda\right)\left(\frac{1}{3}-\lambda\right)\left(\frac{1}{4}-\lambda\right)=0\Rightarrow \lambda_1=\frac{1}{2},\ \lambda_2=\frac{1}{3},\ \lambda_3=\frac{1}{4}.$

对于 $\lambda_1=\frac{1}{2}$. $(A-\lambda_1 I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix} 0 & 1 & 2 \\ 0 & -\frac{1}{6} & 1 \\ 0 & 0 & -\frac{1}{4} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_1=\begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$



对于 $\lambda_2=\frac{1}{3}$. $(A-\lambda_2 I)\vec{x}=0 \Rightarrow \begin{pmatrix} \frac{1}{6} & 1 & 2 \\ 0 & 0 & 1 \\ 0 & 0 & -\frac{1}{12} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_2=\begin{pmatrix} 6 \\ -1 \\ 0 \end{pmatrix}$.



对于 $\lambda_3=\frac{1}{4}$. $(A-\lambda_3 I)\vec{x}=0 \Rightarrow \begin{pmatrix} \frac{1}{4} & 1 & 2 \\ 0 & \frac{1}{12} & 1 \\ 0 & 0 & 0 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix} \Rightarrow \vec{\xi}_3=\begin{pmatrix} 40 \\ -12 \\ 1 \end{pmatrix}$.

取 $P=(\vec{\xi}_1\ \vec{\xi}_2\ \vec{\xi}_3)=\begin{pmatrix} 1 & 6 & 40 \\ 0 & -1 & -12 \\ 0 & 0 & 1 \end{pmatrix}$. 则有: $P^{-1}AP=\Lambda=\begin{pmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{4} \end{pmatrix}$.

$$(P:I)=\left(\begin{array}{ccc|ccc} 1 & 6 & 40 & 1 & 0 & 0 \\ 0 & -1 & -12 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)\to\left(\begin{array}{ccc|ccc} 1 & 0 & -32 & 1 & 6 & 0 \\ 0 & 1 & 12 & 0 & -1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)\to\left(\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 6 & 32 \\ 0 & 1 & 0 & 0 & -1 & -12 \\ 0 & 0 & 1 & 0 & 0 & 1 \end{array}\right)$$

$P^{-1}=\begin{pmatrix} 1 & 6 & 32 \\ 0 & -1 & -12 \\ 0 & 0 & 1 \end{pmatrix}$. $\therefore A=P\Lambda P^{-1}$. $A^n=P\Lambda^n P^{-1}=\begin{pmatrix} 1 & 6 & 40 \\ 0 & -1 & -12 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2^n} & & \\ & \frac{1}{3^n} & \\ & & \frac{1}{4^n} \end{pmatrix}\begin{pmatrix} 1 & 6 & 32 \\ 0 & -1 & -12 \\ 0 & 0 & 1 \end{pmatrix}$

$$\therefore \lim_{n\to\infty}A^n=\lim_{n\to\infty}\begin{pmatrix} 1 & 6 & 40 \\ 0 & -1 & -12 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} \frac{1}{2^n} & & \\ & \frac{1}{3^n} & \\ & & \frac{1}{4^n} \end{pmatrix}\begin{pmatrix} 1 & 6 & 32 \\ 0 & -1 & -12 \\ 0 & 0 & 1 \end{pmatrix}$$

$$=\lim_{n\to\infty}\begin{pmatrix} \frac{1}{2^n} & \frac{6}{2^n}-\frac{6}{3^n} & \frac{32}{2^n}-\frac{12}{3^n}+\frac{40}{4^n} \\ 0 & \frac{1}{3^n} & \frac{12}{3^n}-\frac{12}{4^n} \\ 0 & 0 & \frac{1}{4^n} \end{pmatrix}=\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

7. 设 A 是三阶对称阵, 其特征值为 1, 2, 3, 已知 A 的相应于 1, 2 的特征向量分别为 $\vec{x}_1=(-1,-1,1)^T$, $\vec{x}_2=(1,-2,-1)^T$. 求:

(1) A 的相应于特征值 3 的特征向量;

(2) 方阵 A.

解: (1) 设 $\vec{x}_3=\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$. 则有: $\begin{cases} \vec{x}_3^T\vec{x}_1=0 \\ \vec{x}_3^T\vec{x}_2=0 \end{cases} \Rightarrow \begin{cases} -x_1-x_2+x_3=0 \\ x_1-2x_2-x_3=0 \end{cases} \Rightarrow \vec{x}_3=\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$ —— 对应于 $\lambda_3=3$ 的特征向量.

(2) $\vec{\eta}_1=\frac{\vec{x}_1}{\|\vec{x}_1\|}=\frac{1}{\sqrt{3}}\begin{pmatrix} -1 \\ -1 \\ 1 \end{pmatrix}$. $\vec{\eta}_2=\frac{\vec{x}_2}{\|\vec{x}_2\|}=\frac{1}{\sqrt{6}}\begin{pmatrix} 1 \\ -2 \\ -1 \end{pmatrix}$. $\vec{\eta}_3=\frac{\vec{x}_3}{\|\vec{x}_3\|}=\frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

取 $P=(\vec{\eta}_1\ \vec{\eta}_2\ \vec{\eta}_3)$，则有 $P^TAP=\Lambda=\begin{bmatrix}1&&\\&2&\\&&3\end{bmatrix}$ （合同变换）

$P^{-1}=P^T$.

$$\therefore A=P\Lambda P^T=\begin{bmatrix}-\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{6}}&\frac{1}{\sqrt{2}}\\-\frac{1}{\sqrt{3}}&-\frac{2}{\sqrt{6}}&0\\\frac{1}{\sqrt{3}}&-\frac{1}{\sqrt{6}}&\frac{1}{\sqrt{2}}\end{bmatrix}\begin{bmatrix}1&&\\&2&\\&&3\end{bmatrix}\begin{bmatrix}-\frac{1}{\sqrt{3}}&-\frac{1}{\sqrt{3}}&\frac{1}{\sqrt{3}}\\\frac{1}{\sqrt{6}}&-\frac{2}{\sqrt{6}}&-\frac{1}{\sqrt{6}}\\\frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{2}}\end{bmatrix}=\begin{bmatrix}\frac{13}{6}&-\frac{1}{3}&\frac{5}{6}\\-\frac{1}{3}&\frac{5}{3}&\frac{1}{3}\\\frac{5}{6}&\frac{1}{3}&\frac{13}{6}\end{bmatrix}$$

或：取 $T=(\vec{x}_1\ \vec{x}_2\ \vec{x}_3)=\begin{bmatrix}-1&1&1\\-1&-2&0\\1&-1&1\end{bmatrix}$. 则有 $T^{-1}AT=\Lambda=\begin{bmatrix}1&&\\&2&\\&&3\end{bmatrix}$ （相似变换）

$$(T\,|\,I)=\left(\begin{array}{ccc|ccc}-1&1&1&1&0&0\\-1&-2&0&0&1&0\\1&-1&1&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&-1&1&0&0&1\\0&0&2&1&0&1\\0&-3&1&0&1&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&-1&1&0&0&1\\0&1&-\frac{1}{3}&0&-\frac{1}{3}&-\frac{1}{3}\\0&0&1&\frac{1}{2}&0&\frac{1}{2}\end{array}\right)$$

$$\to\left(\begin{array}{ccc|ccc}1&0&\frac{2}{3}&0&-\frac{1}{3}&\frac{2}{3}\\0&1&-\frac{1}{3}&0&-\frac{1}{3}&-\frac{1}{3}\\0&0&1&\frac{1}{2}&0&\frac{1}{2}\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&0&0&-\frac{1}{3}&-\frac{1}{3}&\frac{1}{3}\\0&1&0&\frac{1}{6}&-\frac{1}{3}&-\frac{1}{6}\\0&0&1&\frac{1}{2}&0&\frac{1}{2}\end{array}\right)\quad T^{-1}=\begin{bmatrix}-\frac{1}{3}&-\frac{1}{3}&\frac{1}{3}\\\frac{1}{6}&-\frac{1}{3}&-\frac{1}{6}\\\frac{1}{2}&0&\frac{1}{2}\end{bmatrix}$$

$$A=T\Lambda T^{-1}=\begin{bmatrix}-1&1&1\\-1&-2&0\\1&-1&1\end{bmatrix}\begin{bmatrix}1&&\\&2&\\&&3\end{bmatrix}\begin{bmatrix}-\frac{1}{3}&-\frac{1}{3}&\frac{1}{3}\\\frac{1}{6}&-\frac{1}{3}&-\frac{1}{6}\\\frac{1}{2}&0&\frac{1}{2}\end{bmatrix}=\begin{bmatrix}\frac{13}{6}&-\frac{1}{3}&\frac{5}{6}\\-\frac{1}{3}&\frac{5}{3}&\frac{1}{3}\\\frac{5}{6}&\frac{1}{3}&\frac{13}{6}\end{bmatrix}$$

8. 设 A 是 n 阶方阵，证明：若 $A\neq 0$，但存在正整数 k，使得 $A^k=0$，则 A 必不能相似于对角阵.

证明：若相似于对角阵 T，使 $T^{-1}AT=\Lambda=\mathrm{diag}(\lambda_1,\lambda_2,\cdots,\lambda_n)$，则：

$$T^{-1}A^kT=\mathrm{diag}(\lambda_1^k,\lambda_2^k,\cdots,\lambda_n^k)=0\quad\therefore\lambda_i^k=0,\ i=1,2,\cdots,n.$$

$\therefore \lambda_i=0$，即 $T^{-1}AT=0$，$\therefore A=0$ 与已知矛盾

$\therefore A$ 不能相似于对角阵.

§6. 二次型及其标准型

习题 P.304

1. 用正交变换将下列二次型化为标准形，并写出所用的变量代换：

(1) $f(x_1,x_2)=4x_1^2+4x_1x_2+x_2^2$

解：$f(x_1,x_2)=(x_1\ x_2)\begin{bmatrix}4&2\\2&1\end{bmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}$. $A=\begin{bmatrix}4&2\\2&1\end{bmatrix}$. $|A-\lambda I|=0\Rightarrow\lambda^2-5\lambda=\lambda(\lambda-5)\Rightarrow\begin{cases}\lambda_1=0\\\lambda_2=5\end{cases}$

对于 $\lambda_1=0$. $(A-\lambda_1 I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix}4&2\\2&1\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \vec{\xi}_1=\begin{pmatrix}1\\-2\end{pmatrix} \Rightarrow \vec{\eta}_1=\frac{1}{\sqrt{5}}\begin{pmatrix}1\\-2\end{pmatrix}$.

对于 $\lambda_2=5$. $(A-\lambda_2 I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix}-1&2\\2&-4\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \vec{\xi}_2=\begin{pmatrix}2\\1\end{pmatrix} \Rightarrow \vec{\eta}_2=\frac{1}{\sqrt{5}}\begin{pmatrix}2\\1\end{pmatrix}$

令 $S=(\vec{\eta}_1,\vec{\eta}_2)=\frac{1}{\sqrt{5}}\begin{bmatrix}1&2\\-2&1\end{bmatrix}$. 则有 $S^TAS=\Lambda=\begin{bmatrix}0&0\\0&5\end{bmatrix}$. 且 $S^T=S^{-1}$.

设 $\begin{pmatrix}y_1\\y_2\end{pmatrix}=S^T\begin{pmatrix}x_1\\x_2\end{pmatrix}=\frac{1}{\sqrt{5}}\begin{bmatrix}1&2\\-2&1\end{bmatrix}^T\begin{pmatrix}x_1\\x_2\end{pmatrix}$. 即: $\begin{cases}y_1=\frac{1}{\sqrt{5}}(x_1-2x_2)\\y_2=\frac{1}{\sqrt{5}}(2x_1+x_2)\end{cases}$

则有: $\hat{f}(y_1,y_2)=(y_1,y_2)\begin{bmatrix}0&0\\0&5\end{bmatrix}\begin{pmatrix}y_1\\y_2\end{pmatrix}=5y_2^2$.

(2) $f(x_1,x_2)=x_1^2+2x_1x_2-x_2^2$

解: $f(x_1,x_2)=(x_1\ x_2)\begin{bmatrix}1&1\\1&-1\end{bmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}$. $A=\begin{bmatrix}1&1\\1&-1\end{bmatrix}$. $|A-\lambda I|=\lambda^2-2=0 \Rightarrow \begin{cases}\lambda_1=\sqrt{2}\\\lambda_2=-\sqrt{2}\end{cases}$

对于 $\lambda_1=\sqrt{2}$. $(A-\lambda_1 I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix}1-\sqrt{2}&1\\1&-1-\sqrt{2}\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \vec{\xi}_1=\begin{pmatrix}-1\\1-\sqrt{2}\end{pmatrix} \Rightarrow \vec{\eta}_1=\frac{1}{\sqrt{4-2\sqrt{2}}}\begin{pmatrix}-1\\1-\sqrt{2}\end{pmatrix}$

对于 $\lambda_2=-\sqrt{2}$. $(A-\lambda_2 I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix}1+\sqrt{2}&1\\1&-1+\sqrt{2}\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}0\\0\end{pmatrix} \Rightarrow \vec{\xi}_2=\begin{pmatrix}-1\\1+\sqrt{2}\end{pmatrix} \Rightarrow \vec{\eta}_2=\frac{1}{\sqrt{4+2\sqrt{2}}}\begin{pmatrix}-1\\1+\sqrt{2}\end{pmatrix}$

令 $S=(\vec{\eta}_1,\vec{\eta}_2)=\begin{bmatrix}\frac{-1}{\sqrt{4-2\sqrt{2}}}&\frac{-1}{\sqrt{4+2\sqrt{2}}}\\\frac{1-\sqrt{2}}{\sqrt{4-2\sqrt{2}}}&\frac{1+\sqrt{2}}{\sqrt{4+2\sqrt{2}}}\end{bmatrix}$. 则有 $S^TAS=\Lambda=\begin{bmatrix}\sqrt{2}&0\\0&-\sqrt{2}\end{bmatrix}$. 且 $S^T=S^{-1}$.

设 $\begin{pmatrix}y_1\\y_2\end{pmatrix}=S^T\begin{pmatrix}x_1\\x_2\end{pmatrix}=\begin{pmatrix}\frac{-1}{\sqrt{4-2\sqrt{2}}}&\frac{1-\sqrt{2}}{\sqrt{4-2\sqrt{2}}}\\\frac{-1}{\sqrt{4+2\sqrt{2}}}&\frac{1+\sqrt{2}}{\sqrt{4+2\sqrt{2}}}\end{pmatrix}\begin{pmatrix}x_1\\x_2\end{pmatrix}$. 即: $\begin{cases}y_1=\frac{1}{\sqrt{4-2\sqrt{2}}}\left[-x_1+(1-\sqrt{2})x_2\right]\\y_2=\frac{1}{\sqrt{4+2\sqrt{2}}}\left[-x_1+(1+\sqrt{2})x_2\right]\end{cases}$

则有: $\tilde{f}(y_1,y_2)=(y_1,y_2)\begin{bmatrix}\sqrt{2}&0\\0&-\sqrt{2}\end{bmatrix}\begin{pmatrix}y_1\\y_2\end{pmatrix}=\sqrt{2}y_1^2-\sqrt{2}y_2^2$.

(3) $f(x_1,x_2,x_3)=3x_1^2+4x_1x_2+8x_1x_3+4x_2x_3+3x_3^2$.

解: $f(x_1,x_2,x_3)=(x_1\ x_2\ x_3)\begin{bmatrix}3&2&4\\2&0&2\\4&2&3\end{bmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}$. $A=\begin{bmatrix}3&2&4\\2&0&2\\4&2&3\end{bmatrix}$.

$|A-\lambda I|=\begin{vmatrix}3-\lambda&2&4\\2&-\lambda&2\\4&2&3-\lambda\end{vmatrix}=-(\lambda+1)^2(\lambda-8)=0 \Rightarrow \lambda_1=\lambda_2=-1,\ \lambda_3=8$.

对于 $\lambda_1=\lambda_2=-1$. $(A-\lambda I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix}4&2&4\\2&1&2\\4&2&4\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \vec{\xi}_1=\begin{pmatrix}-1\\2\\0\end{pmatrix}, \vec{\xi}_2=\begin{pmatrix}-1\\0\\1\end{pmatrix}$

正交化: $\vec{\eta}_1=\begin{pmatrix}-1\\0\\1\end{pmatrix}$. $\vec{\eta}_2=\begin{pmatrix}-1\\2\\0\end{pmatrix}-\frac{(\vec{\xi}_2,\vec{\eta}_1)}{\|\vec{\eta}_1\|^2}\begin{pmatrix}-1\\0\\1\end{pmatrix}=-\frac{1}{2}\begin{pmatrix}1\\-4\\1\end{pmatrix}$.

标准化: $\vec{\varepsilon}_1=\frac{1}{\sqrt{2}}\begin{pmatrix}-1\\0\\1\end{pmatrix}$. $\vec{\varepsilon}_2=\frac{1}{3\sqrt{2}}\begin{pmatrix}1\\-4\\1\end{pmatrix}$

对于 $\lambda_3=8$. $(A-\lambda_3 I)\vec{x}=\vec{0} \Rightarrow \begin{pmatrix}-5&2&4\\2&-8&2\\4&2&-5\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{pmatrix}0\\0\\0\end{pmatrix} \Rightarrow \vec{\xi}_3=\begin{pmatrix}2\\1\\2\end{pmatrix} \Rightarrow \vec{\varepsilon}_3=\frac{1}{3}\begin{pmatrix}2\\1\\2\end{pmatrix}$.

令 $S=(\vec{\varepsilon}_1\ \vec{\varepsilon}_2\ \vec{\varepsilon}_3)=\begin{bmatrix}-\frac{1}{\sqrt{2}}&\frac{1}{3\sqrt{2}}&\frac{2}{3}\\0&-\frac{4}{3\sqrt{2}}&\frac{1}{3}\\\frac{1}{\sqrt{2}}&\frac{1}{3\sqrt{2}}&\frac{2}{3}\end{bmatrix}$. 则有: $S^TAS=\Lambda=\begin{bmatrix}-1&&\\&-1&\\&&8\end{bmatrix}$. 且有 $S^T=S^{-1}$.

设 $\begin{pmatrix}y_1\\y_2\\y_3\end{pmatrix}=S^T\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=\begin{bmatrix}-\frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{2}}\\\frac{1}{3\sqrt{2}}&-\frac{4}{3\sqrt{2}}&\frac{1}{3\sqrt{2}}\\\frac{2}{3}&\frac{1}{3}&\frac{2}{3}\end{bmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}$. 即: $\begin{cases}y_1=\frac{1}{\sqrt{2}}(-x_1+x_3)\\y_2=\frac{1}{3\sqrt{2}}(x_1-4x_2+x_3)\\y_3=\frac{1}{3}(2x_1+x_2+2x_3)\end{cases}$

则有: $\tilde{f}(y_1,y_2,y_3)=(y_1\ y_2\ y_3)\begin{bmatrix}-1&&\\&-1&\\&&8\end{bmatrix}\begin{pmatrix}y_1\\y_2\\y_3\end{pmatrix}=-y_1^2-y_2^2+8y_3^2$.

2. 适当选取直角坐标系，将以下二次曲线化为标准形式，并在原坐标系画出图形:

(1) $2x^2-4xy-y^2-4x+10y-13=0$.

解: $f(x,y)=2x^2-4xy-y^2-4x+10y-13=2(x^2-2xy+y^2)-3y^2-4x+10y-13$

$=2(x-y)^2-3y^2-4(x-y)+6y-13$

$=2(x-y-1)^2-3(y-1)^2-12$

令 $\begin{cases}x'=x-y-1\\y'=y-1\end{cases}$ 则有: $2x'^2-3y'^2-12=0 \Rightarrow \frac{x'^2}{6}-\frac{y'^2}{4}=1$. (双曲线).

$f(x,y)=(x\ y)\begin{bmatrix}2&-2\\-2&-1\end{bmatrix}\begin{pmatrix}x\\y\end{pmatrix}+(-4\ \ 10)\begin{pmatrix}x\\y\end{pmatrix}-13$

$A=\begin{bmatrix}2&-2\\-2&-1\end{bmatrix}$. $|A-\lambda I|=\begin{vmatrix}2-\lambda&-2\\-2&-1-\lambda\end{vmatrix}=(\lambda-3)(\lambda+2)=0$.

$\lambda_1=3$. $\vec{\xi}_1=\begin{pmatrix}2\\-1\end{pmatrix}$. $\vec{\eta}_1=\frac{1}{\sqrt{5}}\begin{pmatrix}2\\-1\end{pmatrix}$.

$\lambda_2=-2$. $\vec{\xi}_2=\begin{pmatrix}1\\2\end{pmatrix}$. $\vec{\eta}_2=\frac{1}{\sqrt5}\begin{pmatrix}1\\2\end{pmatrix}$. $S=\frac{1}{\sqrt5}\begin{bmatrix}2&1\\-1&2\end{bmatrix}$

设 $\cos\theta=\frac{2}{\sqrt5}$. 则有 $\begin{pmatrix}x'\\y'\end{pmatrix}=S^T\begin{pmatrix}x\\y\end{pmatrix}=\frac{1}{\sqrt5}\begin{bmatrix}2&1\\-1&2\end{bmatrix}\begin{pmatrix}x\\y\end{pmatrix}=\begin{pmatrix}\cos\theta&-\sin\theta\\\sin\theta&\cos\theta\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$.

即所有向量旋转 θ 角度. 此时: $f(x',y')=(x'\ y')S^TAS\begin{pmatrix}x'\\y'\end{pmatrix}+(-4\ \ 10)S\begin{pmatrix}x'\\y'\end{pmatrix}-13$

$f(x',y')=(x'\ y')\begin{bmatrix}3&\\&-2\end{bmatrix}\begin{pmatrix}x'\\y'\end{pmatrix}+\frac{1}{\sqrt5}(-18\ \ 16)\begin{pmatrix}x'\\y'\end{pmatrix}-13$

令 $\begin{pmatrix}x''\\y''\end{pmatrix}=\begin{pmatrix}x'\\y'\end{pmatrix}-\frac{1}{\sqrt5}\begin{pmatrix}3\\4\end{pmatrix}$. 则有 $f(x'',y'')=3x''^2-2y''^2-12=0$. (特征值法).

即 $\begin{cases}x''=\frac{1}{\sqrt5}(2x-y-3)\\y''=\frac{1}{\sqrt5}(x+2y-4)\end{cases}$.

(2). $5x^2+12xy-22y^2-12y-19=0$.

解: $f(x,y)=5x^2+12xy-22y^2-12y-19$

$A=\begin{bmatrix}5&6\\6&-22\end{bmatrix}$. $(A|I)=\left(\begin{array}{cc|cc}5&6&1&0\\6&-22&0&1\end{array}\right)\longrightarrow\left(\begin{array}{cc|cc}5&6&1&0\\0&-\frac{146}{5}&-\frac{6}{5}&1\end{array}\right)$

令 $P^T=\begin{pmatrix}1&0\\-\frac{6}{5}&1\end{pmatrix}$ 则有: $P=\begin{pmatrix}1&-\frac{6}{5}\\0&1\end{pmatrix}$. 令 $\begin{pmatrix}x^*\\y^*\end{pmatrix}=P\begin{pmatrix}x'\\y'\end{pmatrix}$.

即 $\begin{cases}x=x'-\frac{6}{5}y'\\y=y'\end{cases}\Rightarrow\begin{cases}x'=x+\frac{6}{5}y\\y'=y\end{cases}$ 成立: $f(x',y')=5x'^2-\frac{146}{5}y'^2-12y'-19$

$=5x'^2-\frac{146}{5}(y'+\frac{15}{73})^2$

令 $\begin{cases}x''=x'\\y''=y'+\frac{15}{73}\end{cases}$ 则有: $f(x'',y'')=5x''^2-\frac{146}{5}y''^2-\frac{1387}{73}$ (双曲线) $-\frac{1387}{73}$.

即: $5x''^2-\frac{146}{5}y''^2-\frac{1387}{73}=0$.

(3) $x^2+2xy+y^2+2x+y=0$.

解: $f(x,y)=x^2+2xy+y^2+2x+y=(x+y)^2+2(x+y)-y$.

令 $\begin{cases}x'=x+y\\y'=y\end{cases}$ 即 $\begin{pmatrix}x'\\y'\end{pmatrix}=\begin{pmatrix}1&1\\0&1\end{pmatrix}\begin{pmatrix}x\\y\end{pmatrix}$. (可逆变换).

$f(x',y')=x'^2+2x'-y'=(x'+1)^2-y'-1$

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$\begin{cases} x'' = x' + 1 \\ y'' = y' \end{cases}$ 则有: $f(x'', y'') = x''^2 - y'' - 1 = 0$. (抛物线)

(4) $40x^2 + 36xy + 25y^2 - 8x - 14y - 1 = 0$

解: $f(x,y) = (x\ y)\begin{bmatrix} 40 & 18 \\ 18 & 25 \end{bmatrix}\begin{pmatrix} x \\ y \end{pmatrix} - 8x - 14y - 1$. $A = \begin{bmatrix} 40 & 18 \\ 18 & 25 \end{bmatrix}$

$(A\ |\ I) = \left(\begin{array}{cc|cc} 40 & 18 & 1 & 0 \\ 18 & 25 & 0 & 1 \end{array}\right) \to \left(\begin{array}{cc|cc} 40 & 18 & 1 & 0 \\ 0 & \frac{169}{10} & -\frac{9}{20} & 1 \end{array}\right)$. $P^T = \begin{pmatrix} 1 & 0 \\ -\frac{9}{20} & 1 \end{pmatrix}$

$\Leftarrow P = \begin{pmatrix} 1 & -\frac{9}{20} \\ 0 & 1 \end{pmatrix}$. 设 $\begin{pmatrix} x \\ y \end{pmatrix} = P\begin{pmatrix} x' \\ y' \end{pmatrix} = \begin{pmatrix} x' - \frac{9}{20}y' \\ y' \end{pmatrix}$

$f(x,y) = (x\ y)A\begin{pmatrix} x \\ y \end{pmatrix} - (8\ \ 14)\begin{pmatrix} x \\ y \end{pmatrix} - 1$

$f(x',y') = (x'\ y')\begin{bmatrix} 40 & 0 \\ 0 & \frac{169}{10} \end{bmatrix}\begin{pmatrix} x' \\ y' \end{pmatrix} - (8\ \ 14)\begin{bmatrix} 1 & -\frac{9}{20} \\ 0 & 1 \end{bmatrix}\begin{pmatrix} x' \\ y' \end{pmatrix} - 1$

$= 40x'^2 + \frac{169}{10}y'^2 - (8\ \ \frac{52}{5})\begin{pmatrix} x' \\ y' \end{pmatrix} - 1 = 40x'^2 + \frac{169}{10}y'^2 - 8x' - \frac{52}{5}y' - 1$

$= 40\left(x'^2 - \frac{1}{5}x'\right) + \frac{169}{10}\left(y'^2 - \frac{104}{169}y'\right) - 1$

$= 40\left(x' - \frac{1}{10}\right)^2 + \frac{169}{10}\left(y' - \frac{52}{169}\right)^2 - 3$

$\Leftarrow \begin{cases} x'' = x' - \frac{1}{10} \\ y'' = y' - \frac{4}{13} \end{cases}$ 则有: $f(x'', y'') = 40x''^2 + \frac{169}{10}y''^2 - 3 = 0$. (椭圆)

3. 利用初等变换将下列二次型化为标准形:

(1) $f(x_1, x_2) = x_1^2 + 4x_1x_2 + 3x_2^2$.

解: $A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \end{bmatrix}$. $(A\ |\ I) = \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 2 & 3 & 0 & 1 \end{array}\right] \to \left[\begin{array}{cc|cc} 1 & 2 & 1 & 0 \\ 0 & -1 & -2 & 1 \end{array}\right]$ $P^T = \begin{bmatrix} 1 & 0 \\ -2 & 1 \end{bmatrix}$

取 $P = \begin{bmatrix} 1 & -2 \\ 0 & 1 \end{bmatrix}$. $\Leftarrow \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = P\begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ 即 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = P^{-1}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} x_1 + 2x_2 \\ x_2 \end{pmatrix}$

$\Leftarrow \begin{cases} y_1 = x_1 + 2x_2 \\ y_2 = x_2 \end{cases}$ 则有: $f(y_1, y_2) = y_1^2 - y_2^2$.

(2) $f(x_1, x_2, x_3) = 5x_1^2 + x_2^2 + 5x_3^2 + 4x_1x_2 - 8x_1x_3 - 4x_2x_3$.

解: $A = \begin{bmatrix} 5 & 2 & -4 \\ 2 & 1 & -2 \\ -4 & -2 & 5 \end{bmatrix}$. $(A\ |\ I) = \left[\begin{array}{ccc|ccc} 5 & 2 & -4 & 1 & 0 & 0 \\ 2 & 1 & -2 & 0 & 1 & 0 \\ -4 & -2 & 5 & 0 & 0 & 1 \end{array}\right] \to \left(\begin{array}{ccc|ccc} 5 & 2 & -4 & 1 & 0 & 0 \\ 0 & \frac{1}{5} & -\frac{2}{5} & -\frac{2}{5} & 1 & 0 \\ 0 & -\frac{2}{5} & \frac{9}{5} & \frac{4}{5} & 0 & 1 \end{array}\right)$

$$\longrightarrow \left(\begin{array}{ccc|ccc} 5 & 2 & -4 & 1 & 0 & 0 \\ 0 & \frac{1}{5} & -\frac{2}{5} & -\frac{2}{5} & 1 & 0 \\ 0 & 0 & 1 & 0 & 2 & 1 \end{array}\right) \quad P^T = \begin{pmatrix} 1 & 0 & 0 \\ -\frac{2}{5} & 1 & 0 \\ 0 & 2 & 1 \end{pmatrix}$$

取 $P = \begin{pmatrix} 1 & -\frac{2}{5} & 0 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{pmatrix}$ 令 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = P \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$ 即 $\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = P^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{bmatrix} 1 & \frac{2}{5} & -\frac{4}{5} \\ 0 & 1 & -2 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

令 $\begin{cases} y_1 = x_1 + \frac{2}{5}x_2 - \frac{4}{5}x_3 \\ y_2 = x_2 - 2x_3 \\ y_3 = x_3 \end{cases}$ 则有: $f(y_1, y_2, y_3) = 5y_1^2 + \frac{1}{5}y_2^2 + y_3^2$.

(3) $f(x_1, x_2, x_3) = x_1^2 - x_2^2 - 2x_1x_2 + 4x_1x_3$

解: $A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & -1 & 0 \\ 2 & 0 & 0 \end{bmatrix}$ $(A \mid I) = \left(\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ -1 & -1 & 0 & 0 & 1 & 0 \\ 2 & 0 & 0 & 0 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 0 & -2 & 2 & 1 & 1 & 0 \\ 0 & 2 & -4 & -2 & 0 & 1 \end{array}\right)$

$$\longrightarrow \left(\begin{array}{ccc|ccc} 1 & -1 & 2 & 1 & 0 & 0 \\ 0 & -2 & 2 & 1 & 1 & 0 \\ 0 & 0 & -2 & -1 & 1 & 1 \end{array}\right) \quad P^T = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ -1 & 1 & 1 \end{pmatrix} \quad 取 P = \begin{pmatrix} 1 & 1 & -1 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

令 $\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = P \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$ 即 $\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = P^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 & -1 & 2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$

令 $\begin{cases} y_1 = x_1 - x_2 + 2x_3 \\ y_2 = x_2 - x_3 \\ y_3 = x_3 \end{cases}$ 则有: $f(y_1, y_2, y_3) = y_1^2 - 2y_2^2 - 2y_3^2$.

(4) $f(x_1, x_2) = 2x_2^2 + 8x_1x_2$

解: $A = \begin{bmatrix} 0 & 4 \\ 4 & 2 \end{bmatrix}$ $(A \mid I) = \left(\begin{array}{cc|cc} 0 & 4 & 1 & 0 \\ 4 & 2 & 0 & 1 \end{array}\right) \longrightarrow \left(\begin{array}{cc|cc} 4 & 2 & 0 & 1 \\ 0 & 4 & 1 & 0 \end{array}\right) \longrightarrow \left(\begin{array}{cc|cc} 2 & 4 & 0 & 1 \\ 4 & 0 & 1 & 0 \end{array}\right)$

$$\longrightarrow \left(\begin{array}{cc|cc} 2 & 4 & 0 & 1 \\ 0 & -8 & 1 & -2 \end{array}\right) \longrightarrow \left(\begin{array}{cc|cc} 2 & 0 & 0 & 1 \\ 0 & -8 & 1 & -2 \end{array}\right) \quad P^T = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$$

取 $P = \begin{pmatrix} 0 & 1 \\ 1 & -2 \end{pmatrix}$ 令 $\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = P \begin{pmatrix} y_1 \\ y_2 \end{pmatrix}$ 即 $\begin{pmatrix} y_1 \\ y_2 \end{pmatrix} = P^{-1} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 0 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 2x_1 + x_2 \\ x_1 \end{pmatrix}$

令 $\begin{cases} y_1 = 2x_1 + x_2 \\ y_2 = x_1 \end{cases}$ 则有: $f(y_1, y_2) = 2y_1^2 - 8y_2^2$.

(5) $f(x_1, x_2, x_3) = 2x_1x_2 + 2x_1x_3 - 6x_2x_3$.

解: $A = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & -3 \\ 1 & -3 & 0 \end{bmatrix}$. $(A \mid I) = \left(\begin{array}{ccc|ccc} 0 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ 1 & -3 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 1 & 1 & -2 & 1 & 1 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ 1 & -3 & 0 & 0 & 0 & 1 \end{array}\right)$

$\to \left(\begin{array}{ccc|ccc} 2 & 1 & -2 & 1 & 1 & 0 \\ 1 & 0 & -3 & 0 & 1 & 0 \\ -2 & -3 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 2 & 1 & -2 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -2 & -3 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 2 & 0 & -2 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{1}{2} & \frac{1}{2} & 0 \\ -2 & -2 & 0 & 0 & 0 & 1 \end{array}\right)$

$\to \left(\begin{array}{ccc|ccc} 2 & 0 & -2 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -2 & -2 & 1 & 1 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & -2 & -2 & 1 & 1 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & -2 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 6 & 3 & -1 & 1 \end{array}\right)$

$\to \left(\begin{array}{ccc|ccc} 2 & 0 & 0 & 1 & 1 & 0 \\ 0 & -\frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 6 & 3 & -1 & 1 \end{array}\right)$. $\therefore P^T = \begin{bmatrix} 1 & 1 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 3 & -1 & 1 \end{bmatrix}$. $P = \begin{bmatrix} 1 & -\frac{1}{2} & 3 \\ 1 & \frac{1}{2} & -1 \\ 0 & 0 & 1 \end{bmatrix}$.

$\because \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = P \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$ $\therefore \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = P^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & -1 \\ -1 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2}x_1 + \frac{1}{2}x_2 - x_3 \\ -x_1 + x_2 + 4x_3 \\ x_3 \end{pmatrix}$.

即: $\begin{cases} y_1 = \frac{1}{2}x_1 + \frac{1}{2}x_2 - x_3 \\ y_2 = -x_1 + x_2 + 4x_3 \\ y_3 = x_3 \end{cases}$ 则有: $f(y_1, y_2, y_3) = 2y_1^2 - \frac{1}{2}y_2^2 + 6y_3^2$.

(6) $f(x_1, x_2, x_3) = -2x_1x_2 + 2x_1x_3 + 4x_2x_3$

解: $A = \begin{bmatrix} 0 & -1 & 1 \\ -1 & 0 & 2 \\ 1 & 2 & 0 \end{bmatrix}$. $(A \mid I) = \left(\begin{array}{ccc|ccc} 0 & -1 & 1 & 1 & 0 & 0 \\ -1 & 0 & 2 & 0 & 1 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} -1 & 0 & 2 & 0 & 1 & 0 \\ 0 & -1 & 1 & 1 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 1 \end{array}\right)$

$\to \left(\begin{array}{ccc|ccc} -2 & -1 & 3 & 1 & 1 & 0 \\ -1 & 0 & 2 & 0 & 1 & 0 \\ 3 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} -2 & -1 & 3 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 3 & 2 & 0 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} -2 & 0 & 3 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 3 & \frac{1}{2} & 0 & 0 & 0 & 1 \end{array}\right)$

$\to \left(\begin{array}{ccc|ccc} -2 & 0 & 3 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{9}{2} & \frac{3}{2} & \frac{3}{2} & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} -2 & 0 & 0 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & \frac{1}{2} & \frac{9}{2} & \frac{3}{2} & \frac{3}{2} & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} -2 & 0 & 0 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & \frac{1}{2} & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 4 & 2 & 1 & 1 \end{array}\right)$

$\to \left(\begin{array}{ccc|ccc} -2 & 0 & 0 & 1 & 1 & 0 \\ 0 & \frac{1}{2} & 0 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 4 & 2 & 1 & 1 \end{array}\right)$. $\therefore P^T = \begin{pmatrix} 1 & 1 & 0 \\ -\frac{1}{2} & \frac{1}{2} & 0 \\ 2 & 1 & 1 \end{pmatrix}$. $P = \begin{pmatrix} 1 & -\frac{1}{2} & 2 \\ 1 & \frac{1}{2} & 1 \\ 0 & 0 & 1 \end{pmatrix}$.

$$\text{记 } \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = P \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}. \quad \text{即 } \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} = P^{-1} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} \frac{1}{2} & \frac{1}{2} & -\frac{3}{2} \\ -1 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$$

$$\text{记 } \begin{cases} y_1 = \frac{1}{2}x_1 + \frac{1}{2}x_2 - \frac{3}{2}x_3 \\ y_2 = -x_1 + x_2 + x_3 \\ y_3 = -x_3 \end{cases} \quad \text{则有: } f(y_1, y_2, y_3) = -2y_1^2 + \frac{1}{2}y_2^2 + 4y_3^2.$$

4. 设 A 为 n 阶实对称矩阵. 证明: 若对于任何 n 维向量 $\vec{x}$, 二次型 $f(x_1, x_2, \cdots x_n) = \vec{x}^T A \vec{x} = 0$. 则 $A = 0$.

证明: $\because$ A 为 n 阶实对称矩阵. $\therefore$ 存在正交阵 S, 使 $S^T A S = \Lambda$. 则有: $A = S \Lambda S^T$.

$$f(x_1, x_2, \cdots x_n) = \vec{x}^T A \vec{x} = \vec{x}^T S \Lambda S^T \vec{x} = (S^T \vec{x})^T \Lambda (S^T \vec{x}).$$

记 $\vec{y} = S^T \vec{x}$. 则有: $f(\vec{y}) = \vec{y}^T \Lambda \vec{y} = 0$.

由 $\vec{x}$ 的任意性, 可依次取 $\vec{y}_1 = (1, 0, \cdots, 0)^T$, $\vec{y}_2 = (0, 1, 0, 0 \cdots 0)^T$, $\cdots$ $\vec{y}_n = (0, 0, \cdots, 0, 1)^T$.

则得: $\lambda_1 = \lambda_2 = \cdots = \lambda_n = 0$. $\therefore \text{rank}(A) = 0$ $\therefore A = 0$.

5. 已知二次型 $f(x_1, x_2, x_3) = 2x_1^2 + 3x_2^2 + 3x_3^2 + 2ax_1x_3$ $(a > 0)$. 通过正交变换可化为标准形 $f(y_1, y_2, y_3) = y_1^2 + 2y_2^2 + 5y_3^2$. 求参数 a 及所用的正交变换矩阵.

解:
$$A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 3 & a \\ 0 & a & 3 \end{bmatrix}. \qquad S^T A S = \Lambda = \begin{bmatrix} 1 & & \\ & 2 & \\ & & 5 \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} 2-\lambda & 0 & 0 \\ 0 & 3-\lambda & a \\ 0 & a & 3-\lambda \end{vmatrix} = (2-\lambda)[(3-\lambda)+a][(3-\lambda)-a] = 0.$$

$\because a > 0$. $\therefore \begin{cases} 3-a=1 \\ 3+a=5 \end{cases} \Rightarrow a = 2$, $\lambda_1 = 1$, $\lambda_2 = 2$, $\lambda_3 = 5$.

对应的 $\vec{x}_1 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$. $\vec{x}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$. $\vec{x}_3 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

标准化: $\vec{\eta}_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$. $\vec{\eta}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}$ $\vec{\eta}_3 = \frac{1}{\sqrt{2}} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. $\therefore S = \begin{bmatrix} 0 & 1 & 0 \\ \frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} & 0 & \frac{1}{\sqrt{2}} \end{bmatrix}$

则有: $S^T A S = \begin{bmatrix} 1 & & \\ & 2 & \\ & & 5 \end{bmatrix}$.

6、设矩阵 $A=\begin{bmatrix}2&0&0\\0&2&1\\0&1&x\end{bmatrix}$ 有一个特征值1. (1) 求 x.

(2) 求矩阵 P. 使得 $(AP)^T(AP)$ 为对角阵.

解: (1) $|A-\lambda I|=\begin{vmatrix}2-\lambda&0&0\\0&2-\lambda&1\\0&1&x-\lambda\end{vmatrix}=(2-\lambda)\left[\lambda^2-(x+2)\lambda+2x-1\right]=0$. 由 $\lambda=1 \Rightarrow x=2$.

(2) $(AP)^T(AP)=P^T(A^TA)P$. $A^TA=A^2=\begin{bmatrix}2&0&0\\0&2&1\\0&1&2\end{bmatrix}\begin{bmatrix}2&0&0\\0&2&1\\0&1&2\end{bmatrix}=\begin{bmatrix}4&0&0\\0&5&4\\0&4&5\end{bmatrix}$.

$|A^2-\lambda I|=\begin{vmatrix}4-\lambda&0&0\\0&5-\lambda&4\\0&4&5-\lambda\end{vmatrix}=(4-\lambda)(\lambda-1)(\lambda-9)=0 \Rightarrow \lambda_1=1,\ \lambda_2=4,\ \lambda_3=9$.

对应的 $\vec{x}_1=\begin{pmatrix}0\\1\\-1\end{pmatrix}$ $\vec{x}_2=\begin{pmatrix}1\\0\\0\end{pmatrix}$ $\vec{x}_3=\begin{pmatrix}0\\1\\1\end{pmatrix}$ 正交阵 $P=\begin{bmatrix}0&1&0\\\frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{2}}\\-\frac{1}{\sqrt{2}}&0&\frac{1}{\sqrt{2}}\end{bmatrix}$.

标准化: $\vec{\eta}_1=\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\-1\end{pmatrix}$ $\vec{\eta}_2=\begin{pmatrix}1\\0\\0\end{pmatrix}$ $\vec{\eta}_3=\frac{1}{\sqrt{2}}\begin{pmatrix}0\\1\\1\end{pmatrix}$

容易验证: $P^TA^2P=P^TA^TAP=(AP)^T(AP)=\begin{bmatrix}1&&\\&4&\\&&9\end{bmatrix}$ 为对角阵.

练习(补充): (正定二次型)

1. 设二次型 $f(x_1,x_2,x_3)=x_1^2+4x_2^2+4x_3^2+2\lambda x_1x_2-2x_1x_3+4x_2x_3$

求当 λ 为何值时, f 为正定二次型

解: $A=\begin{bmatrix}1&\lambda&-1\\\lambda&4&2\\-1&2&4\end{bmatrix}$ $|A_1|=1>0$. $|A_2|=\begin{vmatrix}1&\lambda\\\lambda&4\end{vmatrix}=4-\lambda^2>0$.

$|A_3|=|A|=-4(\lambda+2)(\lambda-1)>0$, 由此可得 $\begin{cases}4-\lambda^2>0\\-4(\lambda+2)(\lambda-1)>0\end{cases} \Rightarrow -2<\lambda<1$.

2、设矩阵 $A=\begin{bmatrix}1&0&1\\0&2&0\\1&0&1\end{bmatrix}$ 矩阵 $B=(kI+A)^2$. 其中 k 为实数, I 为三阶单位阵.

求对角阵 Λ, 使 $B\sim\Lambda$ 相似, 并求 k 为何值时, B 为正定矩阵.

解: $\because A^T=A$. $\therefore B^T=\left[(kI+A)^2\right]^T=\left[(kI+A)^T\right]^2=\left[kI+A^T\right]^2=(kI+A)^2=B$.

$\therefore B$ 也是实对称阵. $\therefore B$ 必可正交相似于对角阵.

$|A-\lambda I|=\begin{vmatrix}1-\lambda&0&1\\0&2-\lambda&0\\1&0&1-\lambda\end{vmatrix}=\lambda(2-\lambda)^2=0 \Rightarrow \lambda_1=0,\ \lambda_2=\lambda_3=2$

存在正交阵 S，使得 $S^TAS=\begin{bmatrix}0&&\\&2&\\&&2\end{bmatrix}$，即 $A=S\begin{bmatrix}0&&\\&2&\\&&2\end{bmatrix}S^T$.

$\therefore B=(kI+A)^2=\left(kSS^T+S\begin{bmatrix}0&&\\&2&\\&&2\end{bmatrix}S^T\right)^2=\left(S\begin{bmatrix}k&&\\&k+2&\\&&k+2\end{bmatrix}S^T\right)^2$

$=S\begin{pmatrix}k^2&&\\&(k+2)^2&\\&&(k+2)^2\end{pmatrix}S^T$. $\overset{\text{记}}{\Sigma}\ S=Q^{-1}$，$S^T=Q$. 则有：$B=Q^{-1}\begin{bmatrix}k^2&&\\&(k+2)^2&\\&&(k+2)^2\end{bmatrix}Q$.

$\therefore$ 所求对角阵 $\Lambda=\begin{bmatrix}k^2&&\\&(k+2)^2&\\&&(k+2)^2\end{bmatrix}$，此时 $B\sim\Lambda$ 相似.

当 $k\neq0$ 且 $k\neq-2$ 时，Λ 的特征值即 B 的特征值均为正数，此时 B 正定.

3. 设 A 为 $m\times n$ 实矩阵，而 $B=\lambda I+A^TA$，其中 λ 为实数，证明：当 $\lambda>0$ 时，B 正定.

证明：$B^T=(\lambda I+A^TA)^T=\lambda I+A^TA=B$. $\therefore B$ 为 n 阶实对称矩阵.

$\forall\vec{x}\in R^n$，有：$\vec{x}^TB\vec{x}=\vec{x}^T(\lambda I+A^TA)\vec{x}=\lambda\vec{x}^T\vec{x}+\vec{x}^TA^TA\vec{x}$

$=\lambda(\vec{x}^T\vec{x})+(A\vec{x})^T(A\vec{x})$.

当 $\vec{x}\neq\vec{0}$ 时，$\vec{x}^T\vec{x}>0$，$(A\vec{x})^T(A\vec{x})\geq0$.

$\therefore$ 当 $\lambda>0$ 时，$\vec{x}^TB\vec{x}=\lambda(\vec{x}^T\vec{x})+(A\vec{x})^T(A\vec{x})>0$. $(\forall\vec{x}\in R^n,\vec{x}\neq\vec{0})$.

$\therefore$ B 正定.

§7. 正定二次型

习题. P. 313.

1. 证明：将正定矩阵 A 化成 $A=LL^T$ 的 L 是唯一的，这里 L 是对角元均为正的下三角矩阵.

（对 A 作 Cholesky 分解的过程中，由 A 的第一行元素唯一确定 L 的第一列元素，再由 A 的第二行元素唯一确定 L 的第二列元素，依次类推，即知 L 是唯一的）

2. 若 A 和 B 为同阶正定阵，对下述论断，或证明其成立，或举例说明其不成立：

(1) $A+B$ 正定. (2) AB 正定.

证明：(1) 成立. $\because A$ 和 B 均正定. $\therefore A^T=A$，$B^T=B\Rightarrow(A+B)^T=A^T+B^T=A+B$.

$\forall\vec{x}\in R^n$，$\vec{x}\neq\vec{0}$，成立：$\vec{x}^TA\vec{x}>0$，$\vec{x}^TB\vec{x}>0\Rightarrow\vec{x}^T(A+B)\vec{x}>0$.

$\therefore A+B$ 正定.

(2) 不成立. 例：$A=\begin{bmatrix}2&1\\1&3\end{bmatrix}$，$B=\begin{bmatrix}5&1\\1&4\end{bmatrix}$，$A,B$ 均正定.

设 $AB=\begin{bmatrix}2&1\\1&3\end{bmatrix}\begin{bmatrix}5&1\\1&4\end{bmatrix}=\begin{bmatrix}11&6\\8&13\end{bmatrix}\neq BA$. AB为对称. ∴ AB为正定.

3. 判断以下矩阵是否正定.

(1) $A=\begin{pmatrix}1&2\\2&1\end{pmatrix}$. 解: $A_1=1>0$. $A_2=|A|=-3<0$. ∴ A为正定.

(2) $A=\begin{pmatrix}2&1&2\\1&2&1\\2&1&3\end{pmatrix}$. 解: $A_1=2>0$. $A_2=\begin{vmatrix}2&1\\1&2\end{vmatrix}=3>0$. $A_3=|A|=3>0$. ∴ A正定.

(3) $A=\begin{pmatrix}2&1&4&6\\1&2&1&5\\4&1&3&4\\6&5&4&2\end{pmatrix}$. 解: $A_1=2>0$. $A_2=\begin{vmatrix}2&1\\1&2\end{vmatrix}=3>0$. $A_3=\begin{vmatrix}2&1&4\\1&2&1\\4&1&3\end{vmatrix}=-1<0$. ∴ A为正定.

(4) $A=\begin{pmatrix}1&0&-1\\0&2&1\\-1&1&3\end{pmatrix}$. 解: $A_1=1>0$. $A_2=\begin{vmatrix}1&0\\0&2\end{vmatrix}=2>0$. $A_3=|A|=3>0$. ∴ A正定.

4. 求非奇矩阵P, 使正定矩阵A为 P^TP 形式:

(1) $A=\begin{pmatrix}1&-1\\-1&2\end{pmatrix}$.

解: $(A\,|\,I)=\left(\begin{array}{cc|cc}1&-1&1&0\\-1&2&0&1\end{array}\right)\to\left(\begin{array}{cc|cc}1&-1&1&0\\0&1&1&1\end{array}\right)$ ∴ $Q^T=\begin{pmatrix}1&0\\1&1\end{pmatrix}$. $Q=\begin{pmatrix}1&1\\0&1\end{pmatrix}$.

则有: $Q^TAQ=\begin{pmatrix}1&0\\0&1\end{pmatrix}\Rightarrow A=(Q^T)^{-1}Q^{-1}=(Q^{-1})^TQ^{-1}$.

∴ $P=Q^{-1}=\begin{bmatrix}1&1\\0&1\end{bmatrix}^{-1}=\begin{bmatrix}1&-1\\0&1\end{bmatrix}$. 此时: $A=P^TP$.

(2) $A=\begin{pmatrix}1&-1&0\\-1&2&-2\\0&-2&5\end{pmatrix}$

解: $(A\,|\,I)=\left(\begin{array}{ccc|ccc}1&-1&0&1&0&0\\-1&2&-2&0&1&0\\0&-2&5&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&-1&0&1&0&0\\0&1&-2&1&1&0\\0&-2&5&0&0&1\end{array}\right)\to\left(\begin{array}{ccc|ccc}1&-1&0&1&0&0\\0&1&-2&1&1&0\\0&0&1&2&2&1\end{array}\right)$

∴ $Q^T=\begin{pmatrix}1&0&0\\1&1&0\\2&2&1\end{pmatrix}$ 即 $Q=\begin{pmatrix}1&1&2\\0&1&2\\0&0&1\end{pmatrix}$. 则有: $Q^TAQ=I$.

$$\Rightarrow A=(-Q^T)^{-1}Q^{-1}=(Q^{-1})^TQ^{-1}. \quad \therefore P=Q^{-1}=\begin{bmatrix}1&1&2\\0&1&2\\0&0&1\end{bmatrix}^{-1}=\begin{bmatrix}1&-1&0\\0&1&-2\\0&0&1\end{bmatrix}$$

5. 试证 $A=\begin{pmatrix}3&-1\\-1&3\end{pmatrix}$ 为正定矩阵，并求正定矩阵 B，使 $B^2=A$.

证明：$A_1=3>0$. $A_2=|A|=8>0$. $\therefore$ A 正定.

$$|A-\lambda I|=\begin{vmatrix}3-\lambda&-1\\-1&3-\lambda\end{vmatrix}=\lambda^2-6\lambda+8=(\lambda-2)(\lambda-4)=0 \Rightarrow \lambda_1=2 \quad \lambda_2=4.$$

对应的：$\vec{x}_1=\begin{pmatrix}1\\1\end{pmatrix}$. $\vec{x}_2=\begin{pmatrix}-1\\1\end{pmatrix}$

标准化：$\vec{\eta}_1=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\1\end{pmatrix}$. $\vec{\eta}_2=\frac{1}{\sqrt{2}}\begin{pmatrix}1\\-1\end{pmatrix}$. $\Rightarrow$ 正交阵：$S=\frac{1}{\sqrt{2}}\begin{bmatrix}1&-1\\1&1\end{bmatrix}$.

成立：$S^TAS=\begin{bmatrix}2&\\&4\end{bmatrix}$.

$$A=S\begin{bmatrix}2&\\&4\end{bmatrix}S^T=S\begin{bmatrix}\sqrt{2}&\\&2\end{bmatrix}S^T\cdot S\begin{bmatrix}\sqrt{2}&\\&2\end{bmatrix}S^T=B^2.$$

$$\therefore B=S\begin{bmatrix}\sqrt{2}&\\&2\end{bmatrix}S^T=\begin{bmatrix}\frac{\sqrt{2}+2}{2}&\frac{\sqrt{2}-2}{2}\\\frac{\sqrt{2}-2}{2}&\frac{\sqrt{2}+2}{2}\end{bmatrix}$$

显然 B 合同于对角阵，且 $\mu_1=\sqrt{2}>0$. $\mu_2=2>0$. $\therefore$ B 正定.

6. 设 A 为正定阵，证明对任何正整数 m，存在正定矩阵 B，使 $B^m=A$.

证明：$\because$ A 正定 $\therefore$ A 的所有特征值 $\lambda_i>0$. 即存在正交阵 S，使得：

$$S^TAS=\Lambda=\mathrm{diag}(\lambda_1,\lambda_2,\cdots\lambda_n) \quad \because \lambda_i>0 \quad \therefore \text{可令 } \mu_i=\sqrt[m]{\lambda_i}, \text{ 则有：}$$

$$A=S\Lambda S^T=\underbrace{S\cdot\mathrm{diag}(\mu_1,\mu_2,\cdots,\mu_n)S^T}_{B}\underbrace{S\cdot\mathrm{diag}(\mu_1,\mu_2,\cdots,\mu_n)S^T}_{B}\cdots\underbrace{S\cdot\mathrm{diag}(\mu_1,\cdots\mu_n)S^T}_{B}$$

$$=B^m, \quad \text{其中 } B=S\cdot\mathrm{diag}(\mu_1,\mu_2,\cdots,\mu_n)\cdot S^T \quad \forall\mu_i>0 \quad \therefore B\text{正定}$$

7. 设 $f(x_1,x_2,\cdots x_n)=\vec{x}^TA\vec{x}$ 是一个二次型，若对于任意实数 $x_1,x_2,\cdots,x_n$ 总有

$$f(x_1,x_2,\cdots,x_n)\le 0$$

且当等成立当且仅当 $x_1=x_2=\cdots=x_n=0$. 则称 $f(x_1,x_2,\cdots x_n)$ 是负定二次型，相应的对称矩阵 A 称为负定矩阵. 证明：A 为负定矩阵的充要条件为 $(-1)^k\det(A_{(1,2,\cdots k)})\ge 0$. $(k=1,\cdots n)$

(略)

8. $f(x_1,x_2,\cdots x_n)=\vec{x}^TA\vec{x}$ 是一个二次型，若对于任意实数 $x_1,x_2,\cdots x_n$ 总有

$f(x_1, x_2, \cdots x_n) \geq 0$.

则称 $f(x_1, x_2, \cdots, x_n)$ 是半正定二次型，相应的对称矩阵称为半正定矩阵。此时半正定矩阵(二次型)具有类似定理 5.7.2 的性质，可以证明。

(证明).

9. 确定 λ 的取值范围，使得下列二次型为正定的：

(1) $5x_1^2 + x_2^2 + \lambda x_3^2 + 4x_1x_2 - 2x_1x_3 - 2x_2x_3$

解：$A = \begin{bmatrix} 5 & 2 & -1 \\ 2 & 1 & -1 \\ -1 & -1 & \lambda \end{bmatrix}$ 若 A 正定，则有：$A_1 = 5 > 0$. $A_2 = \begin{vmatrix} 5 & 2 \\ 2 & 1 \end{vmatrix} = 1 > 0$.

$A_3 = |A| = \begin{vmatrix} 5 & 2 & -1 \\ 2 & 1 & -1 \\ -1 & -1 & \lambda \end{vmatrix} = 5\lambda + 2 + 2 - 1 - 5 - 4\lambda = \lambda - 2 > 0$. $\therefore \lambda > 2$.

(2) $x_1^2 + x_2^2 + 5x_3^2 + 2\lambda x_1x_2 - 2x_1x_3 + 4x_2x_3$

解：$A = \begin{bmatrix} 1 & \lambda & -1 \\ \lambda & 1 & 2 \\ -1 & 2 & 5 \end{bmatrix}$. 若 A 正定，则有：$A_1 = 1 > 0$. $A_2 = \begin{vmatrix} 1 & \lambda \\ \lambda & 1 \end{vmatrix} = 1 - \lambda^2 > 0$.

$A_3 = |A| = \begin{vmatrix} 1 & \lambda & -1 \\ \lambda & 1 & 2 \\ -1 & 2 & 5 \end{vmatrix} = 5 - 4\lambda - 1 - 4 - 5\lambda^2 = -5\lambda^2 - 4\lambda > 0$.

由 $\begin{cases} 1 - \lambda^2 > 0 \\ \lambda(5\lambda + 4) < 0 \end{cases} \Rightarrow -\frac{4}{5} < \lambda < 0$.

10. 设 A 是可逆矩阵，证明 A^TA 是正定矩阵.

证明：$\because (A^TA)^T = A^TA$ $\therefore A^TA$ 对称.

$\forall \vec{x} \in R^n$, $\vec{x} \neq \vec{0}$. 有：$\vec{x}^T(A^TA)\vec{x} = \vec{x}^TA^TA\vec{x} = (A\vec{x})^TA\vec{x}$.

$\because A$ 可逆. $\therefore A\vec{x} = \vec{0} \Leftrightarrow \vec{x} = \vec{0}$

$\therefore \forall \vec{x} \in R^n$, $\vec{x} \neq \vec{0} \Rightarrow A\vec{x} \neq \vec{0}$. $\therefore \vec{x}^T(A^TA)\vec{x} = (A\vec{x})^TA\vec{x} > 0$.

$\therefore A^TA$ 正定.

11. 设 A 为正定矩阵，证明：若 A 同时也正交矩阵，则 A 是单位阵.

证明：$\because A$ 正定. $\therefore$ 存在可逆阵 P，使 $A = P^TP$ (第4题提示. 正定阵合同于单位阵)

$\because A$ 为正交阵. $\therefore A^TA = I$. 即 $P^TPP^TP = (P^TP)^2 = I$. $\therefore A^2 = I$.

$\Rightarrow (A+I)(A-I) = 0$. $\Rightarrow A$ 的特征值为 ± 1. $\because A$ 正定. $\therefore A$ 的特征值为 1

$\therefore A = I$. $(A = S^TIS = S^TS = I)$.

12. 设 A 是一个 n 阶实对称矩阵，证明：当 t 充分大时，$A+tI_n$ 是正定矩阵。

证明：$\because A^T=A$（A 是实对称）。

$\therefore (A+tI_n)^T=A^T+tI_n^T=A+tI_n$。$\therefore A+tI_n$ 为实对称阵。

$\vec{x}^T(A+tI_n)\vec{x}=\vec{x}^TA\vec{x}+t\vec{x}^TI_n\vec{x}=\vec{x}^TA\vec{x}+t\vec{x}^T\vec{x}$

$\forall \vec{x}\in R^n$，$\vec{x}\neq\vec{0}$，当 $t>-\dfrac{\vec{x}^TA\vec{x}}{\|\vec{x}\|^2}$ 时，即有：$\vec{x}^T(A+tI_n)\vec{x}>0$。

$\therefore A+tI_n$ 正定。

13. 设 A,B 是实对称矩阵，而且 A 是正定的，证明：存在 n 阶实可逆矩阵 P，使得 P^TAP 和 P^TBP 都是对角矩阵。

证明：$\because A$ 正定，$\therefore$ 存在可逆阵 Q，使得 $Q^TAQ=I$（正定阵合同于单位阵）。

令 $C=Q^TBQ$，则 $C^T=Q^TB^TQ=Q^TBQ=C$。$\therefore C$ 实对称阵。

$\therefore$ 存在正交阵 S，使得：$S^TCS=\Lambda$，$S^TS=I$。

取 $P=QS$，则 P 可逆（$\because Q,S$ 可逆），且有：

$P^TAP=S^TQ^TAQS=S^TS=I$ 为对角阵。

$P^TBP=S^TQ^TBQS=S^TCS=\Lambda$ 也为对角阵。

14. 求下列矩阵的 Cholesky 分解：

(1) $A=\begin{bmatrix}4&0&-4\\0&1&1\\-4&1&6\end{bmatrix}$

解：$(A\mid I)=\left(\begin{array}{ccc|ccc}4&0&-4&1&0&0\\0&1&1&0&1&0\\-4&1&6&0&0&1\end{array}\right)\longrightarrow\left(\begin{array}{ccc|ccc}4&0&-4&1&0&0\\0&1&1&0&1&0\\0&1&2&1&0&1\end{array}\right)\longrightarrow\left(\begin{array}{ccc|ccc}4&0&-4&1&0&0\\0&1&1&0&1&0\\0&0&1&1&-1&1\end{array}\right)$

$\therefore L=L_1^{-1}\begin{pmatrix}\sqrt{a_{11}}&&\\&\sqrt{a_{22}}&\\&&\sqrt{a_{33}}\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\1&-1&1\end{pmatrix}^{-1}\begin{pmatrix}2&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}1&0&0\\0&1&0\\-1&1&1\end{pmatrix}\begin{pmatrix}2&0&0\\0&1&0\\0&0&1\end{pmatrix}=\begin{pmatrix}2&0&0\\0&1&0\\-2&1&1\end{pmatrix}$

$\therefore A=L\cdot L^T=\begin{bmatrix}2&0&0\\0&1&0\\-2&1&1\end{bmatrix}\begin{bmatrix}2&0&-2\\0&1&1\\0&0&1\end{bmatrix}=\begin{bmatrix}4&0&-4\\0&1&1\\-4&1&6\end{bmatrix}$

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(2) $B=\begin{pmatrix} 9 & 6 & 3 \\ 6 & 5 & 2 \\ 3 & 2 & 2 \end{pmatrix}$.

解: $(B\mid I)=\left(\begin{array}{ccc|ccc} 9 & 6 & 3 & 1 & 0 & 0 \\ 6 & 5 & 2 & 0 & 1 & 0 \\ 3 & 2 & 2 & 0 & 0 & 1 \end{array}\right) \to \left(\begin{array}{ccc|ccc} 9 & 6 & 3 & 1 & 0 & 0 \\ 0 & 1 & 0 & -\frac{2}{3} & 1 & 0 \\ 0 & 0 & 1 & -\frac{1}{3} & 0 & 1 \end{array}\right)$ $L_1=\begin{pmatrix} 1 & 0 & 0 \\ -\frac{2}{3} & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \end{pmatrix}$.

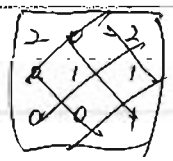
$L=L_1^{-1}\begin{pmatrix} 3 & & \\ & 1 & \\ & & 1 \end{pmatrix}=\begin{pmatrix} 1 & 0 & 0 \\ -\frac{2}{3} & 1 & 0 \\ -\frac{1}{3} & 0 & 1 \end{pmatrix}^{-1}\begin{pmatrix} 3 & & \\ & 1 & \\ & & 1 \end{pmatrix}=\begin{pmatrix} 1 & 0 & 0 \\ \frac{2}{3} & 1 & 0 \\ \frac{1}{3} & 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & & \\ & 1 & \\ & & 1 \end{pmatrix}=\begin{pmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

$\therefore B=LL^T=\begin{pmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}=\begin{pmatrix} 9 & 6 & 3 \\ 6 & 5 & 2 \\ 3 & 2 & 2 \end{pmatrix}$.

15. 利用上题结果，用 Cholesky 分解求解方程组：

(2) $A\vec{x}=\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$.

解: $A=L\cdot L^T$. $A\vec{x}=L\cdot L^T\vec{x}=L\vec{y}=\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$. 设 $\vec{y}=L^T\vec{x}=\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$.

即 

$\begin{pmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 1 & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 1 \\ 2 \end{pmatrix}$. $\therefore \vec{y}=\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$.

$\vec{y}=L^T\vec{x}=\begin{pmatrix} 2 & 0 & -2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}$. $\therefore \vec{x}=\begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}$.

(3) $B\vec{x}=\begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$.

解: $B=L\cdot L^T$. $B\vec{x}=L\cdot L^T\vec{x}=L\vec{y}=\begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$. 设 $\vec{y}=L^T\vec{x}=\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$.

即 $\begin{pmatrix} 3 & 0 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix}\begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}=\begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$ $\therefore \vec{y}=\begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$.

$\vec{y}=L^T\vec{x}=\begin{pmatrix} 3 & 2 & 1 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}=\begin{pmatrix} 0 \\ -1 \\ -1 \end{pmatrix}$ $\therefore \vec{x}=\begin{pmatrix} 1 \\ -1 \\ -1 \end{pmatrix}$.